Complex Tori and Elliptic Curves

Why the Abelian Group Law Must Be the Chord–Tangent Law
via the $\wp$-Embedding

Spring 2026

A textbook developed from midterm and final exam themes: differential forms, residues, compact Riemann surfaces, divisors, Riemann–Roch, complex tori, and the Weierstrass $\wp$-function.

Preface

This book explains one fact in full: the chord–tangent law on a smooth plane cubic is not an arbitrary geometric trick but the abelian group law on the complex torus $\mathbb{C}/\Lambda$, carried onto the cubic by the Weierstrass functions $\wp$ and $\wp'$. Addition on the torus is addition of complex numbers modulo a lattice — associative and commutative for the most elementary reasons — and the embedding $z \mapsto [\wp(z) : \wp'(z) : 1]$ turns it into "draw a line, take the third intersection, reflect." Seen this way, every feature of the chord–tangent rule, including the puzzling reflection step, is forced.

The treatment is meant to be *self-contained*. The genus-one theory is built from a single course in complex analysis: the dimension counts that single out $1, \wp, \wp'$ are derived elementarily from Liouville's theorems and the explicit Weierstrass functions, and the converse of Abel's law is proved constructively with the Weierstrass σ-function. Riemann–Roch is stated as the general theorem for context, but nothing in the book depends on it. Every figure is drawn from coordinates computed exactly by the Sage scripts of appendix C, so the pictures are reproducible rather than schematic, and roughly eighty exercises, all with complete solutions in appendix D, let the reader reconstruct the theory.

Audience and prerequisites. The intended reader is a strong final-year undergraduate or beginning graduate student. We assume one course in single-variable complex analysis (holomorphic and meromorphic functions, Laurent series, residues, Liouville's theorem, the argument principle — all recalled in appendix A), comfort with quotient spaces, and the rudiments of abelian groups and linear algebra. No prior acquaintance with Riemann surfaces, divisors, or Riemann–Roch is needed.

How to read it. The chapters are linear, and fig. 1.1 maps the whole argument on the first pages. A reader meeting the subject for the first time may skim the convergence, Eisenstein-series, and field-generation material of chapter 7 — the most technical in the book — without losing the main thread. The List of Symbols that follows fixes notation, and appendix B records how the material grew out of a course's midterm and final.

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List of Symbols

Number systems and spaces	
$\mathbb{C}, \mathbb{R}, \mathbb{Z}, \mathbb{Q}$	complex, real, integer, rational numbers
$\mathbb{F}$	a field (e.g. a finite field $\mathbb{F}_p$)
$\mathbb{CP}^n$	complex projective space
$\mathbb{H}$	upper half-plane $\{\operatorname{Im}\tau > 0\}$
Lattices and the torus	
$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$	a lattice; ω_1, ω_2 its periods
$\tau = \omega_2/\omega_1$	period ratio, normalized to $\mathbb{H}$
P_Λ	fundamental parallelogram
$T = \mathbb{C}/\Lambda$	the complex torus
O	identity; image of 0; point at infinity of E
$m(\Lambda)$	least length of a nonzero lattice vector
Weierstrass functions	
$\wp,\ \wp'$	Weierstrass function (double pole) and derivative
$\zeta = \sigma'/\sigma$	Weierstrass zeta function, $\zeta' = -\wp$
σ	Weierstrass sigma function
G_k	Eisenstein series of weight k
g_2, g_3	invariants, $g_2 = 60G_4$, $g_3 = 140G_6$
Δ	discriminant $g_2^3 - 27g_3^2$
e_1, e_2, e_3	roots of $4x^3 - g_2x - g_3$; values $\wp(\omega_i/2)$
η_k	quasi-periods, $\eta_k = 2\zeta(\omega_k/2)$
j	the j-invariant $1728\, g_2^3/\Delta$
Functions and fields	
$\mathcal{O},\ \mathcal{M}$	holomorphic, meromorphic functions
$\mathbb{C}(\wp, \wp')$	field of elliptic functions
Divisors and Riemann–Roch	
$\operatorname{Div}(X),\ \operatorname{Div}^0(X)$	divisors; degree-zero divisors
$\operatorname{div}(f)$	divisor of f (a principal divisor)
$\operatorname{ord}_p,\ \operatorname{Res},\ \deg$	order at p, residue, degree
$\operatorname{Prin}(X)$	principal divisors
$\operatorname{Pic}(X),\ \operatorname{Pic}^0(X)$	Picard group; degree-zero Picard group
K	canonical divisor ($K \sim 0$ on T)
$L(D),\ \ell(D)$	Riemann–Roch space and its dimension
$\mathfrak{s},\ \bar{\mathfrak{s}}$	Abel–Jacobi sum map $\operatorname{Div}^0(T) \to T$ and $\operatorname{Pic}^0(T) \to T$
The cubic and its group law	
E	the cubic $Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3$
Φ	Weierstrass embedding $T \to E$
$\oplus,\ \ominus$	group law on E and its inverse (reflection)
$\boxplus$	chord–tangent operation (equal to $\oplus$)

Part I

Tori as Riemann Surfaces

Chapter 1

Introduction and Overview

1.1 A Construction That Should Not Work

Take a smooth cubic curve in the plane, for instance

$$E: \quad y^2 = 4x^3 - g_2 x - g_3,$$

and the following recipe for "adding" two of its points. Given points P and Q on E, draw the straight line through them. A line meets a cubic in three points, so it strikes E at one further point R. Reflect R across the x-axis to obtain a point we call $P \oplus Q$. To add a point to itself, use the tangent line at P in place of the chord. This is the *chord–tangent law*.

Nothing about this recipe looks like it should produce a group. Reflecting a third intersection point is a geometric gesture, not an algebraic operation; there is no visible reason for it to be associative, and no visible reason for inverses to behave. Yet the chord–tangent law *is* an abelian group law, with associativity holding exactly. The standard elementary proofs of associativity are long case analyses or invocations of the Cayley–Bacharach theorem, and they leave the impression that the group structure is a lucky accident of plane geometry.

It is not an accident. These notes explain the following single fact, which makes every group axiom transparent at once.

> The chord–tangent law on a smooth plane cubic is not an arbitrary geometric trick. It is exactly the abelian group law on the complex torus $T = \mathbb{C}/\Lambda$, transported to the cubic by the Weierstrass functions $\wp$ and $\wp'$.

The slogan is

$$\boxed{\text{addition on } \mathbb{C}/\Lambda \quad \Longleftrightarrow \quad \text{chord–tangent addition on } E}$$

where, in homogeneous coordinates $[X : Y : Z]$ on $\mathbb{CP}^2$,

$$E: \quad Y^2 Z = 4X^3 - g_2 X Z^2 - g_3 Z^3.$$

On the torus side, addition is nothing more than addition of complex numbers modulo a lattice Λ; it is associative and commutative for the most elementary of reasons. The entire mystery of the chord–tangent law dissolves once we exhibit a dictionary between the two pictures.

1.2 The Answer Stated First

We state the destination now, before any of the machinery is built, so that every later chapter can be read as a step toward it. The terms appearing here (*lattice*, *Weierstrass function*, *divisor*, *point at infinity*) are defined from scratch in the chapters that follow; forward references point to the precise statements.

Theorem 1.1 (Main theorem)**.** *Let $\Lambda \subset \mathbb{C}$ be a lattice and let $\wp$ be the Weierstrass function of Λ, with derivative $\wp'$. Set*

$$E: \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3 \subset \mathbb{CP}^2,$$

with g_2, g_3 the invariants of Λ. Then the map

$$\Phi : \mathbb{C}/\Lambda \longrightarrow E, \qquad \Phi(z) = \begin{cases} [\wp(z) : \wp'(z) : 1], & z \notin \Lambda, \\ [0:1:0] =: O, & z \in \Lambda, \end{cases}$$

is an isomorphism of compact Riemann surfaces, and it is an isomorphism of groups when E carries the chord–tangent law and $\mathbb{C}/\Lambda$ carries addition modulo Λ. Equivalently: three points of E, counted with intersection multiplicity, are collinear if and only if their Φ-preimages in $\mathbb{C}/\Lambda$ sum to 0.

The map Φ is constructed in chapter 8; that it lands on E comes from the differential equation for $\wp$ in chapter 7; that it is a bijection of Riemann surfaces is chapter 8; and the collinearity statement, which is the heart of the matter, is proved in chapters 10 and 11 using Abel's sum law from chapter 9.

Main conclusion. If a line meets the cubic in three points P, Q, R, counted with intersection multiplicity, then those three points add to zero in the group law transported from the torus:

$$P + Q + R = O.$$

Solving for $P + Q$ gives

$$P + Q = -R.$$

Geometrically, this means: draw the line through P and Q, take the third intersection R, then reflect R across the x-axis.

1.3 The Chain of Ideas

The proof of theorem 1.1 is a chain of structural facts, each of which is of independent interest and each of which occupies one part of this book. The logical skeleton is shown in fig. 1.1.

Read clockwise from the upper left, the chain runs as follows.

1. **The torus carries an obvious group law.** The quotient $\mathbb{C}/\Lambda$ inherits addition from $\mathbb{C}$. Associativity, commutativity, and inverses are immediate because they already hold in $\mathbb{C}$ (chapters 2 and 3).

2. **The torus is a Riemann surface of genus one.** As a smooth compact surface $\mathbb{C}/\Lambda \cong S^1 \times S^1$ is a doughnut; as a Riemann surface it has genus one. This is what eventually forces the cubic equation (chapter 3).

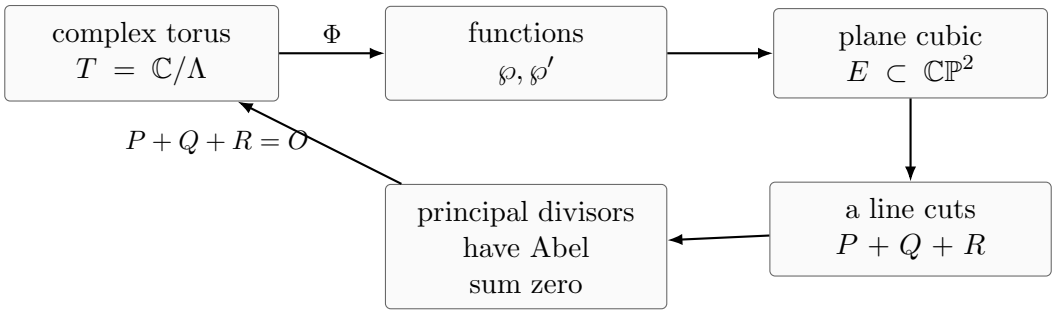


Figure 1.1: The chain of ideas behind theorem 1.1. The torus supplies the group law; the Weierstrass functions $\wp, \wp'$ carry it onto the cubic E; a line meeting E in P, Q, R records the divisor $P+Q+R$; that divisor is principal, so by Abel's sum law its points sum to 0 back on the torus. The loop closes, and the chord–tangent law is forced.

3. **Meromorphic functions on the torus are scarce.** A nonconstant holomorphic function cannot exist (Liouville), and the orders of zeros and poles of a meromorphic function obey rigid global constraints (chapters 4 and 5).

4. **Riemann–Roch counts those functions.** On a genus-one surface the dimension counts are sharp enough to single out a function with a double pole ($\wp$) and one with a triple pole ($\wp'$), and to force a single algebraic relation between them (chapters 6 and 7).

5. **That relation is the cubic.** The differential equation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ says precisely that $(\wp, \wp')$ traces out E. The map Φ of theorem 1.1 is the resulting embedding (chapters 7 and 8).

6. **Lines cut principal divisors.** A line meeting E at P, Q, R defines a meromorphic function on E whose divisor is $P + Q + R - 3O$. Pulled back to the torus, principal divisors have Abel sum zero, which is exactly $P + Q + R = O$ (chapters 9 to 11).

Everything after that — the explicit chord and tangent formulas, the role of the point at infinity as the identity, the two-torsion points, the closed-form addition laws — is the working-out of this one loop (chapters 12 to 16).

1.4 How to Read This Book

The book is organized into six parts that follow the chain above.

Part I — Tori as Riemann Surfaces (chapters 2 and 3).
Lattices, the quotient $\mathbb{C}/\Lambda$, its complex structure, its group law, and its genus.

Part II — Elliptic Functions (chapters 4 and 5).
Doubly periodic meromorphic functions, Liouville's theorems, orders, and divisors.

Part III — Riemann–Roch on Genus One (chapter 6).
The spaces $L(D)$, the canonical divisor, and the dimension counts that make $\wp$ and $\wp'$ inevitable.

Part IV — The Weierstrass Functions (chapters 7 and 8).
Construction of $\wp$, the differential equation, and the embedding Φ into $\mathbb{CP}^2$. The convergence, Eisenstein-series, and field-generation material of chapter 7 is the most technical in the book (graduate level) and may be skimmed on a first reading without losing the main thread.

Part V — The Group Law on the Cubic (chapters 9 to 12).
Abel's sum law, lines as principal divisors, the transport of the group law, and the chord–tangent law case by case.

Part VI — Formulas, Computations, and Synthesis (chapters 13 to 16).
Closed-form addition and doubling formulas, the analytic addition formula, a fully worked numerical example, and pointers onward.

Four appendices support the main text. Appendix A is a self-contained refresher on the complex analysis the book relies on (Liouville's theorem, residues, the argument principle, Laurent series); a reader who is rusty should skim it first. Appendix B records how the present material grew out of a course's midterm and final examinations, for readers who want that bridge. Appendix C documents the Sage scripts that compute the exact coordinates for every figure, so that all pictures are reproducible. Appendix D contains full solutions to every exercise.

Prerequisites. We assume one course in single-variable complex analysis (holomorphic and meromorphic functions, Laurent series, residues, Liouville's theorem, the argument principle), comfort with quotient spaces from point-set topology, and the basics of abelian groups and linear algebra. No prior exposure to Riemann surfaces, divisors, or Riemann–Roch is assumed; these are developed here in the single case of genus one.

Standing notation. The following symbols are fixed throughout. The complex numbers, reals, and integers are $\mathbb{C}$, $\mathbb{R}$, $\mathbb{Z}$; the upper half-plane is $\mathbb{H}$; and $\mathbb{CP}^n$ is complex projective space. A *lattice* is written $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$, and the associated *complex torus* is $T = \mathbb{C}/\Lambda$, with O the image of 0, which will turn out to be the group identity. The Weierstrass function and its derivative are $\wp$ and $\wp'$; the invariants of Λ are g_2, g_3, with discriminant $\Delta = g_2^3 - 27g_3^2$. We write Div for divisors, $\mathrm{div}(f)$ for the divisor of a function, ord_p for order at p, and $L(D)$ for the Riemann–Roch space of a divisor D. Addition on the torus is written $+$; the chord–tangent law on E is written $\oplus$ until theorem 1.1 identifies the two, after which we use $+$ for both.

1.5 Summary

The chord–tangent law looks like an arbitrary geometric construction, but it is the image of ordinary addition on a complex torus under the Weierstrass embedding Φ. Theorem 1.1 is the precise statement, and fig. 1.1 is its proof in outline: the torus gives the group law, $\wp$ and $\wp'$ carry it to the cubic, and the fact that a line cuts a principal divisor closes the loop. The rest of the book builds each link of that chain in turn.

Exercises

Exercise 1.1. In the chord–tangent recipe, a line through two points P, Q of a cubic is asserted to meet the curve in exactly one further point R. Assuming the points are counted with multiplicity and that we work in the projective plane, explain informally why a third intersection point must exist. (The precise tool is Bézout's theorem; see chapter 10.)

Exercise 1.2. Explain why *doubling* a point P (computing $P \oplus P$) uses the tangent line at P rather than a chord. What happens to the chord through P and Q as $Q \to P$?

Exercise 1.3. Recall Liouville's theorem: a bounded entire function is constant. Granting that a meromorphic function on the torus $\mathbb{C}/\Lambda$ with no poles is, in effect, a bounded entire function, explain in one sentence why there is no nonconstant holomorphic function on $\mathbb{C}/\Lambda$. (This is proved carefully in chapter 4.)

Exercise 1.4. Using fig. 1.1 and the chapter list in section 1.4, match each of the five arrows in the diagram to the chapter that establishes it.

Chapter 2

Lattices and the Complex Torus

The object that carries the group law in theorem 1.1 is the quotient $\mathbb{C}/\Lambda$ of the plane by a lattice. This chapter introduces lattices and their fundamental parallelograms, records the one piece of freedom in choosing a basis (the period ratio τ and the action of $\mathrm{SL}_2(\mathbb{Z})$), and then builds the complex structure on $\mathbb{C}/\Lambda$ explicitly, proving that it is a compact Riemann surface. The group law and the genus are taken up in chapter 3.

2.1 Lattices

Definition 2.1. A *lattice* in $\mathbb{C}$ is a subgroup

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

where $\omega_1, \omega_2 \in \mathbb{C}$ are linearly independent over $\mathbb{R}$. The pair (ω_1, ω_2) is called a *basis* of Λ, and ω_1, ω_2 are its *periods*.

Real-linear independence of ω_1 and ω_2 is equivalent to the condition

$$\frac{\omega_2}{\omega_1} \notin \mathbb{R},$$

and it says geometrically that the two periods span a parallelogram of nonzero area. That parallelogram,

$$P_\Lambda = \{\, s\omega_1 + t\omega_2 : 0 \le s, t \le 1 \,\},$$

is the *fundamental parallelogram* of the basis (fig. 2.1). Every point of $\mathbb{C}$ is congruent modulo Λ to exactly one point of the half-open parallelogram $\{s\omega_1 + t\omega_2 : 0 \le s, t < 1\}$, so P_Λ tiles the plane under translation by Λ.

Remark 2.2. A lattice is a *discrete* subgroup of $\mathbb{C}$: the distance

$$m(\Lambda) := \min_{\lambda \in \Lambda \setminus \{0\}} |\lambda|$$

is strictly positive. Indeed, the values $|s\omega_1 + t\omega_2|$ for (s, t) ranging over the compact set $\{\max(|s|, |t|) = 1\}$ attain a positive minimum $c > 0$ by real-linear independence, and any nonzero $\lambda = a\omega_1 + b\omega_2$ with $a, b \in \mathbb{Z}$ has $|\lambda| \ge c \cdot \max(|a|, |b|) \ge c$. This number $m(\Lambda)$ reappears when we build coordinate charts in section 2.5.

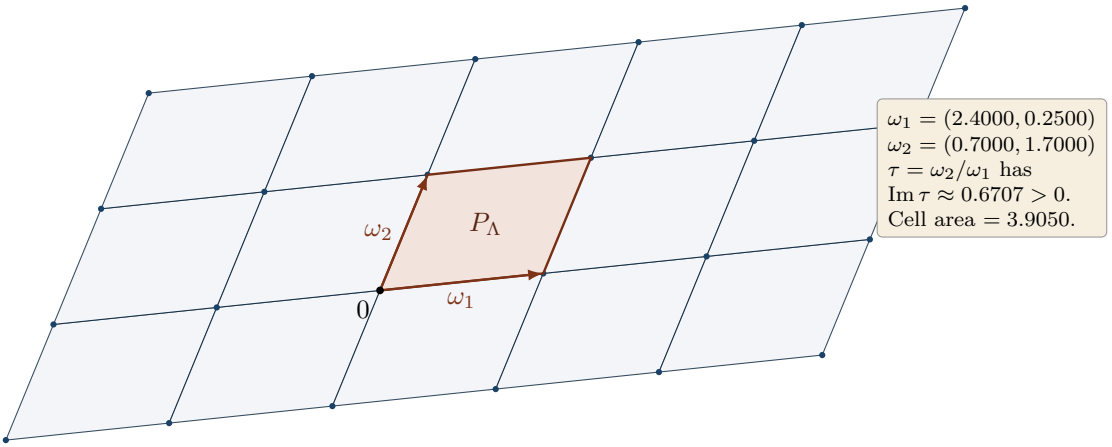


Figure 2.1: A lattice $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ and the fundamental parallelogram P_Λ (shaded). The translates of P_Λ tile the plane. The coordinates shown are computed by `sagemath/01_lattice.sage`.

2.2 Bases and the Period Ratio

A lattice has many bases. If (ω_1, ω_2) is a basis, so is any pair obtained from it by an integer change of variables with determinant ± 1.

Proposition 2.3 (Change of basis)**.** *Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$. A pair (ω_1', ω_2') is another basis of Λ if and only if*

$$\begin{pmatrix} \omega_2' \\ \omega_1' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \omega_2 \\ \omega_1 \end{pmatrix}, \qquad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z}),$$

that is, with $a, b, c, d \in \mathbb{Z}$ and $\det M = ad - bc = \pm 1$.

Proof. If (ω_1', ω_2') is given by an integer matrix M, then $\omega_1', \omega_2' \in \Lambda$, so $\mathbb{Z}\omega_1' + \mathbb{Z}\omega_2' \subseteq \Lambda$. When $\det M = \pm 1$ the inverse M^{-1} is again an integer matrix, so $\omega_1, \omega_2 \in \mathbb{Z}\omega_1' + \mathbb{Z}\omega_2'$ and the inclusion is an equality; the two periods are $\mathbb{R}$-independent because $\det M \neq 0$. Conversely, if both pairs are bases of Λ, then each ω_i' is an integer combination of ω_1, ω_2 and vice versa, giving integer matrices M and N with $MN = I$; hence $\det M \det N = 1$ with both determinants integers, so $\det M = \pm 1$. □

The shape of Λ, as opposed to its scale and rotation, is captured by a single complex number.

Definition 2.4. The *period ratio* of a basis (ω_1, ω_2) is

$$\tau = \frac{\omega_2}{\omega_1} \in \mathbb{C} \setminus \mathbb{R}.$$

After possibly swapping ω_1 and ω_2 we may, and from now on do, choose the basis so that $\operatorname{Im}\tau > 0$; we call such a basis *positively oriented* and regard τ as an element of the upper half-plane $\mathbb{H}$.

Proposition 2.5 (τ transforms by a Möbius map)**.** *If (ω_1', ω_2') is obtained from (ω_1, ω_2) by the matrix $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$ as in proposition 2.3, then the period ratios are related by*

$$\tau' = \frac{a\tau + b}{c\tau + d}, \qquad \operatorname{Im}\tau' = \frac{\det M}{|c\tau + d|^2} \operatorname{Im}\tau.$$

In particular, both bases are positively oriented if and only if $\det M = +1$, i.e. $M \in \mathrm{SL}_2(\mathbb{Z})$.

Proof. Dividing numerator and denominator of $\omega_2'/\omega_1' = (a\omega_2 + b\omega_1)/(c\omega_2 + d\omega_1)$ by ω_1 gives the stated formula for τ'. For the imaginary part, a direct computation with $\tau = x + iy$ gives $\operatorname{Im} \frac{a\tau+b}{c\tau+d} = \frac{(ad-bc)\,y}{|c\tau+d|^2}$, which is the claim. Since $\operatorname{Im}\tau > 0$ and $|c\tau + d|^2 > 0$, the sign of $\operatorname{Im}\tau'$ equals the sign of $\det M$. □

2.3 Lattice Equivalence and Homothety

The Riemann surface $\mathbb{C}/\Lambda$ does not change if we rotate and rescale the whole picture. The precise notion is homothety.

Definition 2.6. Two lattices Λ, Λ' are *homothetic* (or *equivalent*) if $\Lambda' = \alpha\Lambda$ for some $\alpha \in \mathbb{C}^\times$.

Multiplication by α is a holomorphic automorphism of $\mathbb{C}$ carrying Λ onto Λ', so it descends to a biholomorphism $\mathbb{C}/\Lambda \xrightarrow{\sim} \mathbb{C}/\Lambda'$, $[z] \mapsto [\alpha z]$ (chapter 3 checks it also respects the group law). Homothetic lattices therefore give isomorphic tori, and every lattice has a preferred representative in its class.

Proposition 2.7 (Normal form)**.** *Every lattice is homothetic to a lattice of the form*

$$\Lambda_\tau = \mathbb{Z} + \mathbb{Z}\tau, \qquad \tau \in \mathbb{H}.$$

Two such lattices are homothetic, $\Lambda_\tau \sim \Lambda_{\tau'}$, if and only if

$$\tau' = \frac{a\tau + b}{c\tau + d} \quad \textit{for some } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Consequently homothety classes of lattices are in bijection with the orbit space $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$.

Proof. Given $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with a positively oriented basis, scale by $\alpha = 1/\omega_1$: then $\alpha\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ with $\tau = \omega_2/\omega_1 \in \mathbb{H}$. This proves the normal form.

For the equivalence, suppose $\Lambda_{\tau'} = \alpha\Lambda_\tau$. Then $(\alpha, \alpha\tau)$ and $(1, \tau')$ are two positively oriented bases of the same lattice $\Lambda_{\tau'}$, so by propositions 2.3 and 2.5 they differ by a matrix in $\mathrm{SL}_2(\mathbb{Z})$, and the period ratios obey $\tau' = (a\tau + b)/(c\tau + d)$. Conversely, if τ' is the Möbius image of τ under $M \in \mathrm{SL}_2(\mathbb{Z})$, then $c\tau + d$ scales Λ_τ onto $\Lambda_{\tau'}$, so the two are homothetic. The final statement is a restatement: the class of τ modulo $\mathrm{SL}_2(\mathbb{Z})$ determines, and is determined by, the homothety class. □

Remark 2.8. The quotient $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ is the *moduli space* of complex tori: it parametrizes them up to isomorphism. A single complex invariant, the j-invariant $j(\tau)$, gives a bijection of this quotient with $\mathbb{C}$. We will not need the modular theory here; it is sketched among the further directions in chapter 16. What matters now is only that, up to isomorphism, we may always take $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$.

2.4 The Quotient Surface

Fix a lattice Λ. Define an equivalence relation on $\mathbb{C}$ by

$$z \sim w \quad \Longleftrightarrow \quad z - w \in \Lambda,$$

and form the quotient

$$T = \mathbb{C}/\Lambda.$$

A point of T is a coset $[z] = z + \Lambda$, and the *quotient map*

$$\pi : \mathbb{C} \to \mathbb{C}/\Lambda, \qquad z \mapsto [z]$$

is given the quotient topology: $U \subseteq T$ is open exactly when $\pi^{-1}(U)$ is open in $\mathbb{C}$. Geometrically, T is the fundamental parallelogram with opposite edges glued (fig. 2.2).

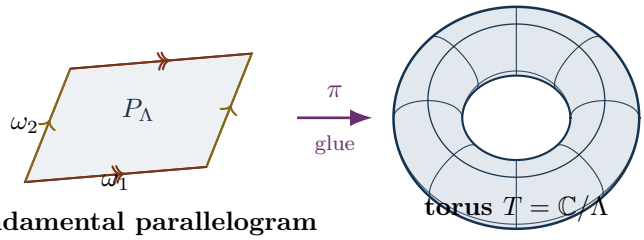


Figure 2.2: Gluing opposite edges of the fundamental parallelogram (matching arrowheads) produces the torus $T = \mathbb{C}/\Lambda$. The torus on the right is rendered from coordinates computed by `sagemath/08_torus_render.sage`.

Proposition 2.9 (Topology of the torus)**.** *The map π is continuous, open, and surjective, and $T = \mathbb{C}/\Lambda$ is a compact, connected, Hausdorff topological space.*

Proof. Continuity and surjectivity are immediate from the definition of the quotient topology. For openness, if $V \subseteq \mathbb{C}$ is open then $\pi^{-1}(\pi(V)) = \bigcup_{\lambda\in\Lambda}(V + \lambda)$ is a union of translates of V, hence open; so $\pi(V)$ is open. Compactness follows because π maps the compact parallelogram P_Λ onto all of T (every coset has a representative in P_Λ), and the continuous image of a compact set is compact; connectedness is inherited from $\mathbb{C}$ the same way.

For the Hausdorff property, let $[z] \neq [w]$. Set $\delta = \frac{1}{3}\min_{\lambda\in\Lambda} |z - w - \lambda| > 0$ (the minimum is positive because $[z] \neq [w]$ and Λ is discrete). The saturated open sets $\pi(D(z,\delta))$ and $\pi(D(w,\delta))$ are disjoint neighborhoods of $[z]$ and $[w]$: if they met, some $z' \in D(z,\delta)$ and $w' \in D(w,\delta)$ would satisfy $z' - w' \in \Lambda$, forcing $|z - w - \lambda| < 2\delta$ for some λ, contradicting the choice of δ. □

2.5 The Complex Structure: Explicit Charts

So far T is only a topological space. We now equip it with a holomorphic atlas and prove it is a Riemann surface. The idea is simple: away from scale $m(\Lambda)$ of remark 2.2, the quotient map π is injective, so its local inverses serve as charts, and the transition maps are translations by lattice vectors — biholomorphic.

Theorem 2.10 ($\mathbb{C}/\Lambda$ is a compact Riemann surface)**.** *The quotient $T = \mathbb{C}/\Lambda$ carries a unique complex structure for which the projection $\pi : \mathbb{C} \to T$ is holomorphic. With this structure T is a compact Riemann surface, and π is a holomorphic covering map and a local biholomorphism.*

Proof. Let $m = m(\Lambda)$ and fix r with $0 < r < m/2$. For any $z_0 \in \mathbb{C}$ the restriction $\pi|_{D(z_0,r)}$ is injective: if $\pi(z) = \pi(z')$ with $z, z' \in D(z_0, r)$, then $z - z' \in \Lambda$ and $|z - z'| < 2r < m$, so $z - z' = 0$. Hence $\pi|_{D(z_0,r)}$ is a continuous open bijection onto the open set $U_{z_0} = \pi(D(z_0, r))$, i.e. a homeomorphism. Define the chart

$$\varphi_{z_0} = \left(\pi|_{D(z_0,r)}\right)^{-1} : U_{z_0} \to D(z_0, r) \subseteq \mathbb{C}.$$

As z_0 ranges over $\mathbb{C}$, the sets U_{z_0} cover T.

Transition maps. Suppose $U_{z_0} \cap U_{z_1} \neq \varnothing$ and let p lie in the overlap, with $\varphi_{z_0}(p) = z$ and $\varphi_{z_1}(p) = z'$. Then $\pi(z) = \pi(z') = p$, so $\lambda := z - z' \in \Lambda$. On the connected component of the overlap containing p, the difference $\varphi_{z_0} - \varphi_{z_1}$ is a continuous Λ-valued function, hence locally constant, so

$$\varphi_{z_0} \circ \varphi_{z_1}^{-1}(w) = w + \lambda$$

is translation by a fixed $\lambda \in \Lambda$. Translations are biholomorphic, so the charts $\{(U_{z_0}, \varphi_{z_0})\}$ form a holomorphic atlas. The underlying space is compact, connected, and Hausdorff by proposition 2.9, so T is a compact Riemann surface.

The projection. In the chart φ_{z_0} the map π reads, for $z \in D(z_0, r)$, as $\varphi_{z_0}(\pi(z)) = z$; that is, π is the identity in these coordinates. Thus π is holomorphic and a local biholomorphism. Each U_{z_0} is evenly covered — $\pi^{-1}(U_{z_0}) = \bigsqcup_{\lambda\in\Lambda}(D(z_0, r) + \lambda)$ is a disjoint union of disks each mapped homeomorphically onto U_{z_0} — so π is a covering map.

Uniqueness. If T carried another complex structure making π holomorphic, then π, being a local homeomorphism, would be a local biholomorphism for it as well, and the two structures would share the local inverses φ_{z_0} as charts; hence they coincide. □

Remark 2.11. Because the charts above use the coordinate z of $\mathbb{C}$ directly, a function $f : T \to \mathbb{C}$ is holomorphic at $[z_0]$ precisely when its lift $f \circ \pi$ is holomorphic at z_0 in the ordinary sense. Holomorphic and meromorphic functions on T are therefore the same thing as Λ-periodic holomorphic and meromorphic functions on $\mathbb{C}$ — the *elliptic functions* of chapter 4. Likewise every translation $z \mapsto z + a$ of $\mathbb{C}$ descends to a biholomorphism of T, a fact we use constantly once the group law is in place.

2.6 Summary

A lattice $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ is a discrete, rank-two subgroup of $\mathbb{C}$; its bases differ by $\mathrm{GL}_2(\mathbb{Z})$, and a positively oriented basis pins down a period ratio $\tau \in \mathbb{H}$ on which $\mathrm{SL}_2(\mathbb{Z})$ acts by Möbius transformations. Up to homothety every lattice is $\mathbb{Z} + \mathbb{Z}\tau$, and homothety classes form the moduli space $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$. The quotient $T = \mathbb{C}/\Lambda$ is a compact Riemann surface: its charts are the local inverses of π, valid on disks of radius below $m(\Lambda)/2$, and the transition maps are lattice translations. With this structure holomorphic functions on T are exactly the Λ-periodic holomorphic functions on $\mathbb{C}$.

Exercises

Exercise 2.1. Show that the area of the fundamental parallelogram P_Λ equals $|\operatorname{Im}(\overline{\omega_1}\,\omega_2)|$, and verify that this number is independent of the choice of basis. (Use proposition 2.3.)

Exercise 2.2. Prove in detail the estimate $|a\omega_1 + b\omega_2| \geq c \max(|a|, |b|)$ from remark 2.2, identifying the constant c as a minimum over a compact set. Conclude that a lattice has only finitely many points in any bounded region.

Exercise 2.3. Let $\Lambda = \mathbb{Z} + \mathbb{Z}i$ (the square lattice, $\tau = i$). Find a matrix in $\mathrm{SL}_2(\mathbb{Z})$ fixing $\tau = i$ under the Möbius action of proposition 2.5, and interpret it as a symmetry of the lattice.

Exercise 2.4. Let $\Lambda' = \alpha\Lambda$ with $\alpha \in \mathbb{C}^\times$. Verify directly that $[z] \mapsto [\alpha z]$ is a well-defined homeomorphism $\mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$, and that in the charts of theorem 2.10 it is biholomorphic.

Exercise 2.5. Show that the radius in theorem 2.10 cannot exceed $m(\Lambda)/2$: exhibit a lattice and a disk of radius $r > m(\Lambda)/2$ on which π fails to be injective. Check, conversely, that the injectivity argument in the proof goes through for every $r \leq m(\Lambda)/2$, so the threshold $m(\Lambda)/2$ is sharp.

Chapter 3

The Torus as Group and as Surface

Chapter 2 made $T = \mathbb{C}/\Lambda$ into a compact Riemann surface. This chapter adds the two structures that make it the right home for the elliptic group law. First, T is an abelian group, and its operations are holomorphic, so T is a complex Lie group; this is the group whose law theorem 1.1 transports to the cubic. Second, T is topologically a torus of genus one, and that single integer is what gives Riemann–Roch its simple shape in chapter 6. We close by separating what is topological (the same for every Λ) from what is complex-analytic (the part that depends on Λ).

3.1 The Group Law

The complex plane is an abelian group under addition, and Λ is a subgroup, so the quotient inherits an operation.

Proposition 3.1. *The formula*

$$[u] + [v] = [u + v]$$

is well defined on $\mathbb{C}/\Lambda$ and makes it an abelian group with identity $0_T = [0]$ and inverses $-[z] = [-z]$.

Proof. Suppose $u' \equiv u$ and $v' \equiv v \pmod{\Lambda}$, say $u' = u + \lambda_1$ and $v' = v + \lambda_2$ with $\lambda_1, \lambda_2 \in \Lambda$. Then $u' + v' = (u + v) + (\lambda_1 + \lambda_2)$, and $\lambda_1 + \lambda_2 \in \Lambda$ because Λ is a subgroup, so $[u' + v'] = [u + v]$; the operation does not depend on representatives. The group axioms are inherited from $(\mathbb{C}, +)$: associativity and commutativity descend immediately, $[0]$ is a two-sided identity, and $[-z]$ is the inverse of $[z]$. □

Equivalently, $\pi : \mathbb{C} \to T$ is a surjective homomorphism of abelian groups with kernel Λ, so $T \cong \mathbb{C}/\Lambda$ as groups — the quotient in the group-theoretic sense agrees with the quotient as a space. Adding two classes is carried out by lifting, adding in $\mathbb{C}$, and reducing back into the fundamental parallelogram (fig. 3.1).

The group law is not merely set-theoretic: it is compatible with the complex structure of chapter 2.

Proposition 3.2 (T is a complex Lie group)**.** *Addition $m : T \times T \to T$, $m([z],[w]) = [z + w]$, and negation $\iota : T \to T$, $\iota([z]) = [-z]$, are holomorphic. Consequently every translation*

$$\tau_a : T \to T, \qquad \tau_a([z]) = [z + a] \quad (a \in \mathbb{C})$$

is a biholomorphic automorphism of T, and the translations act simply transitively on T.

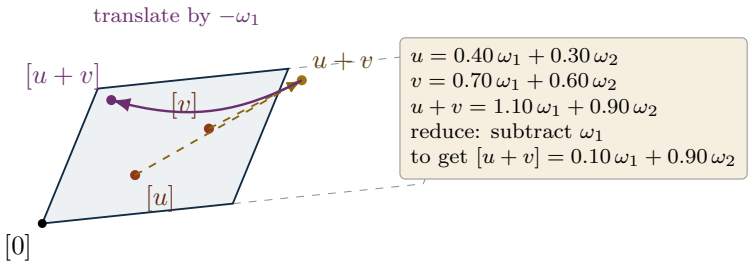


Figure 3.1: Addition on $T = \mathbb{C}/\Lambda$: lift $[u]$ and $[v]$ to $\mathbb{C}$, add to get $u + v$, then reduce modulo Λ back into P_Λ. The arithmetic shown is computed by `sagemath/02_torus_group_law.sage`.

Proof. In the charts of theorem 2.10 the maps m and ι lift to $(z, w) \mapsto z + w$ and $z \mapsto -z$ on open subsets of $\mathbb{C}$, which are holomorphic; since π is a local biholomorphism, m and ι are holomorphic on T. Fixing a, the translation τ_a is holomorphic with holomorphic inverse τ_{-a}, hence biholomorphic. Given $[z], [w] \in T$, the unique translation carrying $[z]$ to $[w]$ is τ_{w-z}, so the action is simply transitive. □

Remark 3.3. Simple transitivity says T looks the same at every point: the translation τ_a is a biholomorphism carrying 0_T to $[a]$. We use this homogeneity constantly — for instance to move a divisor so that one of its points sits at the origin — and it is the reason a single local computation at 0_T often suffices.

3.2 The Torus as a Product of Circles

The abstract law $[u]+[v] = [u+v]$ has a concrete realization that makes commutativity transparent.

The circle group. The unit circle

$$S^1 = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\}$$

is an abelian group under complex multiplication, since $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)} = e^{i\beta}e^{i\alpha}$. The homomorphism $t \mapsto e^{2\pi it}$ from $\mathbb{R}$ onto S^1 has kernel $\mathbb{Z}$, so the first isomorphism theorem gives

$$\mathbb{R}/\mathbb{Z} \;\cong\; S^1.$$

The circle is the real line wound onto itself, with the group law becoming angle addition.

Proposition 3.4 (Torus ≅ product of circles)**.** *Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$. The map*

$$\Psi : \mathbb{C}/\Lambda \longrightarrow S^1 \times S^1, \qquad [\,s\omega_1 + t\omega_2\,] \longmapsto \left(e^{2\pi is},\, e^{2\pi it}\right)$$

is an isomorphism of topological groups.

Proof. Since ω_1, ω_2 are $\mathbb{R}$-independent, every $z \in \mathbb{C}$ has a unique expression $z = s\omega_1 + t\omega_2$ with $s, t \in \mathbb{R}$. The real-linear isomorphism $\phi : \mathbb{R}^2 \to \mathbb{C}$, $(s, t) \mapsto s\omega_1 + t\omega_2$, carries $\mathbb{Z}^2$ onto Λ, so it descends to an isomorphism $\mathbb{R}^2/\mathbb{Z}^2 \xrightarrow{\sim} \mathbb{C}/\Lambda$ of abelian groups. Composing the inverse with the coordinatewise exponential $\mathbb{R}^2/\mathbb{Z}^2 = (\mathbb{R}/\mathbb{Z})^2 \to S^1 \times S^1$ yields Ψ.

Homomorphism. Writing $[z_k] = [s_k\omega_1 + t_k\omega_2]$,

$$\Psi([z_1] + [z_2]) = \left(e^{2\pi i(s_1+s_2)}, e^{2\pi i(t_1+t_2)}\right) = \left(e^{2\pi is_1}e^{2\pi is_2}, e^{2\pi it_1}e^{2\pi it_2}\right) = \Psi([z_1])\,\Psi([z_2]).$$

Injectivity. If $\Psi([z]) = (1,1)$ then $e^{2\pi is} = e^{2\pi it} = 1$, so $s, t \in \mathbb{Z}$ and $z \in \Lambda$, i.e. $[z] = [0]$. *Surjectivity.* The class $[s\omega_1 + t\omega_2]$ maps to $(e^{2\pi is}, e^{2\pi it})$. *Homeomorphism.* Ψ is a continuous bijection between compact Hausdorff spaces, hence a homeomorphism. □

Corollary 3.5. *$T = \mathbb{C}/\Lambda$ is an abelian group, and as a topological group it is independent of Λ: every complex torus is isomorphic to $S^1 \times S^1$.*

Proof. S^1 is abelian, a product of abelian groups is abelian, and Ψ transports this to T. The isomorphism class of $S^1 \times S^1$ does not involve Λ. □

Why commutativity is obvious from the circle picture. A class $[s\omega_1 + t\omega_2]$ corresponds to a pair of angles $(e^{2\pi is}, e^{2\pi it})$, one winding in the ω_1-direction and one in the ω_2-direction. Adding classes rotates the two circles independently,

$$(e^{2\pi is_1}, e^{2\pi it_1})\,(e^{2\pi is_2}, e^{2\pi it_2}) = (e^{2\pi i(s_1+s_2)}, e^{2\pi i(t_1+t_2)}),$$

and rotating by α then β equals rotating by β then α. Commutativity of the group law is just commutativity of adding real angles.

Warning 3.6. Proposition 3.4 is an isomorphism of *topological* groups, not of Riemann surfaces. The product $S^1 \times S^1$ carries no preferred complex structure; the map Ψ uses the real coordinates s, t and is only real-analytic, not holomorphic. The complex structure of T, which remembers the shape τ of chapter 2, is exactly the information that Ψ discards. We return to this in section 3.4.

3.3 Genus and Euler Characteristic

Topologically T is built from the parallelogram P_Λ by gluing opposite edges, and this gluing gives it a cell decomposition from which the Euler characteristic is immediate.

Proposition 3.7. *The torus T has Euler characteristic $\chi(T) = 0$, and hence genus $g = 1$.*

Proof. Identifying opposite edges of P_Λ as in fig. 3.2 produces a CW structure on T: the four corners are all glued to a single point, the two pairs of opposite edges become two edge-loops a, b, and the interior is one 2-cell. Thus

$$V = 1, \qquad E = 2, \qquad F = 1,$$

so

$$\chi(T) = V - E + F = 1 - 2 + 1 = 0.$$

(As a check, $\chi(S^1 \times S^1) = \chi(S^1)\,\chi(S^1) = 0 \cdot 0 = 0$ by the product formula, agreeing with proposition 3.4.) For a compact, connected, orientable surface the Euler characteristic and the genus g — the number of handles — are related by $\chi = 2 - 2g$. Setting $2 - 2g = 0$ gives $g = 1$. □

Remark 3.8. The genus is the one topological number that controls the function theory of a compact Riemann surface. For genus one the Riemann–Roch theorem (chapter 6) takes the especially clean form $\ell(D) = \deg D$ for divisors of positive degree, and it is this that pins down the dimensions of the spaces of functions used to build $\wp$ and $\wp'$. Everything downstream rests on $g = 1$.

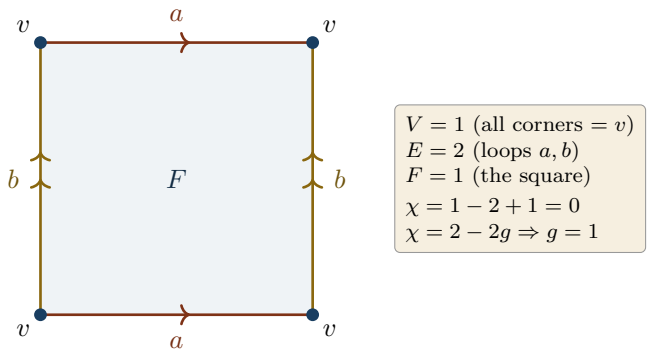


Figure 3.2: The CW structure on the torus. Gluing the two a-edges and the two b-edges of the square sends all four corners to one vertex v, leaves two edge-loops a, b, and one 2-cell F; hence $\chi(T) = 0$ and $g = 1$.

3.4 Topological versus Complex Structure

It is worth separating clearly the two layers we have placed on T.

- **Topological / smooth layer (the same for all Λ).** By corollary 3.5 every complex torus is, as a topological group, the single object $S^1 \times S^1$; by proposition 3.7 it has genus one. No invariant at this level distinguishes one lattice from another — all complex tori are diffeomorphic.
- **Complex-analytic layer (depends on Λ).** The holomorphic structure of theorem 2.10 is extra data. By proposition 2.7 it is recorded, up to isomorphism, by the point $\tau \in \mathbb{H}$ modulo $\mathrm{SL}_2(\mathbb{Z})$; two tori $\mathbb{C}/\Lambda$ and $\mathbb{C}/\Lambda'$ are biholomorphic if and only if their τ-classes agree.

Thus the diffeomorphism type is rigid (one torus) while the biholomorphism type varies in the one-parameter family $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$. The single complex number that separates the classes is the j-invariant of chapter 16. Warning 3.6 is the concrete face of this: the real coordinates (s, t) trivialize T topologically but are blind to the complex structure that the rest of the book exploits.

3.5 Summary

The torus $T = \mathbb{C}/\Lambda$ is an abelian group under $[u] + [v] = [u + v]$, with holomorphic addition and negation, so it is a complex Lie group on which the translations act simply transitively (propositions 3.1 and 3.2). As a topological group it is $S^1 \times S^1$, which makes commutativity visibly the commutativity of adding angles, but this identification is only real-analytic and forgets the complex structure (proposition 3.4 and warning 3.6). A CW decomposition gives $\chi(T) = 0$ and genus $g = 1$ (proposition 3.7), the topological fact behind the simple form of Riemann–Roch. All complex tori share this topology; what varies is the complex structure, recorded by $\tau \in \mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$.

Exercises

Exercise 3.1. Complete the details of proposition 3.2: writing the addition map in two overlapping charts of theorem 2.10, verify explicitly that the transition makes $m : T \times T \to T$ holomorphic, and deduce that $\iota([z]) = [-z]$ is biholomorphic.

Exercise 3.2. Show that the negation map $\iota([z]) = [-z]$ has exactly four fixed points on T, namely

$$[0], \quad \left[\tfrac{\omega_1}{2}\right], \quad \left[\tfrac{\omega_2}{2}\right], \quad \left[\tfrac{\omega_1+\omega_2}{2}\right].$$

These are the *two-torsion* points, the solutions of $2[z] = [0]$; they reappear as the branch points of $\wp$ in chapter 8 and as the two-torsion of the group law in chapter 12.

Exercise 3.3. Fill in the first-isomorphism-theorem argument for $\mathbb{R}/\mathbb{Z} \cong S^1$: check that $t \mapsto e^{2\pi i t}$ is a continuous surjective homomorphism with kernel $\mathbb{Z}$, and that the induced bijection $\mathbb{R}/\mathbb{Z} \to S^1$ is a homeomorphism (use compactness of $[0,1]$).

Exercise 3.4. Recompute $\chi(T)$ using a finer CW structure — for instance subdivide P_Λ into two triangles before gluing — and confirm you again obtain $\chi(T) = 0$, illustrating the invariance of the Euler characteristic.

Exercise 3.5. Show that $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}i)$ and $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}\,2i)$ are homeomorphic (indeed diffeomorphic) as real surfaces. Granting the standard fact that the points $\tau = i$ and $\tau = 2i$ lie in the closed fundamental domain $\{\,|\tau| \geq 1,\ |\operatorname{Re}\tau| \leq \tfrac{1}{2}\,\}$ of $\mathrm{SL}_2(\mathbb{Z})$ and are not identified by it, conclude (using chapter 2) that the two tori are *not* biholomorphic. Explain why this does not contradict corollary 3.5.

Part II

Elliptic Functions

Chapter 4

Elliptic Functions and Liouville's Theorems

By remark 2.11, meromorphic functions on the torus $T = \mathbb{C}/\Lambda$ are the same as Λ-periodic meromorphic functions on $\mathbb{C}$ — the *elliptic functions*. This chapter establishes the rigid constraints these functions obey. There are no nonconstant holomorphic ones, so a nonconstant elliptic function must have poles; and the location and number of its zeros and poles are governed by three theorems of Liouville, all proved by integrating around a single period parallelogram. The third of them, $\sum(\text{zeros}) - \sum(\text{poles}) \in \Lambda$, is the analytic engine of the entire group law: it returns in chapter 9 as Abel's sum law and is what ultimately forces $P + Q + R = O$ for collinear points in theorem 1.1.

4.1 The Field of Elliptic Functions

Definition 4.1. A meromorphic function f on $\mathbb{C}$ is *elliptic* with respect to Λ, or Λ-periodic, if

$$f(z + \lambda) = f(z) \qquad \text{for all } \lambda \in \Lambda.$$

Equivalently, $f(z + \omega_1) = f(z)$ and $f(z + \omega_2) = f(z)$ for a basis ω_1, ω_2.

An elliptic function depends only on the class $[z]$, so it descends to a meromorphic function $T \to \mathbb{CP}^1$; conversely every meromorphic function on T pulls back along π to a Λ-periodic one. The two viewpoints are interchangeable, and we pass between them freely:

$$\{\text{meromorphic functions on } T\} \longleftrightarrow \{\Lambda\text{-elliptic functions on } \mathbb{C}\}.$$

Proposition 4.2. *The Λ-elliptic functions form a field, closed under differentiation: if f, g are elliptic then so are $f \pm g$, fg, f/g (for $g \not\equiv 0$), and f'.*

Proof. Sums, products, and quotients of meromorphic functions are meromorphic, and each inherits Λ-periodicity pointwise, e.g. $(fg)(z + \lambda) = f(z + \lambda)g(z + \lambda) = f(z)g(z)$. Differentiating $f(z + \lambda) = f(z)$ in z gives $f'(z + \lambda) = f'(z)$, so f' is elliptic. □

The central problem of the theory is to describe this field. The answer, reached in chapter 7, is that it is generated by the Weierstrass function $\wp$ and its derivative $\wp'$. Everything in this chapter is a constraint that the generators must satisfy.

4.2 No Nonconstant Holomorphic Elliptic Functions

Theorem 4.3 (Liouville's first theorem)**.** *An elliptic function with no poles is constant.*

Proof. Let f be elliptic and holomorphic on all of $\mathbb{C}$. The closed fundamental parallelogram $\overline{P} = \{s\omega_1 + t\omega_2 : 0 \le s,t \le 1\}$ is compact, so the continuous function $|f|$ attains a maximum M on $\overline{P}$. By periodicity every $z \in \mathbb{C}$ is congruent to a point of $\overline{P}$ and f takes the same value there, so $|f(z)| \le M$ for all z: the entire function f is bounded. By Liouville's theorem (a bounded entire function is constant) f is constant. Equivalently, T is compact, so a holomorphic $f : T \to \mathbb{C}$ attains its maximum modulus and is constant by the maximum modulus principle. □

Thus a nonconstant elliptic function must have at least one pole. The quantitative versions of this statement are Liouville's remaining theorems.

4.3 Integration Around a Period Parallelogram

All three theorems rest on one device. Fix a basis ω_1, ω_2 and a base point $a \in \mathbb{C}$, and let

$$P_a = \{\, a + s\omega_1 + t\omega_2 : 0 \le s,t \le 1 \,\}$$

be the translated fundamental parallelogram, with boundary ∂P_a traversed counterclockwise. Because a nonconstant elliptic function has only finitely many zeros and poles modulo Λ, we may always choose a so that ∂P_a passes through none of them (fig. 4.1). Label the four sides

$$\mathrm{I} : a \to a+\omega_1, \quad \mathrm{II} : a+\omega_1 \to a+\omega_1+\omega_2, \quad \mathrm{III} : a+\omega_1+\omega_2 \to a+\omega_2, \quad \mathrm{IV} : a+\omega_2 \to a.$$

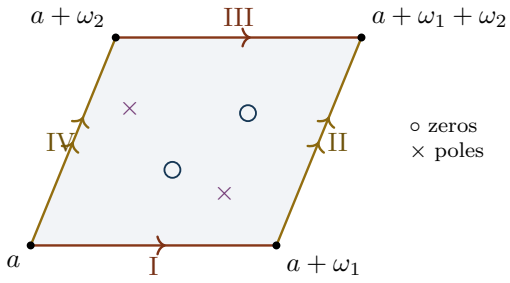


Figure 4.1: A period parallelogram P_a, chosen so its boundary avoids the zeros (∘) and poles (×) of f. For an elliptic integrand the integral over side III cancels that over side I (they are translates by ω_2 traversed oppositely), and II cancels IV; the boundary integral vanishes.

Lemma 4.4 (Opposite sides cancel)**.** *If g is elliptic and has no poles on ∂P_a, then $\oint_{\partial P_a} g(z)\,dz = 0$.*

Proof. Side III is the translate of side I by ω_2, traversed in the opposite direction. Substituting $z = w + \omega_2$ and using $g(w+\omega_2) = g(w)$,

$$\int_{\mathrm{III}} g\,dz = -\int_{\mathrm{I}} g(w+\omega_2)\,dw = -\int_{\mathrm{I}} g(w)\,dw,$$

so $\int_{\mathrm{I}} + \int_{\mathrm{III}} = 0$. Likewise side II is the translate of side IV by ω_1, traversed oppositely, so $\int_{\mathrm{II}} + \int_{\mathrm{IV}} = 0$. Adding gives $\oint_{\partial P_a} g\,dz = 0$. □

4.4 Liouville's Theorems

4.4.1 The sum of residues vanishes

Theorem 4.5 (Liouville's second theorem: residues)**.** *For a nonconstant elliptic function f, the sum of the residues over one period parallelogram is zero:*

$$\sum_{[p]\in T} \operatorname{Res}_p f = 0.$$

Proof. Choose a so that ∂P_a avoids the poles of f. The interior of P_a contains exactly one representative of each pole class. By the residue theorem and lemma 4.4 applied to $g = f$,

$$2\pi i \sum_{p\in P_a} \operatorname{Res}_p f = \oint_{\partial P_a} f(z)\,dz = 0. \qquad \square$$

Corollary 4.6. *No elliptic function has a single simple pole in a period parallelogram. The* order *of a nonconstant elliptic function — the number of poles, counted with multiplicity, in one period parallelogram — is at least* 2*.*

Proof. A function with one simple pole and no others would have a single nonzero residue, contradicting theorem 4.5. So either there are at least two poles, or a single pole of order ≥ 2; in both cases the order is at least 2. $\square$

This is the first hint of the Weierstrass function: the smallest a nonconstant elliptic function can be is order two, realized by a single double pole. That function is $\wp$ (chapter 7).

4.4.2 The counting theorem

Theorem 4.7 (Liouville's third theorem: valence)**.** *A nonconstant elliptic function f of order N has exactly N zeros and N poles in each period parallelogram, counted with multiplicity. More generally, for every $c \in \mathbb{CP}^1$ the equation $f(z) = c$ has exactly N solutions there; $f : T \to \mathbb{CP}^1$ is an N-to-one branched cover.*

Proof. The logarithmic derivative f'/f is elliptic by proposition 4.2, with a simple pole at each zero and pole of f; its residue there is $\operatorname{Res}_p(f'/f) = \operatorname{ord}_p(f)$. Choosing a so ∂P_a avoids these points, lemma 4.4 and the residue theorem give

$$0 = \oint_{\partial P_a} \frac{f'}{f}\,dz = 2\pi i \sum_{p\in P_a} \operatorname{ord}_p(f) = 2\pi i\big(Z - P\big),$$

where Z and P are the numbers of zeros and poles with multiplicity. Hence $Z = P = N$. Applying this to $f - c$ (also elliptic, of order N, with the same poles as f) shows $f - c$ has N zeros, i.e. f takes the value c exactly N times; the case $c = \infty$ is the pole count. $\square$

4.4.3 The Abel relation

The previous theorems used elliptic integrands. The decisive third statement uses $z\,f'(z)/f(z)$, whose factor z is *not* periodic; the failure of cancellation is precisely a lattice vector.

Theorem 4.8 (Liouville's fourth theorem: the Abel relation)**.** *Let f be a nonconstant elliptic function. Listing its zeros $z_1, \dots, z_N$ and poles $p_1, \dots, p_N$ in a period parallelogram with multiplicity,*

$$\sum_{k=1}^{N} z_k \;-\; \sum_{k=1}^{N} p_k \;\in\; \Lambda.$$

Equivalently, in the group T, $\sum_k [z_k] = \sum_k [p_k]$.

Proof. Consider $g(z) = z\,f'(z)/f(z)$. At a zero or pole p of f of order $m = \operatorname{ord}_p(f)$ one has $f'/f \sim m/(z-p)$, so $\operatorname{Res}_p g = m\,p$. With ∂P_a avoiding the zeros and poles, the residue theorem gives

$$\oint_{\partial P_a} g(z)\,dz = 2\pi i \sum_{p \in P_a} \operatorname{ord}_p(f)\,p = 2\pi i\Big(\sum_k z_k - \sum_k p_k\Big). \tag{$*$}$$

Now evaluate the contour integral directly, pairing opposite sides. On side III put $z = w + \omega_2$ with w running along I reversed; since f'/f is ω_2-periodic,

$$\int_{\mathrm{I}} + \int_{\mathrm{III}} = \int_{\mathrm{I}} \big[\, z - (z+\omega_2) \,\big] \frac{f'}{f}\,dz = -\,\omega_2 \int_a^{a+\omega_1} \frac{f'}{f}\,dz.$$

The remaining integral is the change in $\log f$ along side I; because $f(a+\omega_1) = f(a)$, this change is $2\pi i\,k_1$ for some integer k_1 (the winding number of f along the side). Thus $\int_{\mathrm{I}} + \int_{\mathrm{III}} = -2\pi i\,k_1\,\omega_2$. Symmetrically, pairing II with IV and using ω_1-periodicity,

$$\int_{\mathrm{II}} + \int_{\mathrm{IV}} = \omega_1 \int_a^{a+\omega_2} \frac{f'}{f}\,dz = 2\pi i\,k_2\,\omega_1$$

for an integer k_2. Adding,

$$\oint_{\partial P_a} g\,dz = 2\pi i\,(k_2\,\omega_1 - k_1\,\omega_2). \tag{$**$}$$

Comparing $(*)$ and $(**)$ and cancelling $2\pi i$,

$$\sum_k z_k - \sum_k p_k = k_2\,\omega_1 - k_1\,\omega_2 \in \Lambda. \qquad \square$$

The relation behind the group law. Theorem 4.8 says: for any elliptic function, the zeros and poles, added as points of T, are equal. Read backward it is a recipe — if a divisor of points has zero degree and its points sum to 0 in T, it is the divisor of an elliptic function (chapter 6 supplies the converse). When the points are the intersections of the cubic with a line, this congruence becomes $P + Q + R = O$. The chord–tangent law is theorem 4.8 in disguise.

4.5 Consequences and the Road Ahead

The four theorems pin down the coarse shape of every elliptic function:

- it has no poles only if it is constant (theorem 4.3);

- its residues sum to zero, so it cannot have order one (theorem 4.5 and corollary 4.6);
- it attains every value the same number $N \geq 2$ of times (theorem 4.7);
- its zeros and poles satisfy $\sum z_k \equiv \sum p_k \pmod{\Lambda}$ (theorem 4.8).

The minimal case $N = 2$ is therefore the place to look for a generator of the field. A function with a single double pole at $[0]$ would, by theorem 4.7, have two zeros, and by theorem 4.8 those zeros z_1, z_2 would satisfy $z_1 + z_2 \equiv 0$, i.e. $z_2 \equiv -z_1$: the zeros are antipodal. This is exactly the behaviour of $\wp - c$, and it is why $\wp$ is an even function. Whether such a function exists at all is a dimension count, answered by Riemann–Roch in chapter 6; its explicit construction is chapter 7.

4.6 Summary

Elliptic functions are the meromorphic functions on T; they form a differential field. A nonconstant one must have poles (theorem 4.3), of total order $N \geq 2$ (corollary 4.6), and it takes every value exactly N times (theorem 4.7). Integrating f, f'/f, and $z\,f'/f$ around a period parallelogram yields, respectively, the vanishing of the residue sum, the equality of zero and pole counts, and the Abel relation $\sum(\text{zeros}) - \sum(\text{poles}) \in \Lambda$ (theorems 4.5, 4.7 and 4.8). The last is the analytic heart of the group law and reappears as Abel's sum law in chapter 9.

Exercises

Exercise 4.1. Verify in detail that the Λ-elliptic functions form a field closed under d/dz (proposition 4.2). Show in addition that if f is elliptic then so is $f(z + a)$ for any fixed $a \in \mathbb{C}$, and that an even elliptic function ($f(-z) = f(z)$) has f' odd.

Exercise 4.2. Give a complete proof that there is no elliptic function of order one. Then show that an elliptic function of order two has either two distinct simple poles with opposite residues, or a single double pole with residue zero.

Exercise 4.3. Suppose f is elliptic of order two with a single double pole at $[0]$. Using theorems 4.7 and 4.8, show that the two zeros of f are of the form $[w]$ and $[-w]$ for some w. Conclude that $f(z) - f(z_0)$ has zeros at $\pm z_0$ for generic z_0, the signature of an even function.

Exercise 4.4. Let f be elliptic of order N and let $c_1, c_2 \in \mathbb{C}$ be two values. Using theorem 4.8 applied to $f - c_1$ and to $f - c_2$, show that the sum of the N solutions of $f(z) = c_1$ equals, modulo Λ, the sum of the N solutions of $f(z) = c_2$. (The common value is an invariant of f.)

Exercise 4.5. In the proof of theorem 4.8 the integers k_1, k_2 are winding numbers of f along the two sides of P_a. Show directly that $\int_a^{a+\omega_1} (f'/f)\,dz \in 2\pi i\,\mathbb{Z}$ whenever $f(a + \omega_1) = f(a)$ and f has no zeros or poles on the segment, and interpret k_1 as the number of times f winds around 0.

Chapter 5

Divisors and the Degree-Zero Theorem

Chapter 4 controlled the zeros and poles of a single elliptic function. Divisors are the bookkeeping that turns those zero/pole data into an algebraic object one can add and compare. On a compact Riemann surface the divisor of a function has degree zero; on the torus it additionally has Abel sum zero. Packaging these two constraints gives the *Picard group* $\mathrm{Pic}^0(T)$ and a map $\mathrm{Pic}^0(T) \to T$ that is the structural shadow of the group law: when we later put a group law on the cubic, it will be exactly this map.

5.1 Orders and the Divisor of a Function

Let X be a Riemann surface and f a nonzero meromorphic function. In a local coordinate t centered at a point p we may write

$$f(t) = t^m u(t), \qquad u \text{ holomorphic}, \quad u(0) \neq 0.$$

The integer m is the *order* of f at p, written $\mathrm{ord}_p(f) = m$:

$$m > 0 \text{ (zero of order } m), \qquad m = 0 \text{ (finite, nonzero)}, \qquad m < 0 \text{ (pole of order } -m).$$

Remark 5.1. The order does not depend on the chart. If s is another coordinate at p, then $t = s\,h(s)$ with h holomorphic and $h(0) \neq 0$, so $f = s^m\big(h(s)^m u(t(s))\big)$ and the new unit $h^m\,(u \circ t)$ is again nonvanishing at $s = 0$; the exponent m is unchanged. Thus $\mathrm{ord}_p(f)$ is intrinsic.

Definition 5.2. The *divisor* of a nonzero meromorphic function f is the formal sum

$$\mathrm{div}(f) = \sum_{p \in X} \mathrm{ord}_p(f)\, p.$$

On a compact X only finitely many terms are nonzero, so this is a finite sum.

5.2 The Divisor Group and Degree

Definition 5.3. A *divisor* on X is a finite formal integer combination of points,

$$D = \sum_{p \in X} n_p\, p, \qquad n_p \in \mathbb{Z}, \quad n_p = 0 \text{ for all but finitely many } p.$$

Under coefficientwise addition the divisors form the free abelian group $\mathrm{Div}(X)$ on the points of X. The *degree* is the homomorphism

$$\deg : \mathrm{Div}(X) \to \mathbb{Z}, \qquad \deg\Big(\sum_p n_p\, p\Big) = \sum_p n_p.$$

A divisor of the form $\mathrm{div}(f)$ is called *principal*; we write $\mathrm{Prin}(X)$ for the set of principal divisors.

Proposition 5.4. *The principal divisors form a subgroup of* $\mathrm{Div}(X)$*: for nonzero meromorphic* f, g*,*

$$\mathrm{div}(fg) = \mathrm{div}(f) + \mathrm{div}(g), \qquad \mathrm{div}(1/f) = -\,\mathrm{div}(f),$$

and $\mathrm{div}(c) = 0$ *for a nonzero constant* c*.*

Proof. Orders add under multiplication: $\mathrm{ord}_p(fg) = \mathrm{ord}_p(f) + \mathrm{ord}_p(g)$, directly from $f = t^m u$, $g = t^n v$ giving $fg = t^{m+n}uv$. Summing over p gives the first identity; the others are the cases $g = 1/f$ and $f = c$. Hence $\mathrm{Prin}(X)$ is closed under addition and negation, a subgroup. □

5.3 Principal Divisors Have Degree Zero

The single most important property of principal divisors on a compact surface is that they are invisible to the degree.

Theorem 5.5 (Degree-zero theorem)**.** *If* X *is compact and* f *is a nonzero meromorphic function on* X*, then*

$$\deg \mathrm{div}(f) = 0.$$

Proof. Consider the meromorphic differential df/f. Near p, writing $f = t^m u$ with $u(0) \neq 0$,

$$\frac{df}{f} = \frac{d(t^m u)}{t^m u} = m\,\frac{dt}{t} + \frac{du}{u},$$

and the second term is holomorphic at p. Hence df/f has at worst a simple pole at p, with

$$\mathrm{Res}_p\left(\frac{df}{f}\right) = m = \mathrm{ord}_p(f).$$

By the residue theorem on the compact Riemann surface X the sum of all residues of the global differential df/f vanishes, so

$$0 = \sum_p \mathrm{Res}_p\left(\frac{df}{f}\right) = \sum_p \mathrm{ord}_p(f) = \deg \mathrm{div}(f).$$ □

What degree zero means. A meromorphic function on a compact Riemann surface has as many zeros as poles, counted with multiplicity — the divisor form of the argument principle. On the torus this is theorem 4.7 with $N = \deg(\text{pole divisor})$: the two theorems are the same fact seen from the two sides of the $T \leftrightarrow \mathbb{C}$ dictionary.

In particular $\mathrm{Prin}(X) \subseteq \mathrm{Div}^0(X)$, where

$$\mathrm{Div}^0(X) = \ker(\deg) = \{D : \deg D = 0\}$$

is the group of degree-zero divisors.

5.4 Linear Equivalence and the Picard Group

Definition 5.6. Two divisors D, D' are *linearly equivalent*, written $D \sim D'$, if $D - D'$ is principal. The quotient group

$$\operatorname{Pic}(X) = \operatorname{Div}(X) / \operatorname{Prin}(X)$$

is the *Picard group*, and the class of D is denoted $[D]$.

Because principal divisors have degree zero (theorem 5.5), the degree descends to a homomorphism $\deg : \operatorname{Pic}(X) \to \mathbb{Z}$, and its kernel

$$\operatorname{Pic}^0(X) = \operatorname{Div}^0(X) / \operatorname{Prin}(X)$$

is the *degree-zero Picard group*. The size of Pic^0 measures how far divisors of degree zero are from being principal — equivalently, how many meromorphic functions the surface admits. On $\mathbb{CP}^1$ every degree-zero divisor is principal and $\operatorname{Pic}^0 = 0$; on the torus, as we now show, Pic^0 is as large as T itself.

5.5 The Abel–Jacobi Sum Map

Specialize to $X = T = \mathbb{C}/\Lambda$. Here points of T are themselves group elements, so a degree-zero divisor can be *summed*.

Definition 5.7. The *Abel–Jacobi sum map* is

$$\mathfrak{s} : \operatorname{Div}^0(T) \to T, \qquad \mathfrak{s}\Big(\sum_i n_i\,[z_i]\Big) = \sum_i n_i\,[z_i] \in T,$$

where the right-hand sum is taken in the group T.

It is a group homomorphism, immediately from the definition. The content of the previous chapter is exactly that $\mathfrak{s}$ annihilates principal divisors.

Proposition 5.8. *For every nonzero elliptic function f, $\mathfrak{s}(\operatorname{div} f) = 0$. Hence $\mathfrak{s}$ descends to a homomorphism*

$$\bar{\mathfrak{s}} : \operatorname{Pic}^0(T) \to T.$$

Proof. Write $\operatorname{div}(f) = \sum_k [z_k] - \sum_k [p_k]$ with the zeros and poles listed by multiplicity; it has degree zero by theorem 5.5 (or theorem 4.7), so it lies in $\operatorname{Div}^0(T)$. Then $\mathfrak{s}(\operatorname{div} f) = \sum_k [z_k] - \sum_k [p_k]$, which is 0 in T by the Abel relation theorem 4.8. Since $\mathfrak{s}$ kills the subgroup $\operatorname{Prin}(T)$, it factors through $\operatorname{Pic}^0(T) = \operatorname{Div}^0(T)/\operatorname{Prin}(T)$. □

Theorem 5.9 (Abel–Jacobi for the torus)**.** *The map $\bar{\mathfrak{s}} : \operatorname{Pic}^0(T) \to T$ is an isomorphism of groups. In particular every complex torus is its own* Jacobian*: $\operatorname{Pic}^0(T) \cong T$.*

Proof of surjectivity, and the standing of injectivity. Surjectivity. Given any $[a] \in T$, the divisor $[a] - [0]$ has degree zero and $\mathfrak{s}([a] - [0]) = [a] - [0] = [a]$. So $\bar{\mathfrak{s}}$ hits every element of T.

Well-definedness and homomorphism are proposition 5.8.

Injectivity is the converse of Abel's relation: if $D \in \operatorname{Div}^0(T)$ has $\mathfrak{s}(D) = 0$, then D is principal, i.e. $D = \operatorname{div}(f)$ for some elliptic f. This requires *constructing* an elliptic function with prescribed zeros and poles, which we do not yet have the tools for. It follows once the Weierstrass machinery is in place: chapter 6 produces enough functions, and the explicit construction is carried out through chapters 7 to 11. We therefore record injectivity here and complete it there. □

Why this is the whole story. Theorem 5.9 says the degree-zero divisor classes on the torus *are* the points of the torus, via summation. Two divisors are linearly equivalent precisely when their points have the same Abel sum. When we transport the picture to the cubic E in chapters 10 to 11, the group law on E will be defined by linear equivalence of divisors — and by this theorem that is the same as addition on T. The chord–tangent law is the composite

$$E \;\cong\; T \;\xleftarrow{\;\bar{\mathfrak{s}}\;}\; \mathrm{Pic}^0(T),$$

read as a single group.

5.6 Summary

Divisors on X form the free abelian group $\mathrm{Div}(X)$, graded by degree, with the principal divisors $\mathrm{div}(f)$ as a subgroup. On a compact surface every principal divisor has degree zero (theorem 5.5), so degree descends to the Picard group $\mathrm{Pic}(X) = \mathrm{Div}(X)/\mathrm{Prin}(X)$ and cuts out $\mathrm{Pic}^0(X)$. On the torus the Abel–Jacobi map $\mathfrak{s} : \mathrm{Div}^0(T) \to T$ sums a divisor's points; it kills principal divisors by the Abel relation (proposition 5.8), and the induced $\bar{\mathfrak{s}} : \mathrm{Pic}^0(T) \to T$ is an isomorphism (theorem 5.9, with injectivity completed later). This identification $\mathrm{Pic}^0(T) \cong T$ is the algebraic form of the group law that the rest of the book transports to the cubic.

Exercises

Exercise 5.1. Show that $f \mapsto \mathrm{div}(f)$ is a homomorphism from the multiplicative group of nonzero meromorphic functions to $\mathrm{Div}(X)$, and that on a compact X its kernel is exactly the nonzero constants. (Use theorem 5.5 and theorem 4.3 for the torus case.)

Exercise 5.2. Prove carefully that $\mathrm{ord}_p(f)$ is independent of the local coordinate (remark 5.1), and deduce that $\mathrm{ord}_p(f+g) \geq \min(\mathrm{ord}_p f, \mathrm{ord}_p g)$ with equality when the two orders differ.

Exercise 5.3. Using the base point $[0] \in T$, show that

$$\mathrm{Div}(T) \xrightarrow{\sim} \mathbb{Z} \oplus \mathrm{Div}^0(T), \qquad D \mapsto \big(\deg D,\; D - (\deg D)[0]\big),$$

and deduce a splitting $\mathrm{Pic}(T) \cong \mathbb{Z} \oplus \mathrm{Pic}^0(T)$. Combined with theorem 5.9, conclude $\mathrm{Pic}(T) \cong \mathbb{Z} \oplus T$.

Exercise 5.4. Let $[a], [b] \in T$. Using the Abel relation theorem 4.8, prove that the divisor $[a] - [b]$ is principal *if and only if* $[a] = [b]$. Contrast this with $\mathbb{CP}^1$, where *any* two points satisfy $[a] - [b] = \mathrm{div}\big(\frac{z-a}{z-b}\big)$. Why does this recover corollary 4.6?

Exercise 5.5. Take $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ and the divisor $D = [\frac{1}{2}] + [\frac{\tau}{2}] + [\frac{1+\tau}{2}] - 3\,[0]$. Compute $\deg D$ and $\mathfrak{s}(D) \in T$. Is D a candidate for a principal divisor? (Recall the two-torsion points from exercise 3.2.)

Part III

Riemann–Roch on Genus One

Chapter 6

Riemann–Roch and Its Consequences

We can now answer the question left open in chapter 4: how small can the pole divisor of a nonconstant elliptic function be? The answer is a dimension count, and on the torus that count is *elementary* — it follows from the Liouville theorems of chapter 4 together with the explicit functions built in chapter 7, with no appeal to a deep theorem. We derive it that way first, because it is what justifies the functions $1, \wp, \wp'$ and carries the main argument. We then state the *Riemann–Roch* theorem in general, for context. The one auxiliary result it might have supplied — the converse of Abel's law, $\mathrm{Pic}^0(T) \cong T$ — is instead proved constructively in chapter 7 with the σ-function, so Riemann–Roch is needed nowhere and appears only as the general theorem behind the elementary counts.

6.1 The Spaces $L(D)$

We order divisors by $D \geq 0$ iff every coefficient of D is nonnegative (an *effective* divisor), and $D \geq D'$ iff $D - D' \geq 0$.

Definition 6.1. For a divisor D on a compact Riemann surface X,

$$L(D) = \{f \in \mathcal{M}(X) : \mathrm{div}(f) + D \geq 0\} \cup \{0\}, \qquad \ell(D) = \dim_{\mathbb{C}} L(D).$$

The condition $\mathrm{div}(f) + D \geq 0$ says f may have a pole of order at most n_p where $D = \sum n_p\, p$ has $n_p > 0$, and must have a zero of order at least $-n_p$ where $n_p < 0$. For $D = nO$ with $n \geq 0$, $L(nO)$ is the space of functions whose only pole is at O, of order at most n.

Proposition 6.2 (Elementary properties of $L(D)$)**.** *Let D, D' be divisors on a compact Riemann surface X.*

1. *$L(D)$ is a $\mathbb{C}$-vector space, and $\ell(0) = 1$ with $L(0) = \mathbb{C}$.*
2. *If $D \leq D'$ then $L(D) \subseteq L(D')$ and $\ell(D') - \ell(D) \leq \deg(D' - D)$.*
3. *If $\deg D < 0$ then $L(D) = 0$, so $\ell(D) = 0$.*
4. *Linearly equivalent divisors have isomorphic spaces: $D \sim D'$ implies $L(D) \cong L(D')$, so ℓ depends only on the class $[D]$.*

Proof. (1) If $\mathrm{div}(f) + D \geq 0$ and $\mathrm{div}(g) + D \geq 0$ then for $a, b \in \mathbb{C}$, $\mathrm{ord}_p(af + bg) \geq \min(\mathrm{ord}_p f, \mathrm{ord}_p g) \geq -n_p$ (exercise 5.2), so $af + bg \in L(D)$. For $D = 0$, $L(0)$ is the holomorphic functions on X, which are the constants by theorem 4.3 (compactness), so $\ell(0) = 1$.

(2) The inclusion is immediate from $\operatorname{div}(f) + D \le \operatorname{div}(f) + D'$. Writing $D' = D + \sum_j p_j$ with $\deg(D' - D)$ points (with multiplicity), each added point relaxes the pole bound at p_j by one, and the map sending f to its Laurent coefficient of that extra allowed order is linear; its kernel is the smaller space, so the dimension grows by at most one per point.

(3) A nonzero $f \in L(D)$ has $\operatorname{div}(f) \ge -D$, so $0 = \deg \operatorname{div}(f) \ge -\deg D$ by theorem 5.5; thus $\deg D \ge 0$. Contrapositively $\deg D < 0$ forces $L(D) = 0$.

(4) If $D' = D + \operatorname{div}(h)$, then $f \mapsto f/h$ is a linear isomorphism $L(D) \to L(D')$, since $\operatorname{div}(f/h) + D' = \operatorname{div}(f) + D \ge 0$. □

6.2 The Dimension Counts, Elementarily

The counts that the rest of the book uses — $\ell(O) = 1$, $\ell([a]) = 1$, and $\ell(nO) = n$ — need nothing beyond chapter 4 and the explicit functions of chapter 7.

Proposition 6.3 (Dimension counts on the torus)**.** *On $T = \mathbb{C}/\Lambda$:*

1. *$\ell(O) = 1$, and $\ell([a]) = 1$ for every single point $[a]$;*

2. *$\ell(nO) \le n$ for all $n \ge 1$;*

3. *$\ell(nO) = n$ for all $n \ge 1$.*

Proof. (1) A nonzero $f \in L([a])$ has at most a simple pole at $[a]$ and no other poles, so it is an elliptic function of order at most one. By corollary 4.6 there is no elliptic function of order exactly one, so f is holomorphic, hence constant (theorem 4.3); thus $\ell([a]) = 1$. The case $[a] = O$ is the same.

(2) By the point lemma $\ell(D + p) - \ell(D) \in \{0, 1\}$ (exercise 6.1), applied $n - 1$ times starting from $\ell(O) = 1$,

$$\ell(nO) \le \ell(O) + (n - 1) = n.$$

(3) For the matching lower bound we exhibit n independent elements of $L(nO)$. Chapter 7 constructs the elliptic functions $\wp$ (a double pole at O, order 2) and $\wp'$ (a triple pole, order 3). The functions

$$1, \quad \wp^a \ (a \ge 1), \quad \wp^a \wp' \ (a \ge 0)$$

have poles only at O, of orders 0 and $2a$ and $2a + 3$ respectively, so they realize every pole order in $\{0, 2, 3, 4, \ldots, n\}$ — that is n distinct orders, hence n linearly independent members of $L(nO)$. Therefore $\ell(nO) \ge n$, and with (2), $\ell(nO) = n$. □

Remark 6.4. The proof used only the Liouville theorems of chapter 4 (corollary 4.6 and theorem 4.3), the degree-zero theorem (theorem 5.5, via the point lemma), and the existence of $\wp, \wp'$ from chapter 7. No Riemann–Roch is needed. The same argument gives $\ell([a]) = 1$ for any point, which is all the converse of Abel will require of these counts.

6.3 The Canonical Divisor of a Torus

For the general theorem we need the *canonical divisor* K, the divisor of any nonzero meromorphic 1-form. On a compact surface of genus g all such forms are linearly equivalent and $\deg K = 2g - 2$. For the torus the canonical class is trivial.

Proposition 6.5. *On $T = \mathbb{C}/\Lambda$ the holomorphic 1-form dz is translation-invariant and nowhere vanishing, so*

$$K = \operatorname{div}(dz) = 0.$$

Hence the canonical class is trivial, $\deg K = 0$, and the space of holomorphic 1-forms is one-dimensional, $\Omega(T) = \mathbb{C}\,dz$; in particular $g = \dim \Omega(T) = 1$.

Proof. Since $d(z+\lambda) = dz$ for every $\lambda \in \Lambda$, the form dz descends to T; in the charts of theorem 2.10 it is simply dz, which has no zeros or poles, so $\operatorname{div}(dz) = 0$. Any holomorphic 1-form is $f\,dz$ with f holomorphic on T, hence constant (theorem 4.3); thus $\Omega(T) = \mathbb{C}\,dz$ is one-dimensional. The relation $\deg K = 2g - 2$ with $\deg K = 0$ gives $g = 1$, agreeing with proposition 3.7. □

6.4 Riemann–Roch: The General Theorem

For surfaces of any genus the dimension count is governed by one theorem, which we state for context. On the torus it reproduces the counts of proposition 6.3 and extends them to *arbitrary* divisors; we include it for context only. Even the converse of the next section — the one result beyond the elementary counts — is obtained constructively in chapter 7 via the σ-function, so nothing in the book depends on this theorem.

Theorem 6.6 (Riemann–Roch)**.** *Let X be a compact Riemann surface of genus g with canonical divisor K. For every divisor D,*

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

A proof belongs to a course on Riemann surfaces; see, e.g., Miranda [4] or Forster [5]. The term $\ell(K-D)$ is the *index of speciality*; by Serre duality it equals the dimension of holomorphic 1-forms whose divisor is $\geq D$. On the torus $g = 1$ and $K \sim 0$ (proposition 6.5), so $\ell(K - D) = \ell(-D)$ (proposition 6.2(4)) and Riemann–Roch reads

$$\boxed{\ell(D) - \ell(-D) = \deg D}.$$

Proposition 6.7 (Dimensions for arbitrary D, from Riemann–Roch)**.** *On T, for any divisor D: if $\deg D < 0$ then $\ell(D) = 0$; if $\deg D > 0$ then $\ell(D) = \deg D$; and if $\deg D = 0$ then $\ell(D) = 1$ for $D \sim 0$ and 0 otherwise. In particular $\ell(D) = \deg D$ for every divisor of degree ≥ 1, whether or not D is supported at O.*

Proof. For $\deg D < 0$, proposition 6.2(3). For $\deg D > 0$, $-D$ has negative degree so $\ell(-D) = 0$ and the boxed identity gives $\ell(D) = \deg D$. For $\deg D = 0$, the identity gives $\ell(D) = \ell(-D)$; a nonzero $f \in L(D)$ has $\operatorname{div}(f) \geq -D$ of degree 0, forcing $\operatorname{div}(f) = -D$, i.e. $D \sim 0$, so $\ell(D) \leq 1$ with equality iff $D \sim 0$. □

For divisors supported at O this only restates proposition 6.3; its new content, the value for *arbitrary* degree-one divisors, is what Riemann–Roch would contribute — but the next section reaches the same end constructively, so we do not rely on it.

6.5 The Converse of Abel: Completing $\mathrm{Pic}^0(T) \cong T$

We can now finish the injectivity left open in theorem 5.9.

Corollary 6.8. *Every divisor of degree* 1 *on* T *is linearly equivalent to a unique point* $[a]$*. Consequently every degree-zero divisor is linearly equivalent to a unique* $[a]-[0]$*, and the Abel–Jacobi map* $\bar{\mathfrak{s}} : \mathrm{Pic}^0(T) \to T$ *of theorem 5.9 is an isomorphism.*

Proof. The key input is the constructive converse of Abel, theorem 7.16: a degree-zero divisor with Abel sum zero is principal. *Injectivity:* if $\mathfrak{s}(D) = 0$ for $\deg D = 0$, then D is principal, so its class in $\mathrm{Pic}^0(T)$ is trivial; with surjectivity from theorem 5.9, $\bar{\mathfrak{s}}$ is an isomorphism. *Degree-one statement:* for $\deg D = 1$ set $a := \mathfrak{s}(D-[0]) \in T$; then $D-[a]$ has degree 0 and Abel sum $\mathfrak{s}(D-[0]) - a = 0$, so by theorem 7.16 $D-[a]$ is principal and $D \sim [a]$. Uniqueness: if $[a] \sim [a']$ then $[a]-[a']$ is principal of degree 0, so by exercise 5.4 (the forward direction of Abel) $[a] = [a']$. □

Remark 6.9 (Where existence is genuinely used). This corollary is the one result needing more than the elementary counts: the converse of Abel, that a balanced degree-zero divisor is principal. We do *not* import it from Riemann–Roch — theorem 7.16 supplies it constructively, building the required elliptic function explicitly from the Weierstrass σ-function. So even this is proved from within the book. Moreover it is *not* on the critical path to the main theorem: theorem 11.3 uses only the necessity direction of Abel (theorem 4.8) and the elementary counts of proposition 6.3. The isomorphism $\mathrm{Pic}^0(T) \cong T$ is the algebraic *view* of the group law, proved in full but not a step in the chord–tangent argument.

6.6 The Functions Forced

By proposition 6.3, $\ell(nO) = n$ for $n \geq 1$: the dimensions climb by exactly one at each step, so each $L((n{+}1)O)$ contains a function with a pole at O of order exactly $n+1$ not already present.

- $\ell(O) = 1$: $L(O) = \mathbb{C}$. There is *no* nonconstant function with a single simple pole — the count restates corollary 4.6.
- $\ell(2O) = 2$: beyond the constants there is one new function, with a double pole at O. After normalization this is the Weierstrass function $\wp$.
- $\ell(3O) = 3$: one further new function, with a triple pole at O, namely $\wp'$. A basis of $L(3O)$ is $\{1, \wp, \wp'\}$.

Divisor	ℓ	Typical basis	Meaning
O	1	1	no simple-pole function
$2O$	2	$1, \wp$	first nonconstant function
$3O$	3	$1, \wp, \wp'$	projective embedding data

Why $\wp$ and $\wp'$ are inevitable. The dimension counts (proposition 6.3) force the first nonconstant function on the torus to appear with a double pole, and its derivative then has a triple pole. The first three natural functions are therefore

$$1, \qquad \wp, \qquad \wp',$$

and these are exactly the coordinates of the map that embeds the torus as a plane cubic in chapter 8.

Remark 6.10 (The cubic is forced too). Push the count one step further. In $L(6O)$, which has dimension $\ell(6O) = 6$, the seven functions

$$1,\ \wp,\ \wp^2,\ \wp^3,\ \wp',\ \wp\wp',\ \wp'^2$$

all lie (their pole orders at O are $0, 2, 4, 6, 3, 5, 6$, each ≤ 6). Seven vectors in a six-dimensional space are linearly dependent, so they satisfy a linear relation. Matching the order-six poles forces $\wp'^2$ and $\wp^3$ to appear with nonzero coefficients: the relation is the Weierstrass cubic $\wp'^2 = 4\wp^3 - g_2\wp - g_3$, derived explicitly in chapter 7.

6.7 Summary

The space $L(D)$ measures functions with poles bounded by D (proposition 6.2). On the torus the counts that matter — $\ell(O) = 1$, $\ell([a]) = 1$, and $\ell(nO) = n$ — are *elementary*, following from the Liouville theorems and the explicit $\wp, \wp'$ with no Riemann–Roch (proposition 6.3); they force the functions $1, \wp, \wp'$ (section 6.6) and already predict the cubic (remark 6.10). Riemann–Roch (theorem 6.6) is the general theorem behind them; on the torus it collapses to $\ell(D) - \ell(-D) = \deg D$ (proposition 6.7). Even the converse of Abel, giving $\mathrm{Pic}^0(T) \cong T$ (corollary 6.8), is proved constructively in chapter 7 via the σ-function (remark 6.9), so Riemann–Roch is needed nowhere and serves only as the general theorem. The next chapter constructs $\wp$ — and σ — and verifies all of this by hand.

Exercises

Exercise 6.1. Prove the "point lemma": for any divisor D and point p, $\ell(D + p) - \ell(D) \in \{0, 1\}$. (Map $f \in L(D + p)$ to the Laurent coefficient of order $-n_p - 1$ at p; identify the kernel.)

Exercise 6.2. Without invoking Riemann–Roch, show directly from theorem 5.5 that $\ell(D) = 0$ whenever $\deg D < 0$, and that $\ell(D) \leq \deg D + 1$ for all D. (The second is the "Riemann inequality" half of the theorem.)

Exercise 6.3. Show that on T a divisor of degree one is effective if and only if it is a single point, and combine this with proposition 6.3 to give an independent proof that the only effective divisor in the class of $[a]$ is $[a]$ itself.

Exercise 6.4. Using proposition 6.5 and Riemann–Roch with $D = K$, verify the two consistency checks $\ell(K) = g$ and $\deg K = 2g - 2$ for the torus ($g = 1$).

Exercise 6.5. Carry out the counting of remark 6.10 in detail: list the pole orders at O of the seven functions

$$1,\quad \wp,\quad \wp^2,\quad \wp^3,\quad \wp',\quad \wp\wp',\quad \wp'^2,$$

explain why all seven lie in $L(6O)$, and argue that any linear relation among them must have nonzero coefficients on both $\wp'^2$ and $\wp^3$. What goes wrong if either coefficient is zero?

Part IV

The Weierstrass Functions

Chapter 7

The Weierstrass $\wp$-Function

Riemann–Roch told us (chapter 6) that the torus must carry a function with a double pole at O; it did not produce one. This chapter constructs it explicitly — the Weierstrass function $\wp$ — proves the series converges, and derives by hand the relation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ predicted in remark 6.10. The roots of the right-hand side are distinct ($\Delta \neq 0$), so the cubic is smooth; $\wp$ realizes the torus as a double cover of $\mathbb{CP}^1$; and the pair $(\wp, \wp')$ generates the whole field of elliptic functions. Everything the later chapters need about $\wp$ is established here.

7.1 Definition and Convergence

Definition 7.1. For a lattice Λ, the *Weierstrass function* is

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda \in \Lambda \setminus \{0\}} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right).$$

The subtracted $1/\lambda^2$ is what makes the series converge; without it the sum of $1/(z-\lambda)^2$ would diverge. Convergence rests on a counting estimate for the lattice.

Lemma 7.2 (Lattice sum estimate)**.** *The number of lattice points with $|\lambda| \leq R$ is $O(R^2)$, and consequently*

$$\sum_{\lambda \in \Lambda \setminus \{0\}} \frac{1}{|\lambda|^s} < \infty \qquad \text{for every } s > 2.$$

Proof. By exercise 2.2 the lattice points are separated, so disjoint disks of a fixed radius ρ about them lie within $|\lambda| \leq R + \rho$; comparing areas, their number is at most $\pi(R+\rho)^2/(\pi\rho^2) = O(R^2)$. Grouping the nonzero lattice points into annuli $n \leq |\lambda| < n+1$, each annulus contains $O(n)$ points, each contributing $\leq n^{-s}$, so $\sum_\lambda |\lambda|^{-s} \leq \sum_{n \geq 1} O(n)\, n^{-s} = \sum_{n \geq 1} O(n^{1-s})$, which converges for $s > 2$. □

Proposition 7.3. *The series defining $\wp$ converges absolutely and uniformly on every compact subset of $\mathbb{C} \setminus \Lambda$. Hence $\wp$ is meromorphic on $\mathbb{C}$, with a double pole and zero residue at each lattice point and no other poles.*

Proof. Fix $R > 0$ and consider $|z| \leq R$. For $|\lambda| \geq 2R$,

$$\left| \frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right| = \left| \frac{z(2\lambda - z)}{\lambda^2 (z-\lambda)^2} \right| \leq \frac{R\,(2|\lambda| + R)}{|\lambda|^2 (|\lambda| - R)^2} \leq \frac{R \cdot 3|\lambda|}{|\lambda|^2 (|\lambda|/2)^2} = \frac{12R}{|\lambda|^3}.$$

By lemma 7.2 with $s = 3$, $\sum_\lambda |\lambda|^{-3} < \infty$, so by the Weierstrass M-test the tail $\sum_{|\lambda|\geq 2R}$ converges uniformly on $|z| \leq R$; the finitely many remaining terms are meromorphic there. Thus $\wp$ is meromorphic, and near a lattice point λ_0 the term $1/(z-\lambda_0)^2$ supplies a double pole with zero residue while the rest is holomorphic. □

7.2 Periodicity and Evenness

Proposition 7.4. *$\wp$ is even, $\wp(-z) = \wp(z)$, and Λ-periodic, $\wp(z+\lambda) = \wp(z)$. Its derivative*

$$\wp'(z) = -2\sum_{\lambda\in\Lambda} \frac{1}{(z-\lambda)^3}$$

is Λ-periodic and odd, with a triple pole at each lattice point.

Proof. Evenness. Replacing z by $-z$ and reindexing $\lambda \mapsto -\lambda$ (which permutes Λ), $\wp(-z) = \frac{1}{z^2} + \sum_\lambda \big(\frac{1}{(z+\lambda)^2} - \frac{1}{\lambda^2}\big) = \wp(z)$.

The derivative. Differentiating the uniformly convergent series term by term gives $\wp'(z) = -2\sum_\lambda (z-\lambda)^{-3}$, where now the sum runs over *all* $\lambda \in \Lambda$ (the $\lambda = 0$ term is the derivative of $1/z^2$). This sum is manifestly periodic: for $\mu \in \Lambda$, substituting $z \mapsto z+\mu$ permutes the summands $\lambda \mapsto \lambda - \mu$, so $\wp'(z+\mu) = \wp'(z)$. It is odd because $\wp =$ even implies $\wp'$ odd (or directly from the symmetric sum).

Periodicity of $\wp$. For a basis vector ω, the function $g(z) = \wp(z+\omega) - \wp(z)$ has $g'(z) = \wp'(z+\omega) - \wp'(z) = 0$, so g is a constant c. Evaluating at $z = -\omega/2$ (not a lattice point) and using evenness, $c = \wp(\omega/2) - \wp(-\omega/2) = 0$. Hence $\wp(z+\omega) = \wp(z)$ for each basis vector, so $\wp$ is Λ-periodic. □

Thus $\wp$ is an elliptic function of order two (one double pole per period), the minimal order allowed by corollary 4.6, and $\wp'$ is elliptic of order three.

7.3 Eisenstein Series and the Laurent Expansion

Definition 7.5. For an even integer $m \geq 4$ the *Eisenstein series* of weight m is

$$G_m = G_m(\Lambda) = \sum_{\lambda\in\Lambda\setminus\{0\}} \frac{1}{\lambda^m},$$

which converges by lemma 7.2. For odd m the sum vanishes by the symmetry $\lambda \mapsto -\lambda$.

Proposition 7.6. *Near $z = 0$,*

$$\wp(z) = \frac{1}{z^2} + \sum_{k\geq 1} (2k+1)\, G_{2k+2}\, z^{2k} = \frac{1}{z^2} + 3G_4 z^2 + 5G_6 z^4 + O(z^6).$$

Proof. For $0 < |z| < |\lambda|$, expand $\frac{1}{(z-\lambda)^2} = \frac{1}{\lambda^2}\big(1 - z/\lambda\big)^{-2} = \frac{1}{\lambda^2}\sum_{n\geq 0}(n+1)\,(z/\lambda)^n$, so

$$\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} = \sum_{n\geq 1} (n+1)\,\frac{z^n}{\lambda^{n+2}}.$$

Summing over $\lambda \neq 0$ (absolute convergence justifies interchange) and adding $1/z^2$,

$$\wp(z) = \frac{1}{z^2} + \sum_{n \geq 1} (n+1)\, G_{n+2}\, z^n.$$

The odd-n terms drop ($G_{n+2} = 0$ for n odd), leaving the stated even series with $a_2 = 3G_4$, $a_4 = 5G_6$. □

7.4 The Differential Equation

Theorem 7.7 (Weierstrass differential equation)**.** *With $g_2 = 60G_4$ and $g_3 = 140G_6$,*

$$\boxed{\wp'^2 = 4\wp^3 - g_2\wp - g_3.}$$

Proof. From proposition 7.6, $\wp = z^{-2} + 3G_4 z^2 + 5G_6 z^4 + O(z^6)$ and $\wp' = -2z^{-3} + 6G_4 z + 20G_6 z^3 + O(z^5)$. Squaring and cubing,

$$\wp'^2 = 4z^{-6} - 24G_4 z^{-2} - 80G_6 + O(z^2), \qquad 4\wp^3 = 4z^{-6} + 36G_4 z^{-2} + 60G_6 + O(z^2).$$

Subtracting, the order-six poles cancel and

$$\wp'^2 - 4\wp^3 = -60G_4 z^{-2} - 140G_6 + O(z^2).$$

Adding $g_2\wp = 60G_4 z^{-2} + O(z^2)$ and the constant $g_3 = 140G_6$ cancels the remaining negative power and constant:

$$F := \wp'^2 - 4\wp^3 + g_2\wp + g_3 = O(z^2).$$

Now F is elliptic (a polynomial in $\wp, \wp'$), and its only possible poles are at lattice points; the computation shows F is holomorphic at 0, and by periodicity at every lattice point, so F is holomorphic on the compact torus, hence constant (theorem 4.3). Since $F = O(z^2) \to 0$ as $z \to 0$, that constant is 0. Therefore $\wp'^2 = 4\wp^3 - g_2\wp - g_3$. □

The birth of the cubic. The equation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ is not guessed. A specific polynomial in $\wp$ and $\wp'$ was arranged to have no pole at the origin; being elliptic and pole-free it is constant, and the Laurent expansion shows the constant is zero. The map $z \mapsto (\wp(z), \wp'(z))$ therefore lands on the plane cubic $y^2 = 4x^3 - g_2 x - g_3$ — the embedding of chapter 8.

7.5 The Roots, the Discriminant, and the Double Cover

The cubic factors over $\mathbb{C}$ as $4\wp^3 - g_2\wp - g_3 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3)$. The roots have a direct meaning on the torus.

Proposition 7.8. *Let $\frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_1+\omega_2}{2}$ be the three nonzero half-periods and set $e_1 = \wp(\frac{\omega_1}{2})$, $e_2 = \wp(\frac{\omega_2}{2})$, $e_3 = \wp(\frac{\omega_1+\omega_2}{2})$. Then:*

1. $\wp'$ vanishes exactly at the three half-periods, each a simple zero;

2. $4\wp^3 - g_2\wp - g_3 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3)$*;*

3. e_1, e_2, e_3 *are distinct.*

Proof. (1) $\wp'$ is odd and periodic, so at a half-period $\omega/2$, writing $-\omega/2 \equiv \omega/2 \pmod{\Lambda}$, $\wp'(\omega/2) = \wp'(-\omega/2) = -\wp'(\omega/2)$, hence $\wp'(\omega/2) = 0$. That is three zeros; $\wp'$ has order three (triple pole at O), so by theorem 4.7 it has exactly three zeros, and these are simple.

(2) By theorem 7.7, $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ vanishes wherever $\wp' = 0$, i.e. at the half-periods, where $\wp$ takes the values e_1, e_2, e_3. So e_1, e_2, e_3 are the roots of the cubic $4x^3 - g_2x - g_3$, giving the factorization.

(3) Suppose $e_1 = e_2$. The function $\wp - e_1$ has order two (double pole at O), so exactly two zeros. It vanishes at $\omega_1/2$, and there $(\wp - e_1)' = \wp' = 0$, so that zero is double — using up both zeros. Then $\wp - e_1$ cannot also vanish at $\omega_2/2$, contradicting $\wp(\omega_2/2) = e_2 = e_1$. Hence $e_1 \neq e_2$, and similarly all three are distinct. □

Definition 7.9. The *discriminant* of Λ is $\Delta = g_2^3 - 27g_3^2$.

Corollary 7.10. $\Delta = 16\,(e_1-e_2)^2(e_2-e_3)^2(e_1-e_3)^2 \neq 0$. *Equivalently the cubic* $y^2 = 4x^3 - g_2x - g_3$ *is nonsingular.*

Proof. For $4x^3 - g_2x - g_3 = 4(x - e_1)(x - e_2)(x - e_3)$ the elementary symmetric relations give $e_1 + e_2 + e_3 = 0$, and a standard discriminant computation yields $g_2^3 - 27g_3^2 = 16\prod_{i<j}(e_i - e_j)^2$. By proposition 7.8(3) the e_i are distinct, so $\Delta \neq 0$. A plane cubic $y^2 = p(x)$ is singular exactly when p has a repeated root, which $\Delta \neq 0$ excludes. □

The geometric content is that $\wp$ presents the torus as a branched double cover of the sphere (fig. 7.1): a generic value is attained twice ($\wp$ has order two), while the four values e_1, e_2, e_3, ∞ are attained once, at the half-periods and at O. The Riemann–Hurwitz count $\chi(T) = 2\,\chi(\mathbb{CP}^1) -$ (branch points) reads $0 = 2 \cdot 2 - 4$, confirming genus one.

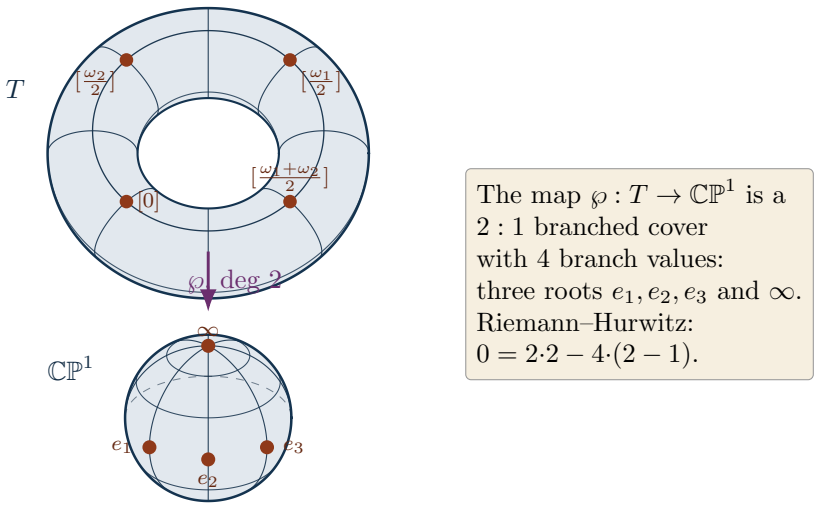


Figure 7.1: $\wp$ as a degree-two branched cover $T \to \mathbb{CP}^1$. The four branch points are the half-periods (mapping to e_1, e_2, e_3) and O (mapping to ∞). Torus and sphere are rendered from `sagemath/08_torus_render.sage` and `sagemath/10_sphere_render.sage`.

7.6 Generation of the Field of Elliptic Functions

We promised in chapter 4 that $\wp, \wp'$ generate all elliptic functions. We can now prove it, and identify the field structure.

Theorem 7.11. *The field of Λ-elliptic functions is $\mathbb{C}(\wp, \wp')$. Moreover $\wp' \notin \mathbb{C}(\wp)$, so $\mathbb{C}(\wp, \wp')$ is a degree-two extension of $\mathbb{C}(\wp)$, with $\wp'$ satisfying the minimal relation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$.*

Proof. Minimality. Every element of $\mathbb{C}(\wp)$ is even, since $\wp$ is even, so a rational function $R(\wp)$ satisfies $R(\wp)(-z) = R(\wp)(z)$. But $\wp'$ is odd and not identically zero, so $\wp' \notin \mathbb{C}(\wp)$. As $\wp'^2 \in \mathbb{C}(\wp)$ by theorem 7.7, the extension $\mathbb{C}(\wp, \wp')/\mathbb{C}(\wp)$ has degree exactly two.

Generation. Let f be elliptic. Split it into even and odd parts,

$$f = f_+ + f_-, \quad f_+(z) = \tfrac{1}{2}\big(f(z) + f(-z)\big), \quad f_-(z) = \tfrac{1}{2}\big(f(z) - f(-z)\big),$$

both elliptic. Then $f_-/\wp'$ is even (odd over odd), so it suffices to show every even elliptic function lies in $\mathbb{C}(\wp)$. Let h be even elliptic. Since $h(-z) = h(z)$, its divisor is invariant under $z \mapsto -z$, so its zeros and poles occur in pairs $\{[a], [-a]\}$ of equal order — except at the four two-torsion points, where $[a] = [-a]$. At a two-torsion point a the order of h is *even*: in the local coordinate $w = z - a$ the involution $z \mapsto -z$ is $w \mapsto -w$ (because $2a \in \Lambda$), and $h(a + w) = h(a - w)$, so h is an even function of w and vanishes (or blows up) to even order.

Now match $\operatorname{div}(h)$ with a rational function of $\wp$. For a non-two-torsion pair where h has order m, the factor $(\wp - \wp(a))^m$ has divisor $m\big([a] + [-a] - 2[O]\big)$. For a two-torsion point a where h has order $2m$, the factor $(\wp - \wp(a))^m$ has divisor $2m\big([a] - [O]\big)$, because $\wp - \wp(a)$ has a *double* zero at the half-period a (there $\wp' = 0$, proposition 7.8). Let $R(\wp)$ be the product of these factors, the exponent positive at zeros of h and negative at its poles. Then $\operatorname{div}(h)$ and $\operatorname{div} R(\wp)$ agree at every point except possibly O; both have degree 0 (theorem 5.5), so their coefficients at O agree too, and $\operatorname{div}(h) = \operatorname{div} R(\wp)$. Hence $h/R(\wp)$ is elliptic with no zeros or poles, so it is constant, and $h \in \mathbb{C}(\wp)$. Therefore $f = f_+ + (f_-/\wp')\,\wp' \in \mathbb{C}(\wp, \wp')$. □

7.7 The σ-Function and the Converse of Abel

The converse of Abel's law — that a degree-zero divisor with Abel sum zero is the divisor of an elliptic function — was deferred in chapters 5 and 6 to an existence theorem. We now discharge it *constructively*, building the function explicitly from a third Weierstrass function.

Definition 7.12. The Weierstrass *sigma function* is

$$\sigma(z) = z \prod_{\lambda \in \Lambda \setminus \{0\}} \Big(1 - \tfrac{z}{\lambda}\Big) \exp\Big(\tfrac{z}{\lambda} + \tfrac{1}{2}\big(\tfrac{z}{\lambda}\big)^2\Big).$$

Proposition 7.13. *The product converges normally on compact sets, so σ is entire; it is odd, $\sigma(-z) = -\sigma(z)$, and its zeros are exactly the lattice points, each simple.*

Proof. Taking logarithms, the factor at λ contributes

$$\log\Big(1 - \tfrac{z}{\lambda}\Big) + \tfrac{z}{\lambda} + \tfrac{1}{2}\big(\tfrac{z}{\lambda}\big)^2 = -\sum_{m \geq 3} \frac{1}{m}\Big(\tfrac{z}{\lambda}\Big)^m = O\big(|\lambda|^{-3}\big)$$

for $|z| \leq R$, $|\lambda| \geq 2R$; by lemma 7.2 with $s = 3$ the sum converges normally, so the product converges to an entire function whose zeros are those of its factors — $z = 0$ from the leading z, and $z = \lambda$ from $1 - z/\lambda$ — each simple. For oddness, re-index $\lambda \mapsto -\lambda$: $\sigma(-z) = -z\prod(1 + z/\lambda)\exp(-z/\lambda + \tfrac{1}{2}(z/\lambda)^2) = -\sigma(z)$. □

The logarithmic derivative $\zeta = \sigma'/\sigma$ is the Weierstrass *zeta function*

$$\zeta(z) = \frac{1}{z} + \sum_{\lambda\neq 0}\Big(\frac{1}{z-\lambda} + \frac{1}{\lambda} + \frac{z}{\lambda^2}\Big),$$

odd, with a simple pole of residue 1 at each lattice point. Differentiating term by term, $\zeta'(z) = -\frac{1}{z^2} + \sum_{\lambda\neq 0}\big(-\frac{1}{(z-\lambda)^2} + \frac{1}{\lambda^2}\big) = -\wp(z)$ (definition 7.1); equivalently $\wp = -(\log\sigma)''$, so this σ is the one attached to the book's $\wp$.

Proposition 7.14 (Quasi-periodicity)**.** *For each basis period ω_k there is a constant $\eta_k = 2\zeta(\omega_k/2)$ with*

$$\zeta(z+\omega_k) = \zeta(z) + \eta_k, \qquad \sigma(z+\omega_k) = -\,\sigma(z)\,\exp\big(\eta_k(z+\tfrac{\omega_k}{2})\big).$$

Proof. Since $\zeta' = -\wp$ is Λ-periodic (proposition 7.4), $\zeta(z+\omega_k)-\zeta(z)$ is constant; evaluating at $z = -\omega_k/2$ and using ζ odd gives the constant $2\zeta(\omega_k/2) =: \eta_k$. As $(\log\sigma)' = \zeta$, $\frac{d}{dz}\log\frac{\sigma(z+\omega_k)}{\sigma(z)} = \eta_k$, so $\sigma(z+\omega_k) = \sigma(z)\,e^{\eta_k z + c_k}$ for a constant c_k. Setting $z = -\omega_k/2$ and using $\sigma(-\omega_k/2) = -\sigma(\omega_k/2) \neq 0$ gives $e^{c_k - \eta_k\omega_k/2} = -1$, which is the stated formula. □

Remark 7.15. The quasi-periods obey the Legendre relation $\eta_1\omega_2 - \eta_2\omega_1 = 2\pi i$ for a positively oriented basis (integrate ζ around a period parallelogram); we will not need it.

Theorem 7.16 (Converse of Abel, constructively)**.** *Let $D = \sum_{i=1}^n [a_i] - \sum_{i=1}^n [b_i]$ be a divisor of degree zero on T with Abel sum zero, $\sum_i a_i \equiv \sum_i b_i \pmod\Lambda$. Then D is principal, $D = \operatorname{div}(F)$, with*

$$F(z) = \frac{\prod_{i=1}^n \sigma(z-a_i)}{\prod_{i=1}^n \sigma(z-b_i)}.$$

Proof. Adjusting one representative by a lattice vector — which changes neither its class nor D — we may assume $\sum_i a_i = \sum_i b_i$ in $\mathbb{C}$. By proposition 7.13 the a_i are simple zeros and the b_i simple poles of F (with multiplicity if repeated), so $\operatorname{div}(F) = D$. For periodicity, by proposition 7.14 each basis period gives

$$\frac{F(z+\omega_k)}{F(z)} = \frac{\prod_i\big(-e^{\eta_k(z-a_i+\omega_k/2)}\big)}{\prod_i\big(-e^{\eta_k(z-b_i+\omega_k/2)}\big)} = \exp\Big(\eta_k\big(\textstyle\sum_i b_i - \sum_i a_i\big)\Big) = 1 :$$

the $(-1)^n$ factors and the $nz + n\omega_k/2$ terms cancel because the counts are equal (degree zero), and the exponent vanishes because $\sum a_i = \sum b_i$ (Abel sum zero). So F is elliptic with $\operatorname{div}(F) = D$, and D is principal. □

Remark 7.17. Theorem 7.16 is the converse of Abel's law; it completes the isomorphism $\operatorname{Pic}^0(T) \cong T$ (theorem 5.9 and corollary 6.8) with no appeal to Riemann–Roch. Together with the elementary counts of proposition 6.3, every load-bearing result in the book is now proved from within it. Riemann–Roch (theorem 6.6) remains as the general theorem, illuminating but unneeded.

7.8 Summary

The Weierstrass function $\wp(z) = z^{-2} + \sum_{\lambda\neq 0}\big((z-\lambda)^{-2} - \lambda^{-2}\big)$ converges (proposition 7.3) to an even elliptic function of order two, with odd derivative $\wp'$ of order three (proposition 7.4). Its Laurent expansion has Eisenstein coefficients G_4, G_6 (proposition 7.6), and a pole-cancellation argument yields the differential equation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ with $g_2 = 60G_4$, $g_3 = 140G_6$

(theorem 7.7). The roots $e_i = \wp(\omega_i/2)$ are the values of $\wp$ at the half-periods and are distinct, so the discriminant $\Delta = g_2^3 - 27g_3^2$ is nonzero and the cubic is smooth (proposition 7.8 and corollary 7.10); $\wp$ exhibits T as a double cover of $\mathbb{CP}^1$ branched over e_1, e_2, e_3, ∞. Finally $\mathbb{C}(\wp, \wp')$ is the full field of elliptic functions, a quadratic extension of $\mathbb{C}(\wp)$ (theorem 7.11). Finally the sigma function σ (definition 7.12), a third Weierstrass function, builds an elliptic function with any prescribed balanced divisor, proving the converse of Abel's law constructively (theorem 7.16) — so $\mathrm{Pic}^0(T) \cong T$ needs no Riemann–Roch. The next chapter turns $(\wp, \wp')$ into the embedding of theorem 1.1.

Exercises

Exercise 7.1. Prove lemma 7.2 in detail, and use it to justify term-by-term differentiation of the $\wp$-series, giving $\wp'(z) = -2\sum_\lambda (z-\lambda)^{-3}$.

Exercise 7.2. From the Laurent expansion derive $a_2 = 3G_4$ and $a_4 = 5G_6$, and confirm $g_2 = 20a_2 = 60G_4$ and $g_3 = 28a_4 = 140G_6$. (Match the z^{-2} and constant terms of $\wp'^2 - 4\wp^3$.)

Exercise 7.3. Give the discriminant computation $g_2^3 - 27g_3^2 = 16\prod_{i<j}(e_i - e_j)^2$ starting from $e_1+e_2+e_3 = 0$, $e_1e_2+e_1e_3+e_2e_3 = -g_2/4$, $e_1e_2e_3 = g_3/4$. Conclude $\Delta \neq 0$ from proposition 7.8(3).

Exercise 7.4. Show every elliptic function f can be written $f = A(\wp) + B(\wp)\,\wp'$ with A, B rational, by splitting f into even and odd parts as in theorem 7.11. Where is $\wp' \notin \mathbb{C}(\wp)$ used?

Exercise 7.5. Differentiate $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ to obtain $\wp'' = 6\wp^2 - \frac{1}{2}g_2$. Expanding both sides in Laurent series, derive a recurrence expressing each Eisenstein coefficient G_{2k} ($k \geq 4$) as a polynomial in G_4 and G_6. (Thus all G_{2k} are determined by G_4, G_6.)

Exercise 7.6. Complete the proof of proposition 7.14: starting from $\zeta(z+\omega_k) = \zeta(z) + \eta_k$ and $(\log\sigma)' = \zeta$, integrate to get $\sigma(z+\omega_k) = \sigma(z)\,e^{\eta_k z + c_k}$, then determine the constant c_k by setting $z = -\omega_k/2$ and using that σ is odd, obtaining $\sigma(z+\omega_k) = -\sigma(z)\,e^{\eta_k(z+\omega_k/2)}$.

Exercise 7.7. In the proof of theorem 7.16, verify the cancellation $F(z+\omega_k)/F(z) = \exp\big(\eta_k(\sum_i b_i - \sum_i a_i)\big)$ in detail, and identify exactly where the two hypotheses are used: that the number of zeros equals the number of poles (degree zero), and that $\sum_i a_i = \sum_i b_i$ (Abel sum zero). Which hypothesis fails, and what goes wrong, for the divisor $[a] - [0]$ with $a \notin \Lambda$?

Chapter 8

The Embedding into $\mathbb{CP}^2$

With $\wp$ and $\wp'$ in hand we can build the map of theorem 1.1: the embedding

$$\Phi : T = \mathbb{C}/\Lambda \longrightarrow E \subset \mathbb{CP}^2, \qquad \Phi([z]) = [\wp(z) : \wp'(z) : 1].$$

This chapter shows Φ is a well-defined holomorphic map onto the smooth cubic E, extends it across the origin to the point at infinity $O = [0 : 1 : 0]$, and proves it is an isomorphism of compact Riemann surfaces — the analytic half of theorem 1.1. Finally we read off the first piece of the group dictionary: the torus inverse $z \mapsto -z$ becomes reflection across the x-axis. The group law itself is transported in chapter 11.

8.1 The Projective Map

For $z \notin \Lambda$ set $\Phi([z]) = [\wp(z) : \wp'(z) : 1]$. Since $\wp, \wp'$ are Λ-periodic (proposition 7.4) this is well-defined on T. The differential equation places the image on the cubic.

Proposition 8.1. *For $z \notin \Lambda$, the point $\Phi([z])$ lies on*

$$E: \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$

Proof. In the affine chart $Z = 1$ the equation reads $y^2 = 4x^3 - g_2x - g_3$. Substituting $x = \wp(z)$, $y = \wp'(z)$ gives exactly the Weierstrass differential equation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ of theorem 7.7, which holds; so $(\wp(z), \wp'(z)) \in E$. □

8.2 Extension at the Origin

At $z = 0$ both $\wp$ and $\wp'$ blow up, so the affine formula fails, but the projective one has a limit. Multiplying all three homogeneous coordinates by z^3,

$$[\wp(z) : \wp'(z) : 1] = [z^3\wp(z) : z^3\wp'(z) : z^3],$$

and using $\wp = z^{-2} + O(z^2)$, $\wp' = -2z^{-3} + O(z)$,

$$z^3\wp(z) \to 0, \qquad z^3\wp'(z) \to -2, \qquad z^3 \to 0 \quad (z \to 0).$$

Hence $\lim_{z\to 0} \Phi([z]) = [0 : -2 : 0] = [0 : 1 : 0]$, and we set

$$\Phi(O) = [0 : 1 : 0] =: O \in E.$$

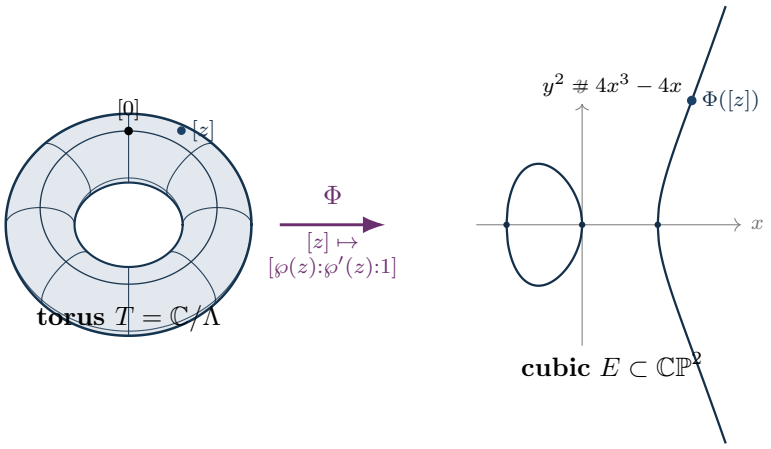


Figure 8.1: The map $\Phi([z]) = [\wp(z) : \wp'(z) : 1]$ carries the torus onto the real locus of the cubic $y^2 = 4x^3 - 4x$ (here $g_2 = 4$, $g_3 = 0$). Curve and sample point are from `sagemath/12_curve_sample.sage`.

Proposition 8.2. *With this value at the origin, $\Phi : T \to E$ is holomorphic.*

Proof. Away from [0], Φ is given by the holomorphic functions $\wp, \wp'$ in the chart $Z = 1$, so it is holomorphic there. Near [0] use the homogeneous representative $\big(z^3\wp(z),\ z^3\wp'(z),\ z^3\big)$: each entry is holomorphic in z (the poles of $\wp, \wp'$ are cancelled by z^3), and they do not vanish simultaneously, since the middle entry tends to $-2 \neq 0$. In the affine chart $Y = 1$ of $\mathbb{CP}^2$ the map is therefore $\big(z^3\wp/(z^3\wp'),\ z^3/(z^3\wp')\big) = \big(\wp/\wp',\ 1/\wp'\big)$, holomorphic at $z = 0$ with value $(0, 0) = O$. So Φ is holomorphic on all of T. □

8.3 The Point at Infinity

Proposition 8.3. *$O = [0 : 1 : 0]$ is the unique point of E on the line at infinity $Z = 0$. It is a smooth point of E, and the line at infinity is the tangent there, meeting E only at O with intersection multiplicity three: O is an inflection point.*

Proof. Setting $Z = 0$ in $Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3$ gives $0 = 4X^3$, so $X = 0$; the only such projective point is $[0 : 1 : 0]$. In the affine chart $Y = 1$, with $F(X, Z) = Z - 4X^3 + g_2XZ^2 + g_3Z^3$, we have $\partial F/\partial Z\,|_{(0,0)} = 1 \neq 0$, so O is smooth and the tangent is $Z = 0$. Substituting $Z = 0$ into E yields $4X^3 = 0$, a triple root at $X = 0$, so $Z = 0$ meets E at O with multiplicity three. □

The flex at O is the geometric reason O will serve as the identity of the group law: a line meeting E three times at O records $3O$, the divisor class that is the identity (chapter 11).

8.4 Why $L(3O)$ Gives the Embedding

The coordinates of Φ are a basis of $L(3O)$. By proposition 6.3, $\ell(3O) = 3$, with basis $\{1, \wp, \wp'\}$: a function with no pole, one with a double pole, one with a triple pole, all at O. Three linearly independent functions define a map to $\mathbb{CP}^2$,

$$[z] \longmapsto [\wp(z) : \wp'(z) : 1],$$

and the single algebraic relation among the three sections — forced by the dimension count of remark 6.10 — is the cubic. Because $\Delta \neq 0$ (corollary 7.10), E is a *smooth* projective plane curve,

hence a compact Riemann surface: near each point one of the partial derivatives of the defining equation is nonzero, and the implicit function theorem provides a holomorphic chart. It is also connected (the polynomial $Y^2 - (4X^3 - g_2X - g_3)$ is irreducible when the cubic has distinct roots).

8.5 Injectivity and the Isomorphism

Lemma 8.4. *If $\wp(z) = \wp(w)$ then $w \equiv z$ or $w \equiv -z \pmod{\Lambda}$.*

Proof. Fix w and consider $\wp - \wp(w)$, elliptic of order two (double pole at $[0]$), so by theorem 4.7 it has exactly two zeros counted with multiplicity. Both w and $-w$ are zeros, since $\wp$ is even; if $w \not\equiv -w$ these are the two distinct zeros, and $\wp(z) = \wp(w)$ forces $z \equiv w$ or $z \equiv -w$. If $w \equiv -w$ (a half-period), then $\wp - \wp(w)$ has a double zero at w, again accounting for both zeros, and $\wp(z) = \wp(w)$ forces $z \equiv w \equiv -w$. In all cases $z \equiv \pm w$. □

Theorem 8.5 (Φ is an isomorphism)**.** *The map $\Phi : T \to E$ is a biholomorphism of compact Riemann surfaces.*

Proof. Injectivity. Suppose $\Phi([z]) = \Phi([w])$ with neither at O. Then $\wp(z) = \wp(w)$ and $\wp'(z) = \wp'(w)$. By lemma 8.4, $w \equiv z$ or $w \equiv -z$. In the second case $\wp'(w) = \wp'(-z) = -\wp'(z)$, and combined with $\wp'(w) = \wp'(z)$ this gives $\wp'(z) = 0$, i.e. z is a half-period (proposition 7.8), where $z \equiv -z$; so $w \equiv z$ anyway. As O is the only point mapping to $[0 : 1 : 0]$ (proposition 8.3), Φ is injective.

Bijectivity and biholomorphy. Φ is a nonconstant holomorphic map between the compact connected Riemann surfaces T and E; its image is open (open mapping theorem) and compact, hence all of the connected curve E, so Φ is surjective. An injective holomorphic map between Riemann surfaces is an immersion with nonvanishing derivative, so Φ is a holomorphic bijection whose inverse is holomorphic. Therefore Φ is a biholomorphism. □

This proves the Riemann-surface assertion of theorem 1.1: the torus and the cubic are the same compact Riemann surface, identified by $(\wp, \wp')$. The remaining assertion — that Φ is a group isomorphism for the chord–tangent law — is the subject of chapters 9 to 11.

8.6 Inverses Become Reflection

Transport the torus inverse $z \mapsto -z$ through Φ. Since $\wp$ is even and $\wp'$ odd,

$$\Phi(-z) = [\wp(-z) : \wp'(-z) : 1] = [\wp(z) : -\wp'(z) : 1],$$

so in affine coordinates $P = (x, y) = (\wp(z), \wp'(z))$ maps to

$$-P = (x, -y).$$

Why reflection appears. Reflection across the x-axis is not an extra convention bolted onto the chord–tangent rule. It *is* the torus inverse $z \mapsto -z$, written in the coordinates $x = \wp(z)$, $y = \wp'(z)$. The points fixed by the reflection are exactly those with $y = 0$ — the images $(e_1, 0), (e_2, 0), (e_3, 0)$ of the half-periods — together with O at infinity. These four are the two-torsion of the group: the points equal to their own inverse.

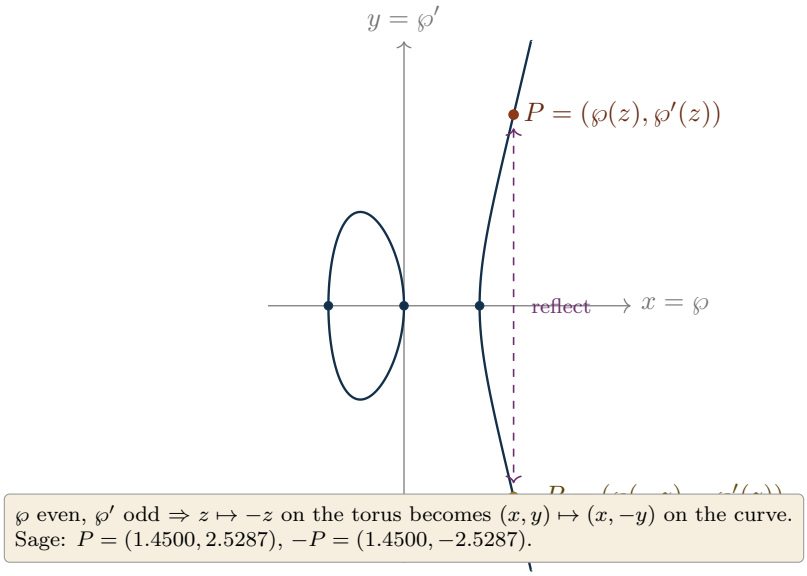


Figure 8.2: The torus inverse $z \mapsto -z$ becomes reflection across the x-axis on the cubic. The fixed points $y = 0$ are the half-periods e_1, e_2, e_3; O is its own inverse at infinity. Curve from `sagemath/12_curve_sample.sage`.

8.7 Summary

The map $\Phi([z]) = [\wp(z) : \wp'(z) : 1]$, extended by $\Phi(O) = [0 : 1 : 0]$, is a holomorphic map of T onto the smooth cubic E (propositions 8.1 and 8.2). The point O is the unique point at infinity and an inflection (proposition 8.3), destined to be the identity. The coordinates are a basis of $L(3O)$, and because $\Delta \neq 0$ the curve E is a compact Riemann surface; Φ is injective (lemma 8.4) and hence a biholomorphism $T \xrightarrow{\sim} E$ (theorem 8.5), the analytic content of theorem 1.1. Under Φ the torus inverse $z \mapsto -z$ is reflection $(x, y) \mapsto (x, -y)$, whose fixed points are the two-torsion. What remains is to see that the *addition* on T becomes the chord–tangent law; that needs Abel's sum law, taken up next.

Exercises

Exercise 8.1. Verify proposition 8.2 in the chart $Y = 1$: show the map near $[0]$ is $z \mapsto (\wp/\wp', 1/\wp')$, compute its value and derivative at $z = 0$, and confirm it is a local holomorphic coordinate. (This shows Φ is an immersion at O.)

Exercise 8.2. Carry out proposition 8.3 in full: confirm O is smooth by computing the gradient of the projective equation there, and verify the intersection multiplicity of $Z = 0$ with E at O is three by substitution. Why does a flex at O make $3O$ a principal divisor?

Exercise 8.3. Prove lemma 8.4 and deduce that $z \mapsto \wp(z)$ realizes $T/\{\pm 1\} \cong \mathbb{CP}^1$, with the four fixed points of $z \mapsto -z$ mapping to the branch values e_1, e_2, e_3, ∞.

Exercise 8.4. Show that Φ sends the four two-torsion points to the points of E fixed by $(x, y) \mapsto (x, -y)$, according to

$$0 \mapsto O, \quad \tfrac{\omega_1}{2} \mapsto (e_1, 0), \quad \tfrac{\omega_2}{2} \mapsto (e_2, 0), \quad \tfrac{\omega_1+\omega_2}{2} \mapsto (e_3, 0),$$

and that these four points are precisely those of E with $y = 0$ or $y = \infty$.

Exercise 8.5. Show the reflection $\iota_E(x, y) = (x, -y)$ is a holomorphic involution of E with quotient $\mathbb{CP}^1$ (via $x = \wp$), and that under Φ it corresponds to $z \mapsto -z$. Conclude that ι_E is the unique nontrivial deck transformation of the double cover $\wp : T \to \mathbb{CP}^1$.

Part V

The Group Law on the Cubic

Chapter 9

Abel's Sum Law

The analytic relation $\sum(\text{zeros}) - \sum(\text{poles}) \in \Lambda$ was proved in chapter 4 (Liouville's fourth theorem, theorem 4.8) and packaged in chapters 5 and 6 as the isomorphism $\mathrm{Pic}^0(T) \cong T$. This chapter states it in the form the cubic needs: a complete criterion for when a divisor is principal, and its specialization to three points. That criterion is the bridge of theorem 1.1 — once we know a line cuts the cubic in a principal divisor (chapter 10), Abel's law turns "collinear" into "sums to O" (chapter 11).

9.1 Statement

Theorem 9.1 (Abel's sum law)**.** *Let $D = \sum_j n_j[z_j]$ be a divisor on $T = \mathbb{C}/\Lambda$. Then D is principal — $D = \mathrm{div}(F)$ for some nonzero elliptic function F — if and only if it satisfies both*

$$\deg D = \sum_j n_j = 0 \qquad \textit{and} \qquad \mathfrak{s}(D) = \sum_j n_j[z_j] = 0 \ \textit{in } T.$$

In representatives, the second condition reads $\sum_j n_j z_j \in \Lambda$.

The two conditions are independent and both necessary: the degree records that a function has as many zeros as poles, and the Abel sum records that their *positions* also balance in the group of T. Neither alone suffices (exercise 9.1).

9.2 Necessity: Principal Implies Both Conditions

Both halves of the forward direction are already theorems, proved by integrating around a period parallelogram in chapter 4.

Proof of necessity in theorem 9.1. Let $D = \mathrm{div}(F)$ and choose a period parallelogram P_a whose boundary avoids the zeros and poles. Integrating F'/F over ∂P_a, opposite sides cancel (lemma 4.4), so $\deg D = \frac{1}{2\pi i}\oint F'/F\,dz = 0$: this is theorem 5.5. For the Abel sum, integrate $z\,F'/F$; its residue at z_j is $n_j z_j$, so the boundary integral is $2\pi i \sum_j n_j z_j$. The factor z is not periodic, and pairing opposite sides leaves

$$\oint_{\partial P_a} z\,\frac{F'}{F}\,dz = \omega_1 \int_{\omega_2\text{-side}} \frac{F'}{F}\,dz - \omega_2 \int_{\omega_1\text{-side}} \frac{F'}{F}\,dz \in 2\pi i\,\Lambda,$$

because each side integral is the change of a branch of $\log F$ over a period, an element of $2\pi i\,\mathbb{Z}$. Dividing by $2\pi i$, $\mathfrak{s}(D) = \sum_j n_j z_j \in \Lambda$: this is Liouville's fourth theorem, theorem 4.8 (the side-pairing detail is exercise 9.3). □

9.3 Sufficiency: The Converse of Abel

The harder direction — that the two balance conditions are *enough* to make a divisor principal — is the converse of Abel, proved constructively by the σ-function in theorem 7.16 and recorded as $\mathrm{Pic}^0(T) \cong T$ in corollary 6.8.

Proof of sufficiency in theorem 9.1. Suppose $\deg D = 0$ and $\mathfrak{s}(D) = 0$. By corollary 6.8 every degree-zero divisor is linearly equivalent to $[a] - [0]$ for a unique $[a] \in T$, and applying $\mathfrak{s}$ shows $a = \mathfrak{s}(D) = 0$. Hence $D \sim [0] - [0] = 0$, i.e. D is principal. (Equivalently, $\bar{\mathfrak{s}} : \mathrm{Pic}^0(T) \xrightarrow{\sim} T$ is an isomorphism (theorem 5.9 and corollary 6.8), so its kernel is exactly the principal divisors.) □

Remark 9.2. The converse is an existence statement: it asserts an elliptic function with prescribed balanced zeros and poles. It is not merely abstract — theorem 7.16 names the function explicitly, $F(z) = \prod_i \sigma(z - a_i) / \prod_i \sigma(z - b_i)$, built from the Weierstrass σ-function; for the divisors arising on the cubic it is the restriction of a line (chapter 10).

Combining the two directions, the principal divisors are precisely the kernel of the combined map

$$(\deg, \mathfrak{s}) : \mathrm{Div}(T) \longrightarrow \mathbb{Z} \oplus T, \qquad \mathrm{Prin}(T) = \ker(\deg, \mathfrak{s}),$$

which is another way to state $\mathrm{Pic}(T) \cong \mathbb{Z} \oplus T$ (exercise 5.3).

9.4 The Three-Point Form

For the chord–tangent law we need Abel's law for divisors supported at three points and O. Writing points of the cubic through Φ, with $P = [p]$, $Q = [q]$, $R = [r]$, the relevant divisors are $[p] + [q] + [r] - 3[O]$.

Corollary 9.3. *For $p, q, r \in \mathbb{C}$, the divisor $[p] + [q] + [r] - 3[0]$ on T is principal if and only if*

$$p + q + r \equiv 0 \pmod{\Lambda}, \qquad \textit{i.e.} \qquad [p] + [q] + [r] = 0 \text{ in } T.$$

More generally $[p] + [q] - [r] - [0]$ is principal iff $[p] + [q] = [r]$ in T.

Proof. The divisor $[p] + [q] + [r] - 3[0]$ has degree $3 - 3 = 0$, so by theorem 9.1 it is principal iff its Abel sum vanishes; and $\mathfrak{s}\big([p] + [q] + [r] - 3[0]\big) = [p] + [q] + [r] - 3[0] = [p] + [q] + [r]$, which is 0 iff $p + q + r \in \Lambda$. The second statement is the same computation for the degree-zero divisor $[p] + [q] - [r] - [0]$, whose Abel sum is $[p] + [q] - [r]$. □

What Abel's law buys. On the torus, "balanced" has two layers. Degree zero (as many zeros as poles) is the argument principle. The Abel sum being zero (their positions cancel in the group) is the genus-one refinement. Together they say: a configuration of points is the zero/pole set of an elliptic function exactly when it sums to O. When those points are where a line meets the cubic, this becomes $P + Q + R = O$, and the chord–tangent law is forced.

9.5 Summary

Abel's sum law (theorem 9.1) is the complete criterion for a divisor on T to be principal: degree zero and Abel sum zero. Necessity is the contour-integral computation of chapter 4 (theorems 4.8 and 5.5); sufficiency is the constructive converse via the σ-function (theorem 7.16), recorded as $\mathrm{Pic}^0(T) \cong T$ (corollary 6.8). Equivalently $\mathrm{Prin}(T) = \ker(\deg, \mathfrak{s})$. In the three-point form (corollary 9.3), $[p] + [q] + [r] - 3[0]$ is principal iff $p + q + r \equiv 0$. The next chapter shows that a line meeting the cubic at P, Q, R produces exactly such a principal divisor, and chapter 11 reads off $P + Q + R = O$.

Exercises

Exercise 9.1. Show the two conditions of theorem 9.1 are independent: exhibit a divisor of degree zero whose Abel sum is nonzero (hence not principal), and a divisor with Abel sum zero but nonzero degree (hence not principal). Conclude that neither condition alone characterizes principal divisors.

Exercise 9.2. Prove corollary 9.3 directly from theorem 9.1, and use it to show that for any P, Q on the cubic there is a unique R with $P + Q + R = O$, namely $R = -(P + Q)$. (This R is the third intersection in the chord construction.)

Exercise 9.3. Fill in the period computation in the proof of necessity: show that pairing opposite sides of ∂P_a for $\oint z\,(F'/F)\,dz$ gives $\omega_1 \int_{\omega_2\text{-side}} (F'/F)\,dz - \omega_2 \int_{\omega_1\text{-side}} (F'/F)\,dz$, and that each side integral lies in $2\pi i\,\mathbb{Z}$. (Compare theorem 4.8 and exercise 4.5.)

Exercise 9.4. Using sufficiency (theorem 9.1), show that for any $a, b \in T$ there is an elliptic function with divisor $[a] + [b] - [a + b] - [0]$. Interpret this function as encoding "$a + b$" and explain why no such function exists with divisor $[a] - [0]$ when $a \neq 0$.

Exercise 9.5. Verify that $(\deg, \mathfrak{s}) : \mathrm{Div}(T) \to \mathbb{Z} \oplus T$ is a surjective group homomorphism with kernel $\mathrm{Prin}(T)$, and deduce $\mathrm{Pic}(T) \cong \mathbb{Z} \oplus T$, recovering exercise 5.3.

Chapter 10

Lines Cut Principal Divisors

Abel's law (chapter 9) decides principality by a balance condition; this chapter supplies the geometry that meets it. A projective line restricted to the cubic is a meromorphic function, and its divisor is $P+Q+R-3O$ where P, Q, R are the three intersection points. Feeding this into Abel's law turns collinearity into the group equation $P+Q+R=O$, and the converse holds too: that is the equivalence on which the chord–tangent law rests.

Throughout, $\Phi : T \xrightarrow{\sim} E$ is the isomorphism of theorem 8.5; for a point $P \in E$ we write $\phi(P) = \Phi^{-1}(P) \in T$ and define the sum of points of E by transport, $P \oplus Q := \Phi(\phi(P) + \phi(Q))$, with identity O. The point of the chapter is to compute this sum geometrically.

10.1 Projective Lines as Functions on the Cubic

A projective line is the zero set of a homogeneous linear form

$$\ell(X, Y, Z) = aX + bY + cZ, \qquad (a, b, c) \neq 0.$$

A single linear form is not a function on $\mathbb{CP}^2$ (it is a section of a line bundle), but the ratio of two linear forms is. Using the line at infinity Z as comparison,

$$\frac{\ell}{Z} \quad \text{is a meromorphic function on } E.$$

Its zeros are where ℓ vanishes on E; its poles are where Z vanishes on E. We make both precise.

10.2 Intersection Multiplicity and Bézout

Definition 10.1. Let $L = \{\ell = 0\}$ be a line and $P \in E$ a point. Choose a linear form m with $m(P) \neq 0$; then ℓ/m is a rational function on E, holomorphic and finite near P. The *intersection multiplicity* of L and E at P is

$$I_P(L, E) = \operatorname{ord}_P\left(\frac{\ell}{m}\right) \geq 0,$$

the order of vanishing of ℓ/m at P in a local coordinate on E. It is independent of m (any two choices differ by a unit at P).

Proposition 10.2 (Bézout for a line and a cubic)**.** *A projective line L meets the cubic E in exactly three points, counted with intersection multiplicity:*

$$L \cdot E := \sum_{P \in L \cap E} I_P(L, E)\, P = P + Q + R, \qquad \deg(L \cdot E) = 3.$$

Proof. Parametrize L by a projective coordinate $s \in \mathbb{CP}^1$: a line is a $\mathbb{CP}^1$, so its points are $[X(s) : Y(s) : Z(s)]$ with X, Y, Z linear in s. Substituting into $F(X, Y, Z) = Y^2Z - 4X^3 + g_2XZ^2 + g_3Z^3$ yields a homogeneous polynomial $F(s)$ of degree 3 in s — a binary cubic form — not identically zero (since E contains no line). Over $\mathbb{C}$, counting in $\mathbb{CP}^1$, it has exactly three roots with multiplicity, and the multiplicity of the root s_0 equals $I_{P_0}(L, E)$ at the corresponding point $P_0 = [X(s_0) : Y(s_0) : Z(s_0)]$. Hence $L \cdot E$ has degree three. □

So the zero divisor of ℓ on E is $\mathrm{div}_0(\ell|_E) = P + Q + R$, the three intersection points.

10.3 The Divisor of the Line at Infinity

The comparison form Z cuts out the line at infinity, which we already analyzed.

Proposition 10.3. $\mathrm{div}_0(Z|_E) = 3O$*: the line at infinity meets E only at O, with multiplicity three.*

Proof. By proposition 8.3, $Z = 0$ meets E solely at $O = [0 : 1 : 0]$ and O is a flex with intersection multiplicity three. Explicitly, in the chart $Y = 1$ with $u = X/Y$, $v = Z/Y$, the curve is $v = 4u^3 - g_2uv^2 - g_3v^3$, so near $(0, 0)$, $v = 4u^3 +$ (higher order); thus u is a local coordinate at O and $Z/Y = v$ vanishes to order three. Hence $\mathrm{ord}_O(Z|_E) = 3$ and $\mathrm{div}_0(Z|_E) = 3O$. □

10.4 The Principal Divisor of a Line

Theorem 10.4. *If a line L meets E in $P + Q + R$ (with multiplicity), then the rational function ℓ/Z on E has divisor*

$$\boxed{\mathrm{div}\left(\frac{\ell}{Z}\right) = P + Q + R - 3O.}$$

In particular $P + Q + R \sim 3O$, so $P + Q + R - 3O$ is principal.

Proof. As a ratio of linear forms, ℓ/Z is meromorphic on E with $\mathrm{div}(\ell/Z) = \mathrm{div}_0(\ell|_E) - \mathrm{div}_0(Z|_E)$. By proposition 10.2, $\mathrm{div}_0(\ell|_E) = P + Q + R$, and by proposition 10.3, $\mathrm{div}_0(Z|_E) = 3O$. Subtracting gives the boxed formula; being the divisor of a function it is principal, and $P + Q + R \sim 3O$. □

The line divisor. Every projective line cuts the cubic in a divisor linearly equivalent to $3O$:

$$P + Q + R \sim 3O, \qquad \text{equivalently} \qquad P + Q + R - 3O \text{ is principal.}$$

Because O is a flex, the line at infinity itself cuts $3O$; every other line cuts a divisor in the same class.

10.5 Collinearity Equals Summing to Zero

Now combine theorem 10.4 with Abel's law. The result is the geometric characterization of the group law.

Theorem 10.5 (Collinearity criterion)**.** *For three points $P, Q, R \in E$ (listed with the multiplicities of a line, so that "collinear" includes tangency), the following are equivalent:*

1. *P, Q, R are collinear, i.e. $P + Q + R = L \cdot E$ for some line L;*

2. *$P + Q + R - 3O$ is principal;*

3. *$\phi(P) + \phi(Q) + \phi(R) = 0$ in T, i.e. $P \oplus Q \oplus R = O$.*

Proof. **(1) ⇒ (2)** is theorem 10.4.

(2) ⇔ (3): transport to T by the isomorphism Φ. The divisor $[\phi(P)] + [\phi(Q)] + [\phi(R)] - 3[\phi(O)] = [\phi(P)] + [\phi(Q)] + [\phi(R)] - 3[0]$ is principal iff, by corollary 9.3, $\phi(P) + \phi(Q) + \phi(R) = 0$ in T.

(2) ⇒ (1): suppose $D = P + Q + R - 3O$ is principal, $D = \operatorname{div}(f)$. Then f has poles only at O, of order at most three, so $f \in L(3O)$. By proposition 6.3, $L(3O)$ has basis $\{1, \wp, \wp'\}$ (pulled back through Φ, these are $1, X/Z, Y/Z$), so $f = (aY + bX + cZ)/Z = \ell/Z$ for a linear form $\ell = aY + bX + cZ$. The zeros of f are then $L \cap E$ for the line $L = \{\ell = 0\}$, so P, Q, R are collinear. □

Remark 10.6. The step **(2) ⇒ (1)** is where $L(3O)$ does its work: *every* function with a pole of order ≤ 3 only at O is the restriction of a line. This is what makes lines, rather than higher curves, the right tool for the group law — the geometry of E in $\mathbb{CP}^2$ sees exactly the class $3O$.

10.6 Summary

A line $L = \{\ell = 0\}$ restricts to the function ℓ/Z on E. It meets E in three points with multiplicity (proposition 10.2), and the line at infinity meets E at $3O$ (proposition 10.3), so $\operatorname{div}(\ell/Z) = P + Q + R - 3O$ is principal (theorem 10.4): every line cuts a divisor $\sim 3O$. Pairing this with Abel's law gives the collinearity criterion (theorem 10.5): three points of E are collinear if and only if they sum to O in the transported group, the converse direction relying on $L(3O) = \langle 1, \wp, \wp' \rangle$. Chapter 11 packages this as the statement that the chord–tangent construction *is* the group law.

Exercises

Exercise 10.1. Show that a vertical line $x = c$ meets $E : y^2 = 4x^3 - g_2 x - g_3$ in the two affine points $(c, \pm y_0)$ and in O at infinity, so $L \cdot E = (c, y_0) + (c, -y_0) + O$. Using theorem 10.5, deduce $(c, y_0) \oplus (c, -y_0) = O$: the two points are inverses, recovering the reflection law of section 8.6.

Exercise 10.2. Prove that $I_P(L, E)$ in definition 10.1 is independent of the auxiliary linear form m with $m(P) \neq 0$. (Two such forms differ by a function that is a unit at P.)

Exercise 10.3. Show that the line L is tangent to E at P if and only if $I_P(L, E) \geq 2$, and that then $L \cdot E = 2P + R'$ for some R'. (This is the tangent case used in doubling, chapter 12.)

Exercise 10.4. Redo the local computation of proposition 10.3 carefully: in the chart $Y = 1$, solve $v = 4u^3 - g_2 uv^2 - g_3 v^3$ for v as a power series in u, and confirm $v = 4u^3 + O(u^5)$, so $\operatorname{ord}_O(Z|_E) = 3$.

Exercise 10.5. Use theorem 10.5 to prove: given any two points $P, Q \in E$, the third intersection R of the line PQ with E satisfies $R = -(P \oplus Q)$ in the group, so $P \oplus Q = -R$ is the reflection of R. Identify which line to use when $P = Q$.

Chapter 11

Transporting the Group Law (Why It Is Forced)

Everything is now in place to prove the main theorem. The isomorphism Φ moves the torus group law onto the cubic; the collinearity criterion of chapter 10 shows that the moved law is the chord–tangent construction. This chapter assembles the two, isolates the logical mechanism that makes the chord–tangent rule unavoidable, and observes that the resulting group law is intrinsic to the pair (E, O) — independent of which lattice and embedding produced it.

11.1 The Transported Group Law

The torus already carries an abelian group law, $[u] + [v] = [u + v]$, with no freedom: it is addition in $\mathbb{C}$ (proposition 3.1). The biholomorphism $\Phi : T \xrightarrow{\sim} E$ of theorem 8.5 carries it to E by the only rule that makes Φ a group isomorphism.

Definition 11.1. The *transported group law* on E is

$$P \oplus Q = \Phi\big(\Phi^{-1}(P) + \Phi^{-1}(Q)\big),$$

with identity $O = \Phi(0) = [0 : 1 : 0]$ and inverse $\ominus P = \Phi(-\Phi^{-1}(P))$. In affine coordinates, by section 8.6, the inverse is reflection, $\ominus(x, y) = (x, -y)$.

By construction $(E, \oplus)$ is an abelian group and Φ is an isomorphism of groups. The substantive question is geometric:

What is $P \oplus Q$ in terms of the cubic alone?

11.2 The Chord–Tangent Construction

Definition 11.2. The *chord–tangent operation* $\boxplus$ on E is defined as follows. For $P, Q \in E$, let L be the line through P and Q — the tangent to E at P if $P = Q$ — and let R be the third point of $L \cap E$ (proposition 10.2). Then

$$P \boxplus Q = \ominus R,$$

the reflection of R. Here $\ominus R$ is the third intersection of the line RO with E, which in affine coordinates is $(x_R, -y_R)$.

This is the recipe of section 1.1: draw the chord (or tangent), take the third intersection, reflect. The main theorem is that it agrees with the transported law.

Theorem 11.3 (Chord–tangent is the group law)**.** *Under the Weierstrass embedding* $\Phi : \mathbb{C}/\Lambda \to E$, $z \mapsto [\wp(z) : \wp'(z) : 1]$*, the transported group law equals the chord–tangent operation:*

$$P \oplus Q = P \boxplus Q \qquad \text{for all } P, Q \in E.$$

Consequently the chord–tangent construction is an abelian group law on E *with identity* O*, and* Φ *is an isomorphism of groups* $(\mathbb{C}/\Lambda, +) \cong (E, \boxplus)$*. This is the group-theoretic assertion of theorem 1.1.*

Proof. Let L be the line through P and Q (tangent if $P = Q$), meeting E in $P + Q + R$ with multiplicity. By the collinearity criterion theorem 10.5, the three points sum to the identity in the transported law:

$$P \oplus Q \oplus R = O, \qquad \text{hence} \qquad P \oplus Q = \ominus R.$$

The group inverse $\ominus R$ is the reflection of R (definition 11.1 and section 8.6), which is exactly $P \boxplus R$'s defining reflection, i.e. $P \oplus Q = \ominus R = P \boxplus Q$. The identity, inverse, and tangent cases are the same statement specialized: when $P = Q$ the line is the tangent and $L \cdot E = 2P + R$, giving $2P = \ominus R$; when $Q = \ominus P$ the line PQ is vertical through O and $R = O$, giving $P \oplus (\ominus P) = O$; when $Q = O$ the line PO meets E at $P, \ominus P, O$, giving $P \oplus O = P$. All are instances of theorem 10.5, worked out one by one in chapter 12. □

11.3 Why the Rule Is Forced

The proof exposes the exact mechanism. The chord–tangent law is not a clever convention; it is the only possible answer, dictated by three inputs, none of which involves a choice.

1. **The torus has a group law.** $[u] + [v] = [u + v]$ is addition in $\mathbb{C}$, forced (proposition 3.1).
2. **The embedding transfers it.** Requiring Φ to be a group isomorphism leaves no freedom in $\oplus$ (definition 11.1).
3. **Lines see that law.** A line cuts E in a divisor $P + Q + R - 3O$ that is principal (theorem 10.4), and Abel's law makes its points sum to O (theorem 10.5). So a line through P, Q determines $R = \ominus(P \oplus Q)$, and the sum is recovered by the one further forced step — negate, i.e. reflect.

No arbitrary convention. The reflection in the chord–tangent rule is often presented as a definition to be accepted. It is not optional. A line gives the symmetric relation $P + Q + R = O$, because line divisors are principal and principal divisors have Abel sum zero. Solving this for $P \oplus Q$ forces

$$P \oplus Q = \ominus R,$$

and $\ominus$ is reflection because the torus inverse $z \mapsto -z$ is $(x, y) \mapsto (x, -y)$. Chord, tangent, third point, reflect: each step is compelled.

11.4 The Group Law Is Intrinsic

A subtle point deserves emphasis. The construction passed through a lattice Λ and an embedding Φ, but the resulting law does not depend on them.

Proposition 11.4. *The chord–tangent law $\boxplus$ depends only on the cubic E and the choice of identity O; it is independent of the lattice Λ and the embedding Φ used to construct it.*

Proof. Definition 11.2 refers only to lines in $\mathbb{CP}^2$, their intersections with E, and the point O — all intrinsic to (E, O). No lattice or $\wp$ appears. Theorem 11.3 shows this intrinsic operation coincides with any transported law $\oplus$ built from a Φ; hence all such transported laws agree with $\boxplus$ and therefore with each other. □

In other words, the abelian group is attached to the geometric object (E, O) itself. The uniformization $\mathbb{C}/\Lambda \cong E$ explains *why* the law is a group and why it is commutative and associative — properties that are opaque from the cubic but transparent on the torus — yet the law it computes is the one a geometer reading only the plane curve would write down.

Remark 11.5. Associativity of $\boxplus$ is the property hardest to see directly on the cubic; the classical proofs invoke the Cayley–Bacharach theorem. Here it is free: $\oplus$ is associative because addition in $\mathbb{C}$ is, and $\boxplus = \oplus$. The torus is, in this sense, the explanation of associativity.

11.5 Summary

Transporting the torus law by Φ gives a group $(E, \oplus)$ with identity O and reflection as inverse (definition 11.1). By the collinearity criterion, this transported law is exactly the chord–tangent construction (theorem 11.3): a line through P, Q meets E again at $R = \ominus(P \oplus Q)$, so $P \oplus Q$ is the reflection of R. The rule is forced by three choiceless facts — the torus group law, its transfer by Φ, and the principality of line divisors (section 11.3) — and the resulting law is intrinsic to (E, O), with associativity inherited from $\mathbb{C}$ (proposition 11.4 and remark 11.5). The remaining chapters make the construction explicit: the pictorial dictionary and the case analysis (chapter 12), then closed-form coordinates (chapter 13).

Exercises

Exercise 11.1. Show that $\oplus$ in definition 11.1 is the unique binary operation on E making Φ a group homomorphism, and verify directly that it is abelian with identity O and inverse $\ominus P = \Phi(-\Phi^{-1}(P))$.

Exercise 11.2. Work out the identity case from theorem 11.3: show the line through P and O meets E at P, $\ominus P$, and O, and deduce $P \oplus O = P$ directly from theorem 10.5.

Exercise 11.3. Explain in one paragraph why associativity of the chord–tangent law is immediate from theorem 11.3, and state precisely which property of $\mathbb{C}$ it reduces to. Why is the direct geometric proof hard?

Exercise 11.4. Suppose two lattices Λ, Λ' and embeddings Φ, Φ' both present the *same* cubic E (with the same O). Using proposition 11.4, show the two transported laws coincide. What does this say about the relationship between Λ and Λ' (cf. proposition 2.7)?

Exercise 11.5. Show that for fixed $A \in E$, the translation $P \mapsto P \oplus A$ is a biholomorphism of E (an automorphism of the Riemann surface), corresponding under Φ to $z \mapsto z + \Phi^{-1}(A)$ on T (proposition 3.2). Deduce that E acts on itself by holomorphic translations, so E is a complex Lie group.

Chapter 12

The Chord–Tangent Law in Cases

Theorem 11.3 proved once and for all that the chord–tangent operation is the transported group law. This chapter makes the statement concrete, walking through every structurally distinct case — adding the identity, two distinct points, doubling, vertical lines and inverses, and two-torsion — and displaying each as a picture. Every case is the single identity $P + Q + R = O$ of theorem 10.5 read in a particular configuration; nothing new is proved, but the geometry becomes visible.

Colour convention. In the figures, rust marks the input points, plum the auxiliary line (chord, tangent, or vertical), teal the third intersection R, and gold the final result obtained by reflecting R across the x-axis. All coordinates are computed by the Sage scripts of appendix C; the running example is $E : y^2 = 4x^3 - 4x$, with $g_2 = 4$, $g_3 = 0$ and roots $e_1 = -1$, $e_2 = 0$, $e_3 = 1$.

12.1 The Pictorial Dictionary

The isomorphism $\Phi : \mathbb{C}/\Lambda \to E$ lets us watch a single point in three synchronized pictures: its position in the fundamental parallelogram, its image on the embedded torus, and its image on the cubic (fig. 12.1). Addition on the left — vector addition modulo Λ — becomes the chord–tangent construction on the right.

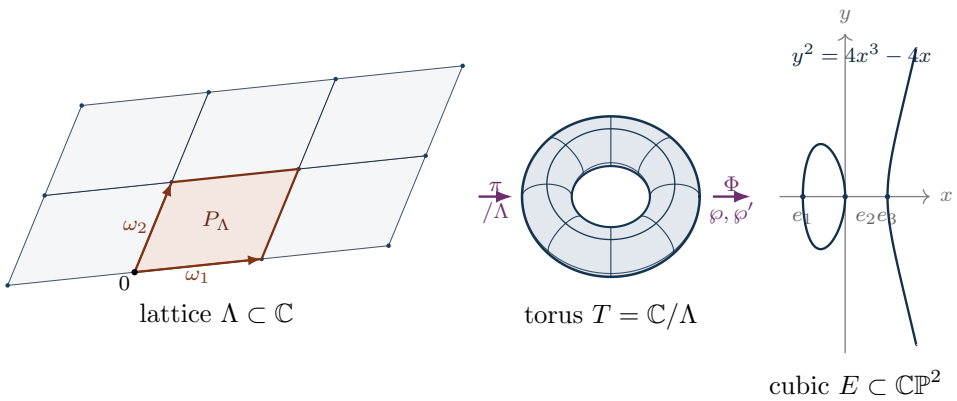


Figure 12.1: The dictionary $\Lambda \rightsquigarrow T \rightsquigarrow E$. A lattice point class $[w] = a\omega_1 + b\omega_2$ appears in the fundamental parallelogram (left), at its embedded position on the torus (middle), and on the cubic $E : y^2 = 4x^3 - 4x$ (right). Computed by `sagemath/11_correspondence.sage` and the rendering scripts.

12.2 Two Distinct Points: the Chord

Let $P, Q \in E$ be distinct finite points with $Q \neq \ominus P$, and let L be the line through them; it is not vertical, so it meets E in a third finite point R. Writing $P = \Phi(u)$, $Q = \Phi(v)$, $R = \Phi(w)$, the intersection divisor is $L \cdot E = P + Q + R$, so $\operatorname{div}(\ell/Z) = P + Q + R - 3O$ is principal (theorem 10.4). Abel's law (corollary 9.3) gives $u + v + w \equiv 0$, hence $w \equiv -(u + v)$ and $R = \Phi(-(u + v))$. Since the inverse is reflection,

$$P \oplus Q = \Phi(u + v) = \ominus\Phi(-(u + v)) = \ominus R = -R.$$

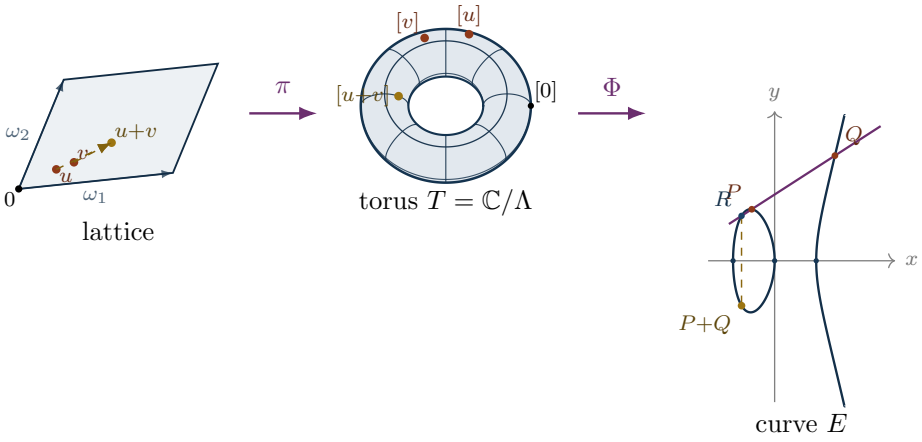


Figure 12.2: Adding two distinct points. On the lattice (left) the gold arrows show $u \to u + v \leftarrow v$; on the torus (middle) both land at $[u + v]$; on the cubic (right) the chord PQ meets E at R, and $P \oplus Q = -R$ is its reflection. Coordinates from `sagemath/11_correspondence.sage`.

Why the sign appears. The line gives $P + Q + R = O$, not $P + Q = R$. Solving forces $P \oplus Q = -R$: the final reflection is exactly this minus sign, traced back to the torus inverse $z \mapsto -z$.

12.3 Doubling: the Tangent

When $P = Q$ the chord degenerates to the tangent line at P, the limit of the chord as $Q \to P$. The tangent meets E at P with intersection multiplicity two (exercise 10.3), so $L \cdot E = 2P + R$. With $P = \Phi(u)$, $R = \Phi(w)$, the principal divisor $2P + R - 3O$ and Abel's law give $2u + w \equiv 0$, so $w \equiv -2u$, $R = \Phi(-2u)$, and

$$2P = \Phi(2u) = \ominus R = -R.$$

12.4 The Identity and Inverses: Vertical Lines

The identity is $O = \Phi(0)$. A line through a finite point $P = (x, y)$ and O is the vertical line $X = xZ$, which also passes through $-P = (x, -y)$. Its intersection divisor is $P + (\ominus P) + O$, so theorem 10.5 gives $P \oplus (\ominus P) \oplus O = O$. Reading this two ways: with the third point $\ominus P$ it says $P \oplus O = P$ (the identity law); directly it says $P \oplus (\ominus P) = O$ (inverses), confirming that $\ominus P = (x, -y)$ is the group inverse.

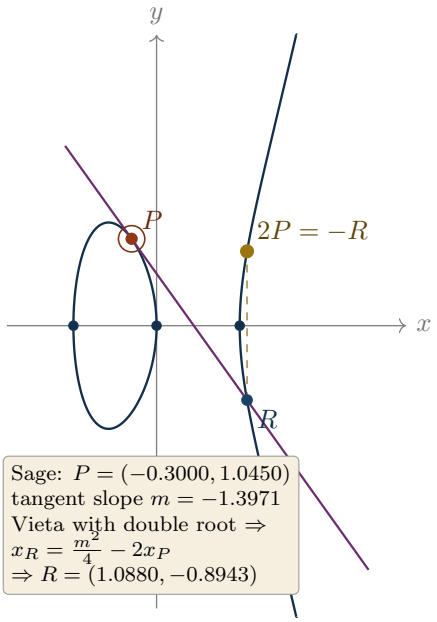


Figure 12.3: Doubling a point. The tangent to E at P (plum) meets E again at R, and $2P = -R$ is its reflection (gold). The slope is $m = (12x_P^2 - g_2)/(2y_P)$; see chapter 13. Coordinates from `sagemath/05_tangent_double.sage`.

12.5 Two-Torsion Points

A point is its own inverse, $P = \ominus P$, exactly when $2P = O$: the *two-torsion*. In affine coordinates $P = \ominus P$ means $(x, y) = (x, -y)$, so $y = 0$; the nonzero two-torsion points are therefore the three x-axis crossings

$$4x^3 - g_2 x - g_3 = 0, \qquad \text{i.e.} \qquad x \in \{e_1, e_2, e_3\},$$

the roots of proposition 7.8. At each the tangent is vertical, so doubling gives $2P = O$ directly. Together with O they form

$$E[2] = \{O, T_1, T_2, T_3\} \cong (\mathbb{Z}/2\mathbb{Z})^2,$$

the images under Φ of the two-torsion of exercise 3.2.

The three nonzero two-torsion points are collinear — they all lie on the x-axis $y = 0$, itself a line meeting E at $(e_1, 0), (e_2, 0), (e_3, 0)$. By theorem 10.5, $T_1 \oplus T_2 \oplus T_3 = O$, which is the statement that the three nonzero elements of $(\mathbb{Z}/2\mathbb{Z})^2$ sum to zero.

12.6 Summary

Each case of the chord–tangent law is one reading of $P + Q + R = O$. For two distinct points the line is a chord and $P \oplus Q = -R$ (section 12.2); for a repeated point it is the tangent and $2P = -R$ (section 12.3); a vertical line through P and O yields the identity law and exhibits $\ominus P = (x, -y)$ (section 12.4); and the points fixed by reflection are the two-torsion $E[2] \cong (\mathbb{Z}/2\mathbb{Z})^2$ (section 12.5). The pictures make the dictionary $\Lambda \rightsquigarrow T \rightsquigarrow E$ literal. Chapter 13 converts these geometric rules into closed-form coordinate formulas.

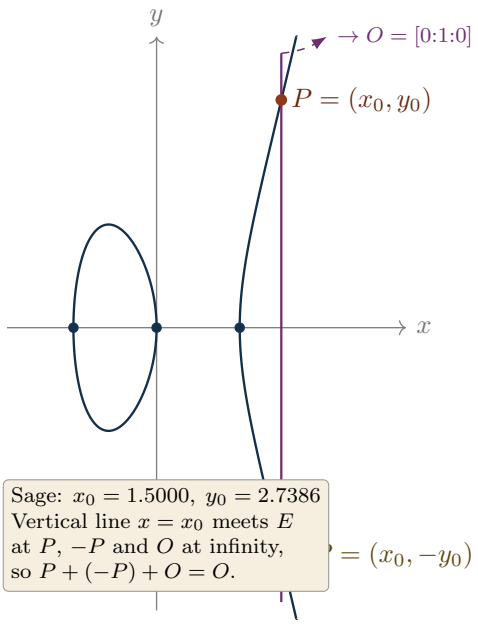


Figure 12.4: Vertical lines give inverses. The line $x = x_0$ meets E at $P = (x_0, y_0)$, $-P = (x_0, -y_0)$, and O at infinity, so $P \oplus (\ominus P) = O$. Coordinates from `sagemath/06_two_torsion.sage`.

Exercises

Exercise 12.1. Let $P = (x_P, y_P)$, $Q = (x_Q, y_Q)$ be distinct with chord $y = mx + k$. Substitute into $y^2 = 4x^3 - g_2x - g_3$ and use Vieta's formulas on the resulting cubic in x to show the third intersection has $x_R = \frac{m^2}{4} - x_P - x_Q$. Then $P \oplus Q = (x_R, -(mx_R + k))$.

Exercise 12.2. For doubling, differentiate $y^2 = 4x^3 - g_2x - g_3$ implicitly to get the tangent slope $m = (12x_P^2 - g_2)/(2y_P)$, and show the doubled point has $x_{2P} = \frac{m^2}{4} - 2x_P$. What happens to m when $y_P = 0$, and how does that match $2P = O$?

Exercise 12.3. Show that the x-axis is the line through the three nonzero two-torsion points, and deduce $T_1 \oplus T_2 \oplus T_3 = O$ from theorem 10.5. Verify this is consistent with $E[2] \cong (\mathbb{Z}/2\mathbb{Z})^2$, where the three nonzero elements sum to zero.

Exercise 12.4. A point P is *three-torsion*, $3P = O$, if and only if P is an inflection point of E (the tangent at P meets E with multiplicity three). Using $L \cdot E = 3P$ and theorem 10.5, prove this, and note O is one such flex (proposition 8.3).

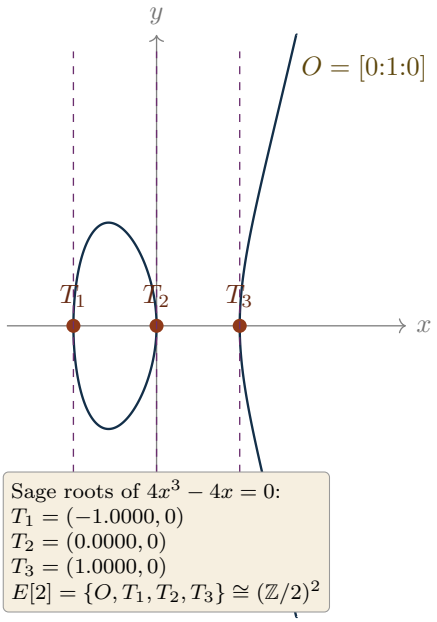


Figure 12.5: The two-torsion. The nonzero two-torsion points T_1, T_2, T_3 are the x-axis crossings $(e_i, 0)$, each with a vertical tangent; with O they form $E[2] \cong (\mathbb{Z}/2\mathbb{Z})^2$. Roots of $4x^3 - 4x$ from `sagemath/06_two_torsion.sage`.

Part VI

Formulas, Computations, and Synthesis

Chapter 13

Explicit Coordinate Formulas

The divisor argument of chapter 11 explains *why* the chord–tangent construction is a group law; for computation we want the law as formulas. This chapter derives the closed forms for addition and doubling on

$$E: \quad y^2 = 4x^3 - g_2x - g_3,$$

tabulates the degenerate cases, and observes that because the formulas are rational, the group law makes sense — and stays a group — over any field, not just $\mathbb{C}$. That observation is what carries elliptic curves into number theory and cryptography (chapter 16).

13.1 Addition of Distinct, Non-Inverse Points

Theorem 13.1 (Addition formula)**.** *Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be finite points of E with $P \neq Q$ and $Q \neq \ominus P$; then $x_1 \neq x_2$. With slope and intercept*

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \qquad b = y_1 - mx_1,$$

the sum $P \oplus Q = (x_3, y_3)$ is

$$\boxed{x_3 = \frac{m^2}{4} - x_1 - x_2, \qquad y_3 = -(mx_3 + b) = m(x_1 - x_3) - y_1.}$$

Proof. The line through P and Q is $y = mx + b$. Substituting into the cubic,

$$(mx + b)^2 = 4x^3 - g_2x - g_3 \quad \Longrightarrow \quad 4x^3 - m^2x^2 - (g_2 + 2mb)x - (g_3 + b^2) = 0.$$

Its three roots are the x-coordinates x_1, x_2, x_R of the intersection points P, Q, R. Vieta's formula for the sum of roots (negative of the x^2-coefficient divided by the leading coefficient) gives $x_1 + x_2 + x_R = m^2/4$, so the third intersection has $x_R = \frac{m^2}{4} - x_1 - x_2$. By theorem 11.3, $P \oplus Q = \ominus R$, and reflection fixes the x-coordinate, so $x_3 = x_R$. The y-coordinate is the reflection of the line value: $y_3 = -(mx_R + b)$. Using $b = y_1 - mx_1$ this is $y_3 = m(x_1 - x_3) - y_1$. □

13.2 Doubling

Theorem 13.2 (Doubling formula)**.** *Let $P = (x_1, y_1)$ with $y_1 \neq 0$. The tangent slope is*

$$m = \frac{12x_1^2 - g_2}{2y_1},$$

and $2P = (x_3, y_3)$ *is*

$$\boxed{x_3 = \frac{m^2}{4} - 2x_1, \qquad y_3 = m(x_1 - x_3) - y_1.}$$

Proof. Differentiating $y^2 = 4x^3 - g_2x - g_3$ implicitly gives $2y\,y' = 12x^2 - g_2$, so the tangent at P has slope $m = (12x_1^2 - g_2)/(2y_1)$ (finite since $y_1 \neq 0$). The tangent meets E at P with multiplicity two, so x_1 is a double root of the cubic of theorem 13.1; Vieta gives $2x_1 + x_R = m^2/4$, hence $x_R = \frac{m^2}{4} - 2x_1$. As before $2P = \ominus R$, so $x_3 = x_R$ and $y_3 = m(x_1 - x_3) - y_1$. □

The doubling formula is the $Q \to P$ limit of the addition formula: the secant slope $(y_2 - y_1)/(x_2 - x_1)$ tends to the tangent slope $(12x_1^2 - g_2)/(2y_1)$, and $x_1 + x_2 + x_R = m^2/4$ becomes $2x_1 + x_R = m^2/4$ (exercise 13.3).

13.3 The Degenerate Cases

The formulas above assume two finite, non-inverse points. The remaining cases are read off directly from the group law (sections 12.4 and 12.5).

Situation	Result	Reason
$P = O$	$P \oplus Q = Q$	O is the identity
$Q = O$	$P \oplus Q = P$	O is the identity
$Q = \ominus P$ (so $x_1 = x_2$, $y_2 = -y_1$)	$P \oplus Q = O$	vertical line
$P = Q$, $y_1 = 0$	$2P = O$	two-torsion (vertical tangent)
$P = Q$, $y_1 \neq 0$	doubling formula	theorem 13.2

The case $Q = \ominus P$ is exactly where the slope $m = (y_2 - y_1)/(x_2 - x_1)$ has a zero denominator: a vertical line has no finite slope, and the sum is the point at infinity O. A program implementing the group law tests for this case separately (exercise 13.4).

13.4 The Group Law Is Algebraic

The formulas of theorems 13.1 and 13.2 are *rational*: x_3 and y_3 are quotients of polynomials in x_1, y_1, x_2, y_2 with coefficients in $\mathbb{Z}[g_2, g_3]$. This has a consequence reaching well beyond the complex numbers.

Proposition 13.3. *Let $k \subseteq \mathbb{C}$ be a subfield containing g_2, g_3. Then the set of k-rational points*

$$E(k) = \{(x, y) \in k^2 : y^2 = 4x^3 - g_2x - g_3\} \cup \{O\}$$

is a subgroup of $(E, \oplus)$.

Proof. The identity O and the inverse $\ominus(x, y) = (x, -y)$ of a k-point are k-points. For closure, if $P, Q \in E(k)$ the addition and doubling formulas produce x_3, y_3 as rational expressions in the coordinates and g_2, g_3, all in k; since k is a field these lie in k, and (x_3, y_3) satisfies the curve equation, so $P \oplus Q \in E(k)$. The degenerate cases yield O or a k-point as well. □

Remark 13.4. Nothing in the formulas used that k sits inside $\mathbb{C}$. Over *any* field k (of characteristic $\neq 2, 3$) with $\Delta = g_2^3 - 27g_3^2 \neq 0$, the same rational maps define an abelian group law on $E(k)$, with the same geometric meaning. This is the foundation of the arithmetic of elliptic curves: the group $E(\mathbb{Q})$ (Mordell–Weil), the groups $E(\mathbb{F}_p)$ used in cryptography, and the reduction maps between them. The complex-analytic uniformization of this book explains *why* the law is associative; the algebraic formulas let it run over any field (chapter 16).

13.5 A Small Worked Example

On $E : y^2 = 4x^3 - 4x$ (so $g_2 = 4$, $g_3 = 0$), take $P = (2, 2\sqrt{6})$, which lies on E since $4 \cdot 8 - 8 = 24 = (2\sqrt{6})^2$. Doubling: the slope is

$$m = \frac{12 \cdot 4 - 4}{2 \cdot 2\sqrt{6}} = \frac{44}{4\sqrt{6}} = \frac{11}{\sqrt{6}},$$

so

$$x_{2P} = \frac{m^2}{4} - 2 \cdot 2 = \frac{121/6}{4} - 4 = \frac{121}{24} - \frac{96}{24} = \frac{25}{24},$$

and $y_{2P} = m(x_1 - x_{2P}) - y_1 = \frac{11}{\sqrt{6}}\left(2 - \frac{25}{24}\right) - 2\sqrt{6} = \frac{11}{\sqrt{6}} \cdot \frac{23}{24} - 2\sqrt{6} = \frac{253}{24\sqrt{6}} - 2\sqrt{6} = \frac{253-288}{24\sqrt{6}} = -\frac{35}{24\sqrt{6}}$. One checks (x_{2P}, y_{2P}) satisfies $y^2 = 4x^3 - 4x$ (exercise 13.5). A fuller symbolic computation, and the comparison with the more common normalization $y^2 = x^3 + ax + b$, is carried out in chapter 15.

13.6 Summary

For distinct non-inverse points the sum has $x_3 = \frac{m^2}{4} - x_1 - x_2$, $y_3 = m(x_1 - x_3) - y_1$ with $m = (y_2 - y_1)/(x_2 - x_1)$ (theorem 13.1); doubling replaces m by the tangent slope $(12x_1^2 - g_2)/(2y_1)$ and $x_1 + x_2$ by $2x_1$ (theorem 13.2). The identity, inverse, and two-torsion cases are degenerate (section 13.3). Because the formulas are rational, the group law is algebraic: $E(k)$ is a group for any field k with $\Delta \neq 0$ (proposition 13.3 and remark 13.4), the gateway to the arithmetic theory.

Exercises

Exercise 13.1. Derive the two equivalent forms of the y-coordinate in theorem 13.1,

$$y_3 = -(mx_3 + b) = m(x_1 - x_3) - y_1,$$

using $b = y_1 - mx_1$. Why is the form $m(x_1 - x_3) - y_1$ preferable numerically?

Exercise 13.2. Verify directly from theorem 13.1 that the addition formula is symmetric under swapping $P \leftrightarrow Q$, so $P \oplus Q = Q \oplus P$: the slope, x_3, and y_3 are all unchanged. (Commutativity, visible in the formula.)

Exercise 13.3. Show that as $Q \to P$ along E (with $y_1 \neq 0$), the secant slope $(y_2 - y_1)/(x_2 - x_1)$ converges to the tangent slope $(12x_1^2 - g_2)/(2y_1)$, and that the addition formula limits to the doubling formula. (Differentiate the curve relation $y_2^2 - y_1^2 = 4(x_2^3 - x_1^3) - g_2(x_2 - x_1)$.)

Exercise 13.4. Explain how the addition formula "detects" the case $Q = \ominus P$: show that $x_1 = x_2$ with $P \neq Q$ forces $y_2 = -y_1$, making the slope undefined, and that the geometric answer is $P \oplus Q = O$. Write pseudocode for the full group law branching on the degenerate cases.

Exercise 13.5. Verify that the doubled point $(x_{2P}, y_{2P}) = \left(\frac{25}{24}, -\frac{35}{24\sqrt{6}}\right)$ of section 13.5 satisfies $y^2 = 4x^3 - 4x$. Then compute $\ominus(2P)$ and confirm it is $(x_{2P}, +\frac{35}{24\sqrt{6}})$.

Chapter 14

The Analytic Addition Formula

The chord computation of chapter 13 has an analytic counterpart: a closed formula for $\wp(u+v)$ in terms of $\wp$ and $\wp'$ at u and v. This chapter proves the Weierstrass addition theorem directly from Abel's law and the differential equation — independently of the geometry — and then shows it reproduces the chord law exactly, with the oddness of $\wp'$ accounting for the reflection. The two derivations meeting is the analytic seal on theorem 1.1.

14.1 The Addition Theorem

Theorem 14.1 (Weierstrass addition formula)**.** *For $u, v \notin \Lambda$ with $u \not\equiv \pm v \pmod{\Lambda}$,*

$$\boxed{\wp(u+v) = -\wp(u) - \wp(v) + \frac{1}{4}\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2.}$$

Proof. Let $A = \dfrac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}$ and $B = \wp'(u) - A\wp(u)$, so that the line $Y = AX + B$ passes through $(\wp(u), \wp'(u))$ and $(\wp(v), \wp'(v))$. Consider the elliptic function

$$h(z) = \wp'(z) - A\,\wp(z) - B.$$

It is elliptic with a single pole at 0, of order three (from $\wp'$), so it has order three and, by theorem 4.7, exactly three zeros counted with multiplicity. Two of them are u and v (where the line meets the curve); let w be the third. By Abel's law (theorem 4.8) applied to $\operatorname{div}(h) = [u]+[v]+[w]-3[0]$,

$$u + v + w \equiv 0 \pmod{\Lambda}, \qquad \text{so} \qquad w \equiv -(u+v).$$

At a zero z of h we have $\wp'(z) = A\wp(z) + B$; squaring and using the differential equation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ (theorem 7.7),

$$4\,\wp(z)^3 - g_2\,\wp(z) - g_3 = \big(A\,\wp(z) + B\big)^2 = A^2\wp(z)^2 + 2AB\,\wp(z) + B^2.$$

Thus $\wp(z)$ is a root of the cubic $4X^3 - A^2X^2 - \cdots = 0$, whose three roots are $\wp(u), \wp(v), \wp(w)$. Vieta's formula for the sum of the roots gives

$$\wp(u) + \wp(v) + \wp(w) = \frac{A^2}{4}.$$

Finally $\wp(w) = \wp(-(u+v)) = \wp(u+v)$ because $\wp$ is even (proposition 7.4). Solving, $\wp(u+v) = \frac{A^2}{4} - \wp(u) - \wp(v)$, which is the boxed formula. □

The proof used only Abel's law, the differential equation, and the evenness of $\wp$ — no reference to the cubic in $\mathbb{CP}^2$. It nevertheless reconstructs the chord rule.

14.2 Agreement with the Chord Law

Set $P = \Phi(u) = (\wp(u), \wp'(u))$, $Q = \Phi(v) = (\wp(v), \wp'(v))$, and $x_1 = \wp(u)$, $y_1 = \wp'(u)$, $x_2 = \wp(v)$, $y_2 = \wp'(v)$. The chord slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\wp'(v) - \wp'(u)}{\wp(v) - \wp(u)} = \frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} = A.$$

The chord formula of theorem 13.1 gives $x_{P\oplus Q} = \frac{m^2}{4} - x_1 - x_2$, and the addition theorem gives $\wp(u+v) = \frac{A^2}{4} - \wp(u) - \wp(v)$; since $m = A$, these are identical:

$$x_{P\oplus Q} = \wp(u+v).$$

The two computations — summing the roots of the intersection cubic, and summing the values of $\wp$ at the zeros of h — are the same calculation viewed through Φ.

14.3 Why the y-Coordinate Has the Reflected Sign

The third intersection of the chord is $R = \Phi(w) = \Phi(-(u+v))$. Its coordinates are

$$R = \big(\wp(-(u+v)),\, \wp'(-(u+v))\big) = \big(\wp(u+v),\, -\wp'(u+v)\big),$$

using $\wp$ even and $\wp'$ odd. Reflecting,

$$\ominus R = \big(\wp(u+v),\, \wp'(u+v)\big) = \Phi(u+v) = P \oplus Q.$$

So the y-coordinate of the sum is $+\wp'(u+v)$, while the third intersection had $-\wp'(u+v)$: the reflection across the x-axis is exactly the oddness of $\wp'$.

Analytic and geometric agreement. The addition formula for $\wp$ is the analytic form of the chord law. The root-sum computation produces the same x-coordinate as $\wp(u+v)$, and the oddness of $\wp'$ supplies the reflection that produces the y-coordinate. The minus sign in "reflect R" and the minus sign in $\wp'(-z) = -\wp'(z)$ are one and the same.

14.4 The Determinant Form

Collinearity of P, Q, and R is a vanishing determinant, which packages the addition theorem symmetrically.

Proposition 14.2. *For u, v generic,*

$$\det \begin{pmatrix} 1 & \wp(u) & \wp'(u) \\ 1 & \wp(v) & \wp'(v) \\ 1 & \wp(u+v) & -\wp'(u+v) \end{pmatrix} = 0.$$

Proof. The three rows are the coordinates $(1, x, y)$ of $P = \Phi(u)$, $Q = \Phi(v)$, and $R = \Phi(-(u+v)) = (\wp(u+v), -\wp'(u+v))$. By theorem 10.5 these three points are collinear — they are the intersection of the chord with E — and three points of the affine plane are collinear precisely when this determinant vanishes. □

Expanding the determinant and solving for $\wp(u+v)$ returns theorem 14.1; the symmetric form makes the role of all three points $u, v, -(u+v)$ manifest.

14.5 The Duplication Formula

Letting $v \to u$ gives the formula for $\wp(2u)$.

Corollary 14.3 (Duplication)**.** *For u with $\wp'(u) \neq 0$,*

$$\wp(2u) = -2\wp(u) + \frac{1}{4}\left(\frac{\wp''(u)}{\wp'(u)}\right)^2 = -2\wp(u) + \frac{1}{4}\left(\frac{6\wp(u)^2 - \frac{1}{2}g_2}{\wp'(u)}\right)^2.$$

Proof. As $v \to u$, the slope $A = \dfrac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}$ is a difference quotient; by l'Hôpital (dividing numerator and denominator by $u - v$ and taking the limit) it tends to $\dfrac{\wp''(u)}{\wp'(u)}$. Theorem 14.1 then gives $\wp(2u) = -2\wp(u) + \frac{1}{4}(\wp''/\wp')^2$. Differentiating $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ yields $\wp'' = 6\wp^2 - \frac{1}{2}g_2$ (exercise 7.5), giving the second form. □

This matches the doubling formula of theorem 13.2: the tangent slope $(12x^2 - g_2)/(2y)$ is exactly $(6\wp^2 - \frac{1}{2}g_2)/\wp' = \wp''/\wp'$, with $x = \wp(u)$, $y = \wp'(u)$.

14.6 Summary

The Weierstrass addition formula (theorem 14.1) gives $\wp(u+v)$ as $-\wp(u) - \wp(v)$ plus a quarter-square of the slope A; it is proved directly from Abel's law and the differential equation, by summing the values of $\wp$ at the three zeros of $h = \wp' - A\wp - B$. It reproduces the chord x-coordinate, and the oddness of $\wp'$ supplies the reflected y-sign (sections 14.2 and 14.3). In symmetric form it is a vanishing 3×3 determinant (proposition 14.2), and the $v \to u$ limit is the duplication formula (corollary 14.3), matching the geometric doubling. Analytic and geometric addition are one operation.

Exercises

Exercise 14.1. Verify the claims about $h(z) = \wp'(z) - A\wp(z) - B$ in the proof of theorem 14.1: that it is elliptic of order three with a triple pole at 0, and that u, v are among its zeros. Why does the case $u \equiv \pm v$ have to be excluded?

Exercise 14.2. Expand the determinant of proposition 14.2 along the first column and solve the resulting linear relation for $\wp(u+v)$, recovering theorem 14.1.

Exercise 14.3. Show that the duplication slope $\wp''(u)/\wp'(u)$ equals the tangent slope $(12x^2 - g_2)/(2y)$ of theorem 13.2 with $x = \wp(u)$, $y = \wp'(u)$, confirming corollary 14.3 matches the geometric doubling formula.

Exercise 14.4. Check that theorem 14.1 is symmetric under $u \leftrightarrow v$, and compute the limit $v \to 0$ to confirm consistency with $\wp(u+0) = \wp(u)$. (The pole of $\wp(v)$ at $v = 0$ cancels against the slope term.)

Exercise 14.5. Apply theorem 14.1 to $\wp(u - v)$ by replacing v with $-v$ (so the slope becomes $(\wp'(u)+\wp'(v))/(\wp(u)-\wp(v))$, using $\wp$ even, $\wp'$ odd). Adding the formulas for $\wp(u+v)$ and $\wp(u-v)$, derive

$$\wp(u+v) + \wp(u-v) = -2\wp(u) - 2\wp(v) + \frac{\wp'(u)^2 + \wp'(v)^2}{2\,(\wp(u) - \wp(v))^2},$$

and conclude, using the differential equation, that the right-hand side is a rational function of $\wp(u)$ and $\wp(v)$.

Chapter 15

Worked Computations and Pitfalls

This chapter is practice. We carry the chord computation through once symbolically, then once with explicit numbers on $E : y^2 = 4x^3 - 4x$, verifying that the result lands back on the curve. We then reconcile our normalization with the textbook form $y^2 = x^3 + ax + b$, and close with a checklist of the mistakes that most often derail a group-law calculation.

15.1 A Complete Symbolic Computation

Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$ be finite points of $E : y^2 = 4x^3 - g_2x - g_3$ with $x_1 \neq x_2$. Define

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \qquad b = y_1 - mx_1, \qquad \text{so the chord is } y = mx + b.$$

Substitute into the cubic and collect terms:

$$(mx + b)^2 = 4x^3 - g_2x - g_3 \implies 4x^3 - m^2x^2 - (g_2 + 2mb)x - (g_3 + b^2) = 0.$$

Because P, Q lie on both curves, x_1, x_2 are two roots; call the third x_3. Matching this against $4(x - x_1)(x - x_2)(x - x_3)$, the x^2-coefficients give $-4(x_1 + x_2 + x_3) = -m^2$, so

$$x_3 = \frac{m^2}{4} - x_1 - x_2.$$

The third intersection is $R = (x_3,\, mx_3 + b)$, and the group sum is $P \oplus Q = \ominus R = (x_3, -(mx_3 + b))$:

$$\boxed{P \oplus Q = \left(\frac{m^2}{4} - x_1 - x_2,\; -m\left(\frac{m^2}{4} - x_1 - x_2\right) - b\right).}$$

This is theorem 13.1 rederived from scratch; the point of repeating it is the method — chord, substitute, Vieta, reflect.

15.2 A Fully Numeric Example

Take $E : y^2 = 4x^3 - 4x$ (so $g_2 = 4$, $g_3 = 0$; roots $e_1 = -1, e_2 = 0, e_3 = 1$), and the two points

$$P = \big(2,\, 2\sqrt{6}\big), \qquad Q = \big(-\tfrac{1}{2},\, \tfrac{\sqrt{6}}{2}\big),$$

both on E: $4\cdot 8-8=24=(2\sqrt{6})^2$ and $4(-\frac{1}{8})-4(-\frac{1}{2})=\frac{3}{2}=(\frac{\sqrt{6}}{2})^2$. Then

$$m=\frac{\frac{\sqrt{6}}{2}-2\sqrt{6}}{-\frac{1}{2}-2}=\frac{-\frac{3}{2}\sqrt{6}}{-\frac{5}{2}}=\frac{3\sqrt{6}}{5},\qquad m^2=\frac{9\cdot 6}{25}=\frac{54}{25}.$$

The x-coordinate of the sum is

$$x_3=\frac{m^2}{4}-x_1-x_2=\frac{54}{100}-2+\tfrac{1}{2}=\frac{27}{50}-\frac{75}{50}=-\frac{24}{25},$$

and

$$y_3=m(x_1-x_3)-y_1=\frac{3\sqrt{6}}{5}\Big(2+\tfrac{24}{25}\Big)-2\sqrt{6}=\frac{3\sqrt{6}}{5}\cdot\frac{74}{25}-2\sqrt{6}=\frac{222\sqrt{6}-250\sqrt{6}}{125}=-\frac{28\sqrt{6}}{125}.$$

Thus

$$P\oplus Q=\left(-\frac{24}{25},\,-\frac{28\sqrt{6}}{125}\right),$$

and one verifies it lies on E: $\big(\frac{28\sqrt{6}}{125}\big)^2=\frac{4704}{15625}$ while $4x_3^3-4x_3=4\big(-\frac{24}{25}\big)^3-4\big(-\frac{24}{25}\big)=\frac{4704}{15625}$ as well (exercise 15.1). The arithmetic is exact; the same steps run over any field once $\sqrt{6}$ is whatever element squares to 6 there.

15.3 Comparison with the Standard Form $y^2=x^3+ax+b$

Many texts normalize the cubic so its leading coefficient is one. The change of variable is a vertical rescale.

Proposition 15.1. *The substitution $X=x$, $Y=y/2$ carries $E: y^2=4x^3-g_2x-g_3$ to*

$$E':\quad Y^2=X^3+aX+b,\qquad a=-\frac{g_2}{4},\quad b=-\frac{g_3}{4},$$

an isomorphism of elliptic curves. Under it the addition formula $x_{P\oplus Q}=m^2/4-x_1-x_2$ becomes the familiar $X_{P\oplus Q}=M^2-X_1-X_2$, where $M=m/2$ is the slope in the (X,Y) coordinates.

Proof. Setting $y=2Y$ in $y^2=4x^3-g_2x-g_3$ gives $4Y^2=4X^3-g_2X-g_3$, i.e. $Y^2=X^3-\frac{g_2}{4}X-\frac{g_3}{4}$. The map $(x,y)\mapsto(x,y/2)$ is invertible and regular, hence an isomorphism fixing O. A line $y=mx+b$ becomes $Y=\frac{m}{2}X+\frac{b}{2}$, so the slope transforms as $M=m/2$; substituting into $x_3=m^2/4-x_1-x_2=(m/2)^2-x_1-x_2=M^2-X_1-X_2$ gives the standard formula. □

So the factor $\frac{1}{4}$ in our addition law is not a special feature of $\wp$: it is exactly the leading coefficient 4 of $4x^3$, and it disappears the moment one rescales y to make the cubic monic. We kept the 4 because it is what the differential equation $\wp'^2=4\wp^3-g_2\wp-g_3$ produces naturally.

15.4 Common Pitfalls

1. **Forgetting the point at infinity.** The group law is incomplete without $O=[0:1:0]$. It is the identity, and it is where every vertical line meets E; dropping it breaks closure (section 12.4).

2. **Calling the third intersection the sum.** A line gives $P\oplus Q\oplus R=O$, so $P\oplus Q=\ominus R$, *not* R. The final reflection is mandatory (section 12.2).

3. **Ignoring multiplicity.** For $P = Q$ use the tangent, which meets E at P with multiplicity two: $L \cdot E = 2P + R$. The divisor rule is unchanged; only the line changes (section 12.3).

4. **Treating reflection as a drawing trick.** The reflection $(x, y) \mapsto (x, -y)$ is the inverse map; it is $(x, y) \mapsto (x, -y)$ precisely because $\wp$ is even and $\wp'$ is odd (section 14.3).

5. **Using $\ell(nO) = n$ without the genus.** Those dimension counts are genus-one facts, resting on the absence of an order-one elliptic function (corollary 4.6 and proposition 6.3). They fail on curves of other genus.

6. **Confusing affine and projective equations.** The affine $y^2 = 4x^3 - g_2x - g_3$ omits O; the compact curve is the projective $Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3$. Statements about the group law live on the projective curve (proposition 8.3).

7. **Forgetting $\Delta \neq 0$.** If $g_2^3 - 27g_3^2 = 0$ the cubic is singular, not an elliptic curve, and the "group law" degenerates. Smoothness is what corollary 7.10 guarantees from the distinctness of the e_i.

15.5 Summary

The chord computation, done symbolically, gives $P \oplus Q$ in closed form (section 15.1); done numerically on $y^2 = 4x^3 - 4x$ it produces $P \oplus Q = (-\frac{24}{25}, -\frac{28\sqrt{6}}{125})$, exactly on the curve (section 15.2). The rescale $y = 2Y$ converts to the monic form $Y^2 = X^3 + aX + b$ and explains the factor $\frac{1}{4}$ (proposition 15.1). The recurring pitfalls — omitting O, forgetting the reflection, mishandling multiplicity or the genus, or working with a singular cubic — all trace back to one principle: the group law lives on the smooth projective curve and is governed by $P \oplus Q \oplus R = O$.

Exercises

Exercise 15.1. Verify that $P \oplus Q = (-\frac{24}{25}, -\frac{28\sqrt{6}}{125})$ of section 15.2 satisfies $y^2 = 4x^3 - 4x$, and compute $\ominus(P \oplus Q)$.

Exercise 15.2. Repeat the symbolic computation of section 15.1 for doubling: starting from the tangent at $P = (x_1, y_1)$ with slope $m = (12x_1^2 - g_2)/(2y_1)$, derive $2P = (\frac{m^2}{4} - 2x_1,\ -m(\frac{m^2}{4} - 2x_1) - b)$.

Exercise 15.3. Convert $E : y^2 = 4x^3 - 4x$ to standard form $Y^2 = X^3 + aX + b$ via $Y = y/2$ (find a, b), recompute $P \oplus Q$ from section 15.2 in the (X, Y) coordinates, and check it matches after mapping back by $y = 2Y$.

Exercise 15.4. For each of pitfalls 1, 2, and 7 in section 15.4, give a concrete calculation where the mistake produces a wrong or undefined answer, and show how the correct principle fixes it.

Exercise 15.5. With $P = (2, 2\sqrt{6})$, $Q = (-\frac{1}{2}, \frac{\sqrt{6}}{2})$, and a third point S of your choice on $E : y^2 = 4x^3 - 4x$, verify numerically that $(P \oplus Q) \oplus S = P \oplus (Q \oplus S)$. (Associativity is guaranteed by remark 11.5; this is a consistency check on the formulas.)

Chapter 16

Synthesis and Further Directions

We have reached the destination announced in chapter 1. This closing chapter recollects the argument as a single chain, then sketches the directions the subject opens onto — the moduli of elliptic curves, isogenies and torsion, the arithmetic over $\mathbb{Q}$ and finite fields, and the link to modular forms — each of which begins exactly where this book ends.

16.1 The Complete Chain

The proof of theorem 1.1 is the following chain, each link a chapter.

1. A lattice $\Lambda \subset \mathbb{C}$ gives a compact genus-one Riemann surface $T = \mathbb{C}/\Lambda$ (chapters 2 and 3).
2. T carries the intrinsic abelian group law $[u] + [v] = [u + v]$ (proposition 3.1).
3. A genus-one dimension count forces the first nonconstant function to have a double pole — the Weierstrass $\wp$ (proposition 6.3 and chapter 7).
4. The functions $1, \wp, \wp'$ span $L(3O)$ and define a map $\Phi : T \to \mathbb{CP}^2$ (section 8.4).
5. The differential equation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ places the image on the cubic E (theorem 7.7 and proposition 8.1).
6. The origin $0 \in T$ maps to $O = [0 : 1 : 0]$, and Φ is a biholomorphism (proposition 8.2 and theorem 8.5).
7. The inverse $z \mapsto -z$ becomes $(x, y) \mapsto (x, -y)$ (section 8.6).
8. A line cuts E in $P + Q + R$; since $Z = 0$ cuts $3O$, the divisor $P + Q + R - 3O$ is principal (theorem 10.4).
9. Principal divisors on T have Abel sum zero, so $P + Q + R = O$ (theorems 9.1 and 10.5).
10. Hence $P \oplus Q = -R$: the chord–tangent law (theorem 11.3).

Three descriptions of one group have emerged, all canonically isomorphic (fig. 16.1): the *analytic* group $(\mathbb{C}/\Lambda, +)$, the *geometric* group $(E, \boxplus)$ of chord–tangent, and the *algebraic* group $\mathrm{Pic}^0(E)$ of degree-zero divisor classes.

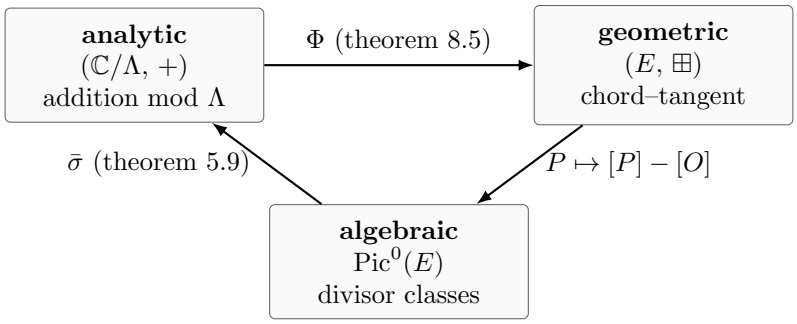


Figure 16.1: Three views of one group. The Weierstrass embedding Φ identifies the analytic and geometric groups; the Abel–Jacobi map $\bar{\sigma}$ identifies the algebraic group $\mathrm{Pic}^0(E)$ with the torus; and $P \mapsto [P] - [O]$ closes the triangle. All three are the same abelian group.

Final takeaway. The chord–tangent law is the visible projective shadow of the simple addition law $[u] + [v] = [u + v]$ on the complex torus. The Weierstrass embedding turns addition into line intersection because line intersections give principal divisors, and principal divisors have zero Abel sum. The torus explains the group; the cubic displays it; the divisor classes name it.

16.2 Further Directions

Everything below takes $(E, O) \cong (\mathbb{C}/\Lambda, 0)$ as its starting point.

16.2.1 The j-Invariant and Moduli

Proposition 2.7 reduced the classification of complex tori to $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$. A single number resolves it: the *j-invariant*

$$j = 1728\,\frac{g_2^3}{g_2^3 - 27g_3^2} = 1728\,\frac{g_2^3}{\Delta}.$$

Two elliptic curves over $\mathbb{C}$ are isomorphic iff they have the same j, and $j : \mathbb{H}/\mathrm{SL}_2(\mathbb{Z}) \to \mathbb{C}$ is a bijection: the moduli space of elliptic curves is the j-line $\mathbb{C}$. As a function of τ, $j(\tau)$ is the basic modular function, invariant under $\mathrm{SL}_2(\mathbb{Z})$. The lemniscatic curve $y^2 = 4x^3 - 4x$ of our examples has $j = 1728$ (exercise 16.1), a curve with extra symmetry (complex multiplication by $\mathbb{Z}[i]$).

16.2.2 Isogenies and Torsion

An *isogeny* is a nonconstant holomorphic group homomorphism $E \to E'$; analytically it is induced by $z \mapsto \alpha z$ for some $\alpha \in \mathbb{C}$ with $\alpha\Lambda \subseteq \Lambda'$. The most important example is multiplication by n,

$$[n] : E \to E, \qquad P \mapsto \underbrace{P \oplus \cdots \oplus P}_{n},$$

a rational map of degree n^2 whose kernel is the n-torsion. Over $\mathbb{C}$,

$$E[n] = \{P : nP = O\} = \tfrac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^2,$$

generalizing the $E[2] \cong (\mathbb{Z}/2\mathbb{Z})^2$ of section 12.5. The torsion carries the Weil pairing $E[n] \times E[n] \to \mu_n$, a perfect alternating pairing that pervades the arithmetic theory.

16.2.3 Elliptic Curves over $\mathbb{Q}$ and Finite Fields

Because the group law is given by rational formulas (proposition 13.3), it runs over any field. Two cases drive modern number theory.

Over $\mathbb{Q}$, the *Mordell–Weil theorem* states that $E(\mathbb{Q})$ is a finitely generated abelian group,

$$E(\mathbb{Q}) \cong \mathbb{Z}^r \oplus E(\mathbb{Q})_{\text{tors}},$$

with r the *rank*. The torsion is completely classified (Mazur); the rank is deep, conjecturally governed by the L-function through the Birch–Swinnerton-Dyer conjecture.

Over a finite field $\mathbb{F}_p$, the group $E(\mathbb{F}_p)$ is finite with

$$\#E(\mathbb{F}_p) = p + 1 - a_p, \qquad |a_p| \le 2\sqrt{p}$$

(Hasse's bound). The difficulty of inverting the map $n \mapsto nP$ in $E(\mathbb{F}_p)$ — the elliptic curve discrete logarithm problem — is the basis of elliptic-curve cryptography.

16.2.4 Modular Forms and Modularity

The Eisenstein series G_4, G_6 that produced g_2, g_3 (definition 7.5) are the first *modular forms*: under $\Lambda \mapsto \alpha\Lambda$ they scale by α^{-4}, α^{-6}, so as functions of τ they satisfy $G_k(\frac{a\tau+b}{c\tau+d}) = (c\tau+d)^k G_k(\tau)$. The discriminant $\Delta = g_2^3 - 27g_3^2$ is a cusp form of weight 12, and $j = 1728g_2^3/\Delta$ has weight 0 (exercise 16.3). The arithmetic of elliptic curves over $\mathbb{Q}$ is bound to modular forms by the *modularity theorem* (Wiles, Taylor–Wiles, Breuil–Conrad–Diamond–Taylor): every elliptic curve over $\mathbb{Q}$ arises from a weight-two modular form. Its most famous consequence is Fermat's Last Theorem.

16.2.5 Where to Read Next

Silverman's *The Arithmetic of Elliptic Curves* [6] is the standard entry to the arithmetic theory; Silverman–Tate, *Rational Points on Elliptic Curves* [7], is a gentler start. For the complex-analytic and modular side, see Diamond–Shurman, *A First Course in Modular Forms* [8], and Serre, *A Course in Arithmetic* [9]. Each of these opens at the point reached here: the elliptic curve is a complex torus, and its group law is addition.

16.3 Summary

The chord–tangent law is addition on $\mathbb{C}/\Lambda$ transported by $(\wp, \wp')$, and the same group appears in three guises — analytic, geometric, algebraic (fig. 16.1). From this single fact the subject branches: the j-invariant classifies the curves (section 16.2.1); isogenies and torsion organize the maps between them (section 16.2.2); the rational group law carries the theory over $\mathbb{Q}$ and $\mathbb{F}_p$ into number theory and cryptography (section 16.2.3); and the Eisenstein series tie it to modular forms and, through modularity, to Fermat (section 16.2.4). The torus was the key throughout: it makes a mysterious geometric construction into the most natural thing in the world, the addition of points.

Exercises

Exercise 16.1. Compute the j-invariant of $E : y^2 = 4x^3 - 4x$ (so $g_2 = 4$, $g_3 = 0$). Confirm $\Delta = 64$ and $j = 1728$. What is j for a curve with $g_2 = 0$ (so $y^2 = 4x^3 - g_3$)?

Exercise 16.2. Show that over $\mathbb{C}$, $E[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^2$, and that multiplication by n, realized as $z \mapsto nz$ on $\mathbb{C}/\Lambda$, is an n^2-to-one map, so $[n]$ has degree n^2.

Exercise 16.3. Using $G_k(\alpha\Lambda) = \alpha^{-k}G_k(\Lambda)$, show that under $\Lambda \mapsto \alpha\Lambda$ one has $g_2 \mapsto \alpha^{-4}g_2$, $g_3 \mapsto \alpha^{-6}g_3$, $\Delta \mapsto \alpha^{-12}\Delta$, and hence j is invariant. Interpret these as the weights $4, 6, 12, 0$.

Exercise 16.4. Take $E : y^2 = x^3 + 1$ over $\mathbb{F}_5$. List its affine points, add the point at infinity, and find $\#E(\mathbb{F}_5)$. Check that it satisfies the Hasse bound $|\#E(\mathbb{F}_5) - 6| \le 2\sqrt{5}$, and identify the group structure.

Part VII

Appendices

Appendix A

Complex-Analysis Toolkit

This appendix collects, for reference, the results from a first course in complex analysis that the main text uses. Proofs are in any standard text — Ahlfors [1], Stein–Shakarchi [2], Conway [3] — and we state the facts precisely and note where each is invoked.

A.1 Holomorphic and Meromorphic Functions

A function f on an open $U \subseteq \mathbb{C}$ is *holomorphic* if it is complex-differentiable at every point; equivalently it is given near each point by a convergent power series. It is *meromorphic* on U if it is holomorphic off a discrete set of poles. Near a point p a meromorphic $f \not\equiv 0$ has a *Laurent expansion*

$$f(z) = \sum_{k \geq m} c_k (z - p)^k, \qquad c_m \neq 0,$$

and the integer $m = \operatorname{ord}_p(f)$ is the *order*: $m > 0$ a zero, $m < 0$ a pole of order $-m$, $m = 0$ a nonzero finite value. The *residue* is the coefficient $\operatorname{Res}_p f = c_{-1}$. These notions are intrinsic to a Riemann surface because they are unchanged by holomorphic coordinate changes (remark 5.1), which is what lets us speak of orders and residues on $T = \mathbb{C}/\Lambda$ throughout chapters 4 to 5.

A.2 Liouville's Theorem and the Maximum Modulus Principle

Theorem A.1 (Liouville)**.** *A bounded entire function (holomorphic on all of $\mathbb{C}$) is constant.*

Theorem A.2 (Maximum modulus)**.** *A nonconstant holomorphic function on a connected open set has no interior local maximum of $|f|$. Consequently a holomorphic function on a compact connected Riemann surface is constant.*

These are the engine of theorem 4.3: a holomorphic elliptic function is bounded (by periodicity and compactness of the fundamental parallelogram), hence constant. The same statements force $L(0) = \mathbb{C}$ and $\Omega(T) = \mathbb{C}\,dz$ in chapter 6.

A.3 Residues and the Residue Theorem

Theorem A.3 (Residue theorem in the plane)**.** *If f is meromorphic on a domain containing a positively oriented simple closed curve γ and its interior, with no poles on γ, then*

$$\frac{1}{2\pi i}\oint_\gamma f(z)\,dz = \sum_{p \text{ inside}} \operatorname{Res}_p f.$$

On a compact Riemann surface there is no boundary, and the global form is a vanishing theorem.

Theorem A.4 (Sum of residues on a compact surface)**.** *For a meromorphic 1-form ω on a compact Riemann surface X, $\sum_{p\in X} \operatorname{Res}_p \omega = 0$.*

The planar version drives the period-parallelogram computations of chapter 4 (the boundary integral vanishes because opposite sides cancel, leaving $\sum \text{Res} = 0$); the compact version is the clean statement behind $\deg \operatorname{div}(f) = 0$ applied to df/f in theorem 5.5. Both are global Stokes-type statements (appendix B).

A.4 The Argument Principle and Winding Numbers

Theorem A.5 (Argument principle)**.** *If f is meromorphic inside and on a positively oriented simple closed curve γ, with no zeros or poles on γ, then*

$$\frac{1}{2\pi i}\oint_\gamma \frac{f'(z)}{f(z)}\,dz = Z - P,$$

the number of zeros minus the number of poles inside, with multiplicity. The left side is the winding number of $f\circ\gamma$ about 0.

Applied on a period parallelogram, this gives the valence theorem theorem 4.7 (#zeros = #poles for an elliptic function) and, through $\oint z\,f'/f$, the Abel relation theorem 4.8; the integrality of the winding number is the $2\pi i\,\mathbb{Z}$ in those proofs (exercise 4.5).

A.5 Pointers

The Weierstrass M-test (uniform convergence of a series dominated by a convergent series of constants) justifies the convergence of the $\wp$-series in proposition 7.3. The implicit function theorem provides holomorphic charts on a smooth plane curve, making E a Riemann surface in section 8.4. A nonconstant holomorphic map between compact Riemann surfaces is a finite branched cover and is surjective; this underlies the degree-two cover $\wp : T \to \mathbb{CP}^1$ (section 7.5) and the surjectivity of Φ in theorem 8.5.

Appendix B

The Exam-to-Lecture Bridge

These notes grew out of a course whose midterm and final assembled exactly the tools the main argument needs. This appendix records that lineage, for readers who want to see where each ingredient came from; each item points to the chapter that develops it in full.

B.1 From the Midterm

1. **Differential forms and Stokes' theorem.** Integral theorems are best read as statements about differential forms. The residue theorem on a compact Riemann surface (theorem A.4) is another global Stokes-type identity — the vanishing of a sum of local data — and it is the form used in chapter 5.

2. **Cauchy–Green and residues.** That residues are global obstructions — local singular data constrained by a global sum — is the philosophy behind Abel's law (chapters 4 and 9): the zeros and poles of an elliptic function cannot be prescribed freely.

3. **The torus as a quotient.** The midterm computed invariants of real and complex tori. The complex torus $T = \mathbb{C}/\Lambda$ is the source of the group law here (chapters 2 and 3).

4. **Euler characteristic of a torus.** The midterm found $\chi(T) = 0$; with $\chi = 2 - 2g$ this gives genus $g = 1$ (proposition 3.7), exactly the genus that makes Riemann–Roch take its elliptic-curve form.

B.2 From the Final

1. **Compact Riemann surfaces and meromorphic maps.** The final used that nonconstant holomorphic maps between compact Riemann surfaces are finite branched coverings. Here $\wp, \wp'$ are meromorphic on the compact surface T, and $\wp : T \to \mathbb{CP}^1$ is the degree-two cover of section 7.5.

2. **Principal divisors have degree zero.** Proved on the final via the logarithmic derivative df/f; it is the degree half of Abel's law (theorem 5.5).

3. **Riemann–Roch.** On a genus-one surface it predicts the dimensions $\ell(nO) = n$ that force $1, \wp, \wp'$ to be the coordinate functions — counts the present treatment derives elementarily (proposition 6.3 and section 6.6).

4. **The torus and $\wp$.** The final's $\wp$-problem derived $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ (theorem 7.7), which is precisely the equation of the plane cubic.

Exam-to-lecture bridge. The midterm supplies the geometric and differential-form intuition; the final supplies the compact Riemann surface tools. Together they explain why the algebraic-looking chord–tangent rule is forced by complex analysis: the torus has genus one, principal divisors balance, and Riemann–Roch produces $\wp$ and $\wp'$.

Appendix C

The Sage-to-TikZ Figure Pipeline

Every figure in this book is drawn with exact coordinates, not freehand. The geometry — lattice points, the torus and sphere silhouettes, the elliptic curve samples, the chord and tangent intersections — is computed by a set of SageMath scripts that emit LaTeX macros, which the TikZ figures then reference. This appendix documents the pipeline so the figures can be regenerated and checked.

C.1 Overview

The pipeline lives in the `sagemath/` directory (a sibling of `book/`). Each script `NN_name.sage` computes coordinates and writes them to `coords_name.tex` as `\newcommand` definitions; the driver `build_coords.sh` runs all scripts and concatenates their output into a single `coords.tex`. The document `\input`s that file, so a figure can write, e.g., `\fill (\chordPx,\chordPy)` and get the precise Sage-computed point.

C.2 The Scripts

Script	Output `coords_*`	Role
`01_lattice`	`lattice`	lattice basis and cell data
`02_torus_group_law`	`group`	group-law points $u, v, u{+}v$
`03_elliptic_curve`	`curve`	the roots e_1, e_2, e_3
`04_chord_tangent`	`chord`	chord example $P, Q, R, -R$
`05_tangent_double`	`tangent`	doubling example $P, R, 2P$
`06_two_torsion`	`torsion`	two-torsion + vertical-line points
`07_branched_cover`	`branch`	branched-cover data
`08_torus_render`	`torus`	torus arcs (precise torus)
`09_torus_markers`	`torus_markers`	markers on the torus
`10_sphere_render`	`sphere`	sphere arcs (the $\mathbb{CP}^1$ image)
`11_correspondence`	`correspondence`	three-panel markers
`12_curve_sample`	`curve_sample`	arcs of $y^2 = 4x^3 - 4x$

The macros are named after their script: e.g. `01_lattice` emits `\latwax`, `\latomegaone`; `04_chord_tangent` emits `\chordPx`, `\chordQx`, `\chordRx`, `\chordSx`; `11_correspondence` emits the `\cor...` family; and the three `\drawprecise...` macros consume `coords_torus`, `coords_sphere`, and `coords_curve_sample`.

The running example throughout is $E : y^2 = 4x^3 - 4x$ ($g_2 = 4$, $g_3 = 0$, roots $-1, 0, 1$), with a fixed lattice whose basis is recorded in `coords_lattice.tex`.

C.3 The Convenience Macros

Three painting macros, defined in `book/preamble.tex`, encapsulate the most intricate figures so a chapter can drop them into a scope:

- `\drawprecisetorus` — the tilted torus (outer/inner silhouettes, meridians, latitudes split into visible/hidden arcs), from `coords_torus`;
- `\drawprecisesphere` — the $\mathbb{CP}^1$ sphere matching the torus style, from `coords_sphere`;
- `\drawprecisecurve` — the real locus of $y^2 = 4x^3 - 4x$ (compact oval plus unbounded branch, with the three branch points), from `coords_curve_sample`.

C.4 Building and Consuming the Coordinates

The driver `sagemath/build_coords.sh` runs every `0?_*.sage` in order and concatenates the per-script files into `coords.tex`. The book's preamble loads

```
\input{\sagedir/coords.tex}
\input{\sagedir/coords_torus.tex}
\input{\sagedir/coords_curve_sample.tex}
```

where `\sagedir` points at `../sagemath`. The aggregated `coords.tex` contains the lattice, group, curve, chord, tangent, torsion, branch, torus-marker, sphere, correspondence, and curve-sample macros; the precise torus rendering `coords_torus.tex` is generated by `08_torus_render` but loaded separately.

C.5 Regenerating the Figures

With SageMath installed, regenerate all coordinates and rebuild the book by

```
cd sagemath && ./build_coords.sh
cd ../book && pdflatex main.tex && pdflatex main.tex
```

The committed `coords*.tex` files let the book compile *without* Sage; they need to be regenerated only if a `.sage` script changes (for instance to use a different lattice or curve). If a new figure needs precise geometry, add a script `sagemath/NN_name.sage` that writes `coords_name.tex`, list that file in `build_coords.sh`, and `\input` it (or rely on the aggregated `coords.tex`). Coordinates should never be hand-typed: keeping them under Sage control is what makes the figures exact and the book reproducible.

Appendix D

Solutions to Exercises

This appendix gives full solutions to every exercise. Solutions are grouped by chapter and reference each exercise by its number.

D.1 Chapter 1

Exercise 1.1. Write the line and the cubic in homogeneous coordinates on $\mathbb{CP}^2$. Substituting the linear equation of the line into the cubic gives a single homogeneous cubic equation in one projective parameter along the line, i.e. a degree-three form. Over $\mathbb{C}$, and counting multiplicity, such a form has exactly three roots, so the line meets the cubic in three points counted with multiplicity. Two of them are P and Q; the third is R. Passing to the projective plane is what guarantees the count is exact — it supplies the points at infinity where, for example, a vertical line meets the curve. The precise statement is Bézout's theorem (chapter 10).

Exercise 1.2. As $Q \to P$ along the curve, the secant line through P and Q approaches the tangent line at P. The chord construction needs two intersection points to start from; when the two points coincide, "meeting the curve at P twice" is exactly the condition that the line be tangent there, i.e. that it meet E at P with intersection multiplicity at least two. So doubling $P \oplus P$ uses the tangent at P, whose third intersection with E is then reflected. Intersection multiplicity is made precise in chapter 10, and the tangent (doubling) law in chapter 12.

Exercise 1.3. A holomorphic function on $\mathbb{C}/\Lambda$ lifts to a holomorphic function on $\mathbb{C}$ that is periodic with respect to Λ; its values are determined by those on the (compact) fundamental parallelogram, so it is bounded, hence by Liouville's theorem constant. Therefore the only holomorphic functions on $\mathbb{C}/\Lambda$ are the constants (chapter 4).

Exercise 1.4. Reading the arrows of fig. 1.1:

- $T \xrightarrow{\Phi} (\wp, \wp')$: the Weierstrass functions are built on the torus in chapter 7.
- $(\wp, \wp') \to E$: the differential equation embeds the torus as the cubic, chapters 7 and 8.
- $E \to$ (a line cuts $P + Q + R$): intersecting E with a line, chapter 10.
- (line) $\to$ (principal divisors have Abel sum zero): a line gives a principal divisor, chapter 10.
- (Abel sum zero) $\to T$ with label $P + Q + R = O$: Abel's sum law, chapter 9, transported in chapter 11.

D.2 Chapter 2

Exercise 2.1. Writing $\omega_1 = x_1 + iy_1$ and $\omega_2 = x_2 + iy_2$, the parallelogram spanned by the vectors (x_1, y_1) and (x_2, y_2) has area $|x_1y_2 - x_2y_1|$. A short computation gives $\operatorname{Im}(\overline{\omega_1}\,\omega_2) = x_1y_2 - y_1x_2$, so the area is $|\operatorname{Im}(\overline{\omega_1}\,\omega_2)|$, which is nonzero exactly because ω_1, ω_2 are $\mathbb{R}$-independent. Under a change of basis by $M \in \mathrm{GL}_2(\mathbb{Z})$ (proposition 2.3) the area scales by $|\det M| = 1$, so it is a lattice invariant, the *covolume* of Λ.

Exercise 2.2. The function $(s, t) \mapsto |s\omega_1 + t\omega_2|$ is continuous and, by $\mathbb{R}$-independence, strictly positive on the compact set $K = \{(s, t) \in \mathbb{R}^2 : \max(|s|, |t|) = 1\}$; let $c > 0$ be its minimum. For integers a, b not both zero, put $n = \max(|a|, |b|) \geq 1$; then $(a/n, b/n) \in K$, so $|a\omega_1 + b\omega_2| = n\,|(a/n)\omega_1 + (b/n)\omega_2| \geq cn = c\max(|a|, |b|)$. In particular $m(\Lambda) \geq c > 0$. A bounded region $\{|\lambda| \leq R\}$ then forces $\max(|a|, |b|) \leq R/c$, leaving finitely many (a, b), so finitely many lattice points.

Exercise 2.3. For $\Lambda = \mathbb{Z} + \mathbb{Z}i$, the matrix $M = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ sends $\tau = i$ to $\frac{0 \cdot i - 1}{1 \cdot i + 0} = -1/i = i$, so it fixes i. It corresponds to the basis change $(\omega_1, \omega_2) = (1, i) \mapsto (\omega_1', \omega_2') = (i, -1) = (i \cdot 1,\ i \cdot i)$, i.e. multiplication of the lattice by i: the order-four rotational symmetry of the square lattice. (Its square $-I$ acts trivially on τ, reflecting that -1 also preserves Λ.)

Exercise 2.4. If $z \sim w$ in $\mathbb{C}/\Lambda$, i.e. $z - w \in \Lambda$, then $\alpha z - \alpha w \in \alpha\Lambda = \Lambda'$, so $[z] \mapsto [\alpha z]$ is well defined; the inverse is $[w] \mapsto [\alpha^{-1}w]$, so it is a bijection, and it is continuous both ways because $\pi' \circ (\alpha\cdot) = (\text{map}) \circ \pi$ with $z \mapsto \alpha z$ continuous. In the charts of theorem 2.10 the map lifts to $z \mapsto \alpha z$, which is holomorphic with holomorphic inverse; hence it is biholomorphic.

Exercise 2.5. Take $\Lambda = \mathbb{Z} + \mathbb{Z}i$, so $m(\Lambda) = 1$ and $m(\Lambda)/2 = 1/2$. For any $r > 1/2$ the open disk $D(0, r)$ contains both $-1/2$ and $1/2$, and these differ by $1 \in \Lambda$; hence $\pi(-1/2) = \pi(1/2)$ while $-1/2 \neq 1/2$, so π is not injective on $D(0, r)$. Conversely, for $r \leq 1/2$ any two points $z, z' \in D(z_0, r)$ satisfy $|z - z'| < 2r \leq 1 = m(\Lambda)$; a nonzero lattice vector has length at least $m(\Lambda)$, so $z - z' \in \Lambda$ forces $z = z'$ and π is injective. Thus injectivity holds exactly up to radius $m(\Lambda)/2$, and the threshold is sharp. (The proof's choice $r < m(\Lambda)/2$ is a convenient strict bound; any $r \leq m(\Lambda)/2$ would serve.)

D.3 Chapter 3

Exercise 3.1. Cover $T \times T$ by products $U_{z_0} \times U_{w_0}$ of the charts $\varphi_{z_0}, \varphi_{w_0}$ from theorem 2.10. In the source product chart the points are (z, w) with $z \in D(z_0, r)$, $w \in D(w_0, r)$, and $m([z], [w]) = [z + w]$; reading the target through the chart $\varphi_{z_0+w_0}$ (valid near $[z_0 + w_0]$) the map is $(z, w) \mapsto z + w$, a holomorphic function of two complex variables. Since π is a local biholomorphism, m is holomorphic. For ι, the local expression through φ_{z_0} and φ_{-z_0} is $z \mapsto -z$, holomorphic with holomorphic inverse $z \mapsto -z$; hence ι is biholomorphic.

Exercise 3.2. A point $[z]$ is fixed by ι iff $[-z] = [z]$, i.e. $2z \in \Lambda$, i.e. $z \in \frac{1}{2}\Lambda$. Modulo Λ, the quotient $\frac{1}{2}\Lambda/\Lambda \cong (\frac{1}{2}\mathbb{Z}/\mathbb{Z})^2 \cong (\mathbb{Z}/2\mathbb{Z})^2$ has exactly four elements, represented by 0, $\frac{\omega_1}{2}$, $\frac{\omega_2}{2}$, $\frac{\omega_1+\omega_2}{2}$. These are precisely the solutions of $2[z] = [0]$, the two-torsion subgroup $T[2] \cong (\mathbb{Z}/2\mathbb{Z})^2$.

Exercise 3.3. The map $e : \mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, satisfies $e(t_1 + t_2) = e(t_1)e(t_2)$, so it is a homomorphism; it is continuous and surjective (every unit-modulus number is $e^{2\pi i t}$ for some t), and $e(t) = 1 \iff t \in \mathbb{Z}$, so $\ker e = \mathbb{Z}$. The first isomorphism theorem gives a continuous group isomorphism $\bar{e} : \mathbb{R}/\mathbb{Z} \to S^1$. Finally $\mathbb{R}/\mathbb{Z}$ is the continuous image of the compact interval $[0, 1]$, hence compact, and S^1 is Hausdorff, so the continuous bijection $\bar{e}$ is a homeomorphism.

Exercise 3.4. Cut P_Λ along a diagonal into two triangles before gluing. After identification one finds, for example, $V = 1$ (all corners still collapse to one vertex), $E = 3$ (the two boundary loops a, b plus the diagonal d), and $F = 2$ (the two triangles). Then $\chi = V - E + F = 1 - 3 + 2 = 0$, the same value: the Euler characteristic is independent of the chosen CW structure.

Exercise 3.5. Both lattices are of the form $\mathbb{Z}+\mathbb{Z}\tau$ with $\tau = i$ and $\tau = 2i$, so by corollary 3.5 each torus is homeomorphic (and diffeomorphic) to $S^1 \times S^1$; composing the two homeomorphisms gives a homeomorphism between them. For the complex structure, proposition 2.7 says $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}i)$ and $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\,2i)$ are biholomorphic iff i and $2i$ are related by an element of $\mathrm{SL}_2(\mathbb{Z})$. Granting that both lie in the closed fundamental domain and are not identified there, they are inequivalent, so the tori are not biholomorphic. There is no contradiction with corollary 3.5: that corollary asserts only a *topological* isomorphism, and warning 3.6 stresses that the homeomorphism Ψ is not holomorphic. Homeomorphic surfaces can carry inequivalent complex structures.

D.4 Chapter 4

Exercise 4.1. For f, g elliptic, each of $f \pm g, fg, f/g$ is meromorphic, and periodicity is pointwise: $(fg)(z+\lambda) = f(z+\lambda)g(z+\lambda) = f(z)g(z)$, and similarly for the others; the constants $0, 1$ are elliptic, so we have a field. Differentiating $f(z+\lambda) = f(z)$ gives $f'(z+\lambda) = f'(z)$, so f' is elliptic. For fixed a, $(f(\cdot + a))(z + \lambda) = f(z + a + \lambda) = f(z + a)$, so $f(z + a)$ is elliptic. If $f(-z) = f(z)$, differentiate: $-f'(-z) = f'(z)$, i.e. $f'(-z) = -f'(z)$, so f' is odd.

Exercise 4.2. An order-one function would have, in a period parallelogram, a single simple pole and no other poles; the residue there is nonzero (the defining feature of a simple pole), contradicting $\sum \mathrm{Res} = 0$ (theorem 4.5). Hence no order-one elliptic function exists. For order two the total pole multiplicity is 2: either two simple poles at $p_1 \neq p_2$ with residues r_1, r_2, and theorem 4.5 forces $r_1 + r_2 = 0$; or a single pole of order two, with Laurent expansion $c_{-2}(z-p)^{-2} + c_{-1}(z-p)^{-1} + \cdots$ whose residue is c_{-1}, and since it is the only pole, $\sum \mathrm{Res} = c_{-1} = 0$.

Exercise 4.3. By theorem 4.7 the order-two function f has exactly two zeros z_1, z_2 (with multiplicity). Its only pole is the double pole at $[0]$, so theorem 4.8 reads $z_1 + z_2 - (0 + 0) \in \Lambda$, i.e. $z_2 \equiv -z_1 =: -w$. The zeros are $[w], [-w]$. Now $f - f(z_0)$ is again elliptic of order two with the same double pole at $[0]$; one zero is z_0, so by the same argument its second zero is $-z_0$. Thus f takes each value at a pair $\pm z_0$, which is the hallmark of an even function $f(-z) = f(z)$.

Exercise 4.4. The function $f - c_1$ is elliptic of order N with the same poles $p_1, \ldots, p_N$ as f and with zeros the solutions $z_1^{(1)}, \ldots$ of $f = c_1$. By theorem 4.8, $\sum_k z_k^{(1)} - \sum_k p_k \in \Lambda$. The same for c_2: $\sum_k z_k^{(2)} - \sum_k p_k \in \Lambda$. Subtracting the two congruences, the pole sums cancel and $\sum_k z_k^{(1)} \equiv \sum_k z_k^{(2)} \pmod{\Lambda}$. Hence the sum of the solution set of $f = c$ is, modulo Λ, independent of c; its common value (equal to $\sum p_k \bmod \Lambda$) is an invariant of f.

Exercise 4.5. Let $\gamma(s) = a+s\omega_1$, $s \in [0,1]$, avoid the zeros and poles of f. Then $\int_a^{a+\omega_1}(f'/f)\,dz = \int_0^1 \frac{d}{ds}\log f(\gamma(s))\,ds = \log f(a+\omega_1) - \log f(a)$ for a continuous branch of $\log f$ along γ. The real part is $\log|f(a+\omega_1)| - \log|f(a)| = 0$ because $f(a+\omega_1) = f(a)$; the imaginary part is the net change of argument, $2\pi k_1$, where $k_1 \in \mathbb{Z}$ is the number of times the image curve $s \mapsto f(\gamma(s))$ winds around 0. Hence the integral equals $2\pi i\, k_1$, and k_1 is precisely that winding number.

D.5 Chapter 5

Exercise 5.1. By proposition 5.4, $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$, so div is a homomorphism from $(\mathcal{M}(X)^\times, \cdot)$ to $(\operatorname{Div}(X), +)$. If $\operatorname{div}(f) = 0$ then f has no zeros or poles, so it is holomorphic and nonvanishing on the compact X; by theorem 4.3 (or the maximum modulus principle) f is a nonzero constant. Conversely $\operatorname{div}(c) = 0$. Hence $\ker\operatorname{div} = \mathbb{C}^\times$.

Exercise 5.2. Independence of coordinate is remark 5.1: a coordinate change $t = s\,h(s)$, $h(0) \neq 0$, turns $f = t^m u$ into $f = s^m\,(h^m\,u\circ t)$ with a nonvanishing unit, so the exponent m is unchanged. For the sum, write $f = t^a u$, $g = t^b v$ with $a = \operatorname{ord}_p f \leq b = \operatorname{ord}_p g$ and units u, v. Then $f+g = t^a(u + t^{b-a}v)$; the factor $u + t^{b-a}v$ is holomorphic, so $\operatorname{ord}_p(f+g) \geq a = \min(a,b)$, and if $a < b$ it equals $u(0) \neq 0$ at $t = 0$, giving $\operatorname{ord}_p(f+g) = a$.

Exercise 5.3. The map $D \mapsto (\deg D,\ D - (\deg D)[0])$ is a homomorphism, since deg is and the second slot is linear in D; its second component has degree $\deg D - \deg D = 0$, so it lands in $\mathbb{Z} \oplus \operatorname{Div}^0(T)$. The inverse is $(n, E) \mapsto E + n[0]$, so it is an isomorphism. Principal divisors have degree 0 and lie in the Div^0 slot, so quotienting gives $\operatorname{Pic}(T) \cong \mathbb{Z} \oplus \operatorname{Pic}^0(T)$; with $\operatorname{Pic}^0(T) \cong T$ from theorem 5.9, $\operatorname{Pic}(T) \cong \mathbb{Z} \oplus T$.

Exercise 5.4. If $[a] = [b]$ then $[a]-[b] = 0 = \operatorname{div}(1)$ is principal. Conversely, if $[a]-[b] = \operatorname{div}(f)$ is principal, then it lies in $\operatorname{Div}^0(T)$ and theorem 4.8 (via proposition 5.8) gives $\mathfrak{s}([a]-[b]) = [a]-[b] = 0$ in T, i.e. $[a] = [b]$. On $\mathbb{CP}^1$ the function $(z-a)/(z-b)$ has divisor $[a]-[b]$ for any $a \neq b$, so the statement fails there. The torus version recovers corollary 4.6: a principal divisor $[a]-[b]$ with $a \neq b$ would be the divisor of an order-one elliptic function (one simple zero, one simple pole), which does not exist.

Exercise 5.5. The divisor $D = [\frac{1}{2}] + [\frac{\tau}{2}] + [\frac{1+\tau}{2}] - 3[0]$ has $\deg D = 3 - 3 = 0$. Its Abel sum is

$$\mathfrak{s}(D) = \left[\tfrac{1}{2} + \tfrac{\tau}{2} + \tfrac{1+\tau}{2}\right] = \left[\tfrac{2+2\tau}{2}\right] = [\,1+\tau\,] = [0],$$

since $1 + \tau \in \Lambda$. So D has degree zero and Abel sum zero, hence is a candidate for a principal divisor — and indeed it is: the three nonzero half-periods $\frac{1}{2}, \frac{\tau}{2}, \frac{1+\tau}{2}$ are exactly the two-torsion points of exercise 3.2, and $D = \operatorname{div}(\wp')$, the divisor of the Weierstrass derivative (simple zeros at the half-periods, a triple pole at $[0]$), as we will see in chapter 7.

D.6 Chapter 6

Exercise 6.1. Let $D = \sum n_q\, q$. An $f \in L(D+p)$ may have a pole at p of order up to $n_p + 1$. In a coordinate t at p, the map $\lambda : L(D+p) \to \mathbb{C}$ sending f to its Laurent coefficient of order $-(n_p+1)$ (the most negative now allowed) is linear. Its kernel is exactly the functions whose pole order at p is at most n_p, i.e. $L(D)$. Hence $L(D+p)/L(D) \hookrightarrow \mathbb{C}$, so $\ell(D+p) - \ell(D) = \dim\operatorname{im}\lambda \in \{0, 1\}$.

Exercise 6.2. If $\deg D < 0$ and $0 \neq f \in L(D)$, then $\operatorname{div}(f) \geq -D$ has degree 0 (theorem 5.5) but $\deg(-D) > 0$, impossible; so $L(D) = 0$. For the inequality, induct on $\deg D$ downward using the point lemma: any D can be written $D = D_0 + \sum_{j=1}^{k} p_j$ with $\deg D_0 \leq 0$ and $k = \deg D - \deg D_0$; since $\ell(D_0) \leq 1$ (it is 0 for negative degree, and for degree 0 a nonzero $f \in L(D_0)$ has $\operatorname{div}(f) = -D_0$, so $D_0 \sim 0$ and $\ell(D_0) = 1$) and each added point raises ℓ by at most one (exercise 6.1), $\ell(D) \leq 1 + (\deg D - \deg D_0) \leq \deg D + 1$.

Exercise 6.3. A divisor of degree one is $\sum n_q\, q$ with $\sum n_q = 1$; it is effective (≥ 0) iff all $n_q \geq 0$, which with sum 1 means exactly one coefficient equals 1 and the rest 0, i.e. a single point. By proposition 6.3, $\ell([a]) = 1$, so $L([a]) = \mathbb{C}$ (constants); the effective divisors linearly equivalent to $[a]$ are the $\operatorname{div}(f) + [a]$ for $0 \neq f \in L([a])$, i.e. f constant, giving only $[a]$ itself.

Exercise 6.4. By proposition 6.5, $K \sim 0$, so $\ell(K) = \ell(0) = 1 = g$. Taking $D = K$ in Riemann–Roch: $\ell(K) - \ell(K - K) = \deg K + 1 - g$, i.e. $\ell(K) - \ell(0) = \deg K + 1 - g$. With $\ell(K) = \ell(0) = 1$ and $g = 1$ this reads $0 = \deg K + 1 - 1$, so $\deg K = 0 = 2g - 2$, consistent.

Exercise 6.5. The pole orders at O are

$$1{:}0, \quad \wp{:}2, \quad \wp^2{:}4, \quad \wp^3{:}6, \quad \wp'{:}3, \quad \wp\wp'{:}5, \quad \wp'^2{:}6,$$

all ≤ 6 and with no other poles, so all seven lie in $L(6O)$. Since $\ell(6O) = 6 < 7$, they are linearly dependent: $\sum c_i F_i = 0$ nontrivially. The only two functions with a pole of order exactly 6 are $\wp^3$ and $\wp'^2$; their leading Laurent coefficients at O are the only order-6 contributions, so for the order-6 pole to cancel, the coefficients of $\wp^3$ and $\wp'^2$ must be nonzero and balance. If either were zero, the other would contribute an uncancelled order-six pole, contradicting $\sum c_i F_i = 0$ (a holomorphic, indeed zero, function). Hence both appear, and solving for the relation gives $\wp'^2 = 4\wp^3 - g_2\wp - g_3$.

D.7 Chapter 7

Exercise 7.1. Lattice points are uniformly separated (exercise 2.2), so disjoint disks of radius $\rho = \frac{1}{2}m(\Lambda)$ about them are contained in the disk of radius $R + \rho$ once $|\lambda| \leq R$; comparing areas bounds their number by $(R + \rho)^2/\rho^2 = O(R^2)$. Splitting into annuli $n \leq |\lambda| < n + 1$ (each with $O(n)$ points) gives $\sum_\lambda |\lambda|^{-s} \leq \sum_n O(n)\, n^{-s} = \sum_n O(n^{1-s}) < \infty$ for $s > 2$. The differentiated terms $\frac{d}{dz}\big((z-\lambda)^{-2} - \lambda^{-2}\big) = -2(z-\lambda)^{-3}$ satisfy $|-2(z-\lambda)^{-3}| \leq 16/|\lambda|^3$ for $|z| \leq R$, $|\lambda| \geq 2R$, which is summable ($s = 3$); uniform convergence of the differentiated series justifies term-by-term differentiation, and adding the $\lambda = 0$ term $\frac{d}{dz}z^{-2} = -2z^{-3}$ gives $\wp'(z) = -2\sum_{\lambda\in\Lambda}(z-\lambda)^{-3}$.

Exercise 7.2. From proposition 7.6, $\wp = z^{-2} + 3G_4 z^2 + 5G_6 z^4 + O(z^6)$, so $a_2 = 3G_4$, $a_4 = 5G_6$. Using $\wp' = -2z^{-3} + 2a_2 z + 4a_4 z^3 + \cdots$, one computes $\wp'^2 - 4\wp^3 = -20a_2 z^{-2} - 28a_4 + O(z^2)$. Choosing $g_2 = 20a_2$ and $g_3 = 28a_4$ cancels the z^{-2} term and the constant; substituting $a_2 = 3G_4$, $a_4 = 5G_6$ gives $g_2 = 60G_4$, $g_3 = 140G_6$.

Exercise 7.3. The cubic $4x^3 - g_2x - g_3 = 4\big(x^3 - \frac{g_2}{4}x - \frac{g_3}{4}\big)$ is monic after dividing by 4, with $p = -g_2/4$, $q = -g_3/4$. The discriminant of a monic cubic $x^3 + px + q$ is $\prod_{i<j}(e_i - e_j)^2 = -4p^3 - 27q^2$. Hence

$$\prod_{i<j}(e_i - e_j)^2 = -4\Big(-\tfrac{g_2}{4}\Big)^3 - 27\Big(-\tfrac{g_3}{4}\Big)^2 = \frac{g_2^3}{16} - \frac{27g_3^2}{16} = \frac{g_2^3 - 27g_3^2}{16},$$

so $\Delta = g_2^3 - 27g_3^2 = 16\prod_{i<j}(e_i - e_j)^2$. By proposition 7.8(3) the e_i are distinct, so $\Delta \neq 0$.

Exercise 7.4. Write $f = f_+ + f_-$ with $f_\pm(z) = \frac{1}{2}(f(z) \pm f(-z))$, both elliptic, f_+ even and f_- odd. By the generation step of theorem 7.11, even elliptic functions lie in $\mathbb{C}(\wp)$, so $f_+ = A(\wp)$ for a rational A. Since $\wp'$ is odd, $f_-/\wp'$ is even, hence $f_-/\wp' = B(\wp)$ for a rational B; thus $f_- = B(\wp)\,\wp'$ and $f = A(\wp) + B(\wp)\,\wp'$. The fact $\wp' \notin \mathbb{C}(\wp)$ guarantees this representation is genuine and unique: if $A_1(\wp) + B_1(\wp)\wp' = A_2(\wp) + B_2(\wp)\wp'$ then $(B_1 - B_2)(\wp)\,\wp' = (A_2 - A_1)(\wp) \in \mathbb{C}(\wp)$ forces $B_1 = B_2$ (else $\wp' \in \mathbb{C}(\wp)$) and then $A_1 = A_2$.

Exercise 7.5. Differentiating $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ gives $2\wp'\wp'' = (12\wp^2 - g_2)\wp'$, and dividing by $2\wp'$ (not identically zero), $\wp'' = 6\wp^2 - \frac{1}{2}g_2$. Write $\wp = z^{-2} + \sum_{n\geq 1} c_n z^{2n}$ with $c_n = (2n+1)G_{2n+2}$. Then $\wp'' = 6z^{-4} + \sum_{n\geq 1} 2n(2n-1)c_n z^{2n-2}$, while $6\wp^2 = 6z^{-4} + 12\sum_{n\geq 1} c_n z^{2n-2} + 6\sum_{n\geq 2}\big(\sum_{i+j=n} c_i c_j\big) z^{2n-2}$. Matching the coefficient of z^{2n-2} for $n \geq 3$ (the cases $n = 1, 2$ absorb g_2) gives $2n(2n-1)c_n = 12c_n + 6\sum_{i+j=n} c_i c_j$, i.e.

$$c_n = \frac{3}{(2n+3)(n-2)} \sum_{\substack{i+j=n \\ i,j\geq 1}} c_i c_j \qquad (n \geq 3).$$

Since $c_1 = 3G_4$ and $c_2 = 5G_6$, this recurrence determines every c_n, hence every G_{2n+2}, as a polynomial in G_4 and G_6; for instance $c_3 = \frac{3}{9\cdot 1}c_1^2 = \frac{1}{3}c_1^2$, giving $G_8 = \frac{3}{7}G_4^2$.

Exercise 7.6. Because $\zeta' = -\wp$ is periodic, $g(z) := \zeta(z + \omega_k) - \zeta(z)$ has $g' = 0$, so $g \equiv \eta_k$ is constant; at $z = -\omega_k/2$, oddness of ζ gives $\eta_k = \zeta(\omega_k/2) - \zeta(-\omega_k/2) = 2\zeta(\omega_k/2)$. Since $(\log\sigma)' = \sigma'/\sigma = \zeta$, the function $\log\frac{\sigma(z+\omega_k)}{\sigma(z)}$ has derivative $\zeta(z+\omega_k) - \zeta(z) = \eta_k$, so it equals $\eta_k z + c_k$ for some constant c_k, i.e. $\sigma(z+\omega_k) = \sigma(z)\,e^{\eta_k z + c_k}$. Evaluate at $z = -\omega_k/2$: the left side is $\sigma(\omega_k/2)$, the right $\sigma(-\omega_k/2)e^{-\eta_k\omega_k/2 + c_k} = -\sigma(\omega_k/2)e^{-\eta_k\omega_k/2 + c_k}$ (oddness), and $\sigma(\omega_k/2) \neq 0$ since $\omega_k/2 \notin \Lambda$; hence $e^{c_k - \eta_k\omega_k/2} = -1$ and $\sigma(z+\omega_k) = -\sigma(z)\,e^{\eta_k(z+\omega_k/2)}$.

Exercise 7.7. Using $\sigma(z + \omega_k - c) = -\sigma(z-c)\,e^{\eta_k(z-c+\omega_k/2)}$ for each shift $c \in \{a_i, b_i\}$,

$$\frac{F(z+\omega_k)}{F(z)} = \frac{\prod_{i=1}^n\big(-e^{\eta_k(z-a_i+\omega_k/2)}\big)}{\prod_{i=1}^n\big(-e^{\eta_k(z-b_i+\omega_k/2)}\big)}.$$

The numerator is $(-1)^n e^{\eta_k(nz - \sum a_i + n\omega_k/2)}$ and the denominator $(-1)^n e^{\eta_k(nz - \sum b_i + n\omega_k/2)}$; the factor $(-1)^n$ cancels because the two products have the *same number* of factors (degree zero), and the nz and $n\omega_k/2$ terms cancel, leaving $\exp\big(\eta_k(\sum b_i - \sum a_i)\big)$. This is 1 because $\sum a_i = \sum b_i$ (Abel sum zero). Thus degree zero is used to cancel $(-1)^n$ and the z-dependence, and Abel sum zero to kill the residual exponent. For $[a] - [0]$ with $a \notin \Lambda$ the counts match (degree zero), so the candidate $\sigma(z-a)/\sigma(z-0)$ has the right zeros and poles, but $\sum a_i - \sum b_i = a \notin \Lambda$, so $F(z+\omega_k)/F(z) = e^{\eta_k a} \neq 1$ in general: F is not elliptic, consistent with there being no order-one elliptic function (corollary 4.6).

D.8 Chapter 8

Exercise 8.1. Near $[0]$, the homogeneous representative $(z^3\wp, z^3\wp', z^3)$ has middle entry tending to $-2 \neq 0$, so divide by it: in the chart $Y = 1$ the map is $z \mapsto (X, Z) = (\wp/\wp', 1/\wp')$. From $\wp = z^{-2} + O(z^2)$, $\wp' = -2z^{-3} + O(z)$: $\wp/\wp' = \frac{z^{-2}+\cdots}{-2z^{-3}+\cdots} = -\frac{1}{2}z + O(z^3)$ and $1/\wp' = -\frac{1}{2}z^3 + O(z^5)$. Both are holomorphic at $z = 0$ with value $(0,0) = O$, and $X = \wp/\wp'$ has derivative $-\frac{1}{2} \neq 0$ there, so it is a local holomorphic coordinate and Φ is an immersion at O (indeed z and X agree to first order up to the factor $-\frac{1}{2}$).

Exercise 8.2. Write the curve as $G = Y^2Z - 4X^3 + g_2XZ^2 + g_3Z^3 = 0$. At $O = [0:1:0]$, $\nabla G = (-12X^2 + g_2Z^2,\ 2YZ,\ Y^2 + 2g_2XZ + 3g_3Z^2) = (0,0,1) \neq 0$, so O is smooth with tangent $Z = 0$. Substituting $Z = 0$ into G gives $-4X^3 = 0$, a triple root, so $Z = 0$ meets E at O with multiplicity three; O is a flex. Hence the line $Z = 0$ cuts the divisor $3O$ on E, so $3O = \mathrm{div}(\mathrm{line}/\mathrm{line})$ is principal — concretely $3O \sim 3O$, and any two lines cut linearly equivalent divisors, making $3O$ the class that will be the group identity.

Exercise 8.3. Lemma 8.4 is proved in the text: $\wp - \wp(w)$ has order two, with zeros $w, -w$, so $\wp(z) = \wp(w) \Rightarrow z \equiv \pm w$. Thus the fibers of $\wp : T \to \mathbb{CP}^1$ are the orbits $\{z, -z\}$ of $z \mapsto -z$, giving a bijection $T/\{\pm 1\} \to \mathbb{CP}^1$. The fixed points of $z \mapsto -z$ are the four two-torsion points (exercise 3.2); there the two sheets come together, so they map to the branch values, namely $\wp(\omega_i/2) = e_i$ and $\wp(0) = \infty$.

Exercise 8.4. At $z = 0$, $\Phi(0) = O$. At a half-period $\omega_i/2$, $\wp'(\omega_i/2) = 0$ (proposition 7.8) and $\wp(\omega_i/2) = e_i$, so $\Phi(\omega_i/2) = (e_i, 0)$. The points of E fixed by $(x, y) \mapsto (x, -y)$ satisfy $y = -y$, i.e. $y = 0$ (the three $(e_i, 0)$) or the point at infinity O; these are exactly the four images, matching the two-torsion.

Exercise 8.5. The map $\iota_E(x, y) = (x, -y)$ preserves $y^2 = 4x^3 - g_2x - g_3$ (the right side depends only on x), fixes O, and satisfies $\iota_E^2 = \mathrm{id}$, so it is a holomorphic involution. Its invariant function is $x = \wp$, and two points share an x-value iff they are $\pm$-related, so the quotient E/ι_E is the x-line $\mathbb{CP}^1$. Under Φ, ι_E pulls back to the map with $\wp(z) \mapsto \wp(z)$, $\wp'(z) \mapsto -\wp'(z) = \wp'(-z)$, i.e. $z \mapsto -z$. Since the double cover $\wp : T \to \mathbb{CP}^1$ has deck group $\{\pm 1\}$ of order two, its unique nontrivial deck transformation is $z \mapsto -z$, which is ι_E.

D.9 Chapter 9

Exercise 9.1. Take $D_1 = [a] - [0]$ with $a \notin \Lambda$: $\deg D_1 = 0$ but $\mathfrak{s}(D_1) = [a] \neq 0$, so D_1 is not principal (a principal divisor of this shape would be an order-one elliptic function, excluded by corollary 4.6). Take $D_2 = 2[0]$: $\mathfrak{s}(D_2) = 2[0] = 0$ but $\deg D_2 = 2 \neq 0$, so D_2 is not principal. Thus each condition can hold while the other fails, so neither alone characterizes principality.

Exercise 9.2. Corollary 9.3 is immediate from theorem 9.1: $[p]+[q]+[r]-3[0]$ has degree 0, and its Abel sum is $[p]+[q]+[r]$, zero iff $p+q+r \in \Lambda$. For existence/uniqueness of R: given $P = [p]$, $Q = [q]$, the equation $P + Q + R = O$ in the group T has the unique solution $R = -(P + Q) = [-(p + q)]$. Through the isomorphism Φ this R is a well-defined point of the cubic, the reflection of the third intersection.

Exercise 9.3. Label the sides of P_a as in section 4.3. Pairing side I with III (translates by ω_2, reversed) and using ω_2-periodicity of F'/F, $\int_{\mathrm{I}} + \int_{\mathrm{III}} = -\omega_2 \int_a^{a+\omega_1} (F'/F)\, dz$; pairing II with IV gives $+\omega_1 \int_a^{a+\omega_2} (F'/F)\, dz$. Each remaining integral is the change of a branch of $\log F$ along a side; since $F(a + \omega_k) = F(a)$, the real part returns to its start and the imaginary part changes by 2π times an integer (the winding number, exercise 4.5), so each integral lies in $2\pi i\,\mathbb{Z}$. Hence $\oint z(F'/F)\, dz = \omega_1 \cdot 2\pi i\, k_2 - \omega_2 \cdot 2\pi i\, k_1 \in 2\pi i\, \Lambda$.

Exercise 9.4. The divisor $D = [a] + [b] - [a+b] - [0]$ has $\deg D = 2 - 2 = 0$ and Abel sum $\mathfrak{s}(D) = [a] + [b] - [a+b] - [0] = (a+b) - (a+b) = 0$ in T. By sufficiency in theorem 9.1, D is principal, so some elliptic F has $\operatorname{div}(F) = D$; its zeros a, b and poles $a+b, 0$ encode the group sum $a+b$. By contrast $[a] - [0]$ with $a \neq 0$ has Abel sum $[a] \neq 0$, so it is *not* principal — there is no elliptic function with a single simple zero and a single simple pole, consistent with corollary 4.6.

Exercise 9.5. Both deg and $\mathfrak{s}$ are homomorphisms $\operatorname{Div}(T) \to \mathbb{Z}$ and $\operatorname{Div}(T) \to T$, so $(\deg, \mathfrak{s})$ is a homomorphism to $\mathbb{Z} \oplus T$. It is surjective: $(n, [a]) = (\deg, \mathfrak{s})\big(n[0] + ([a] - [0])\big)$, since that divisor has degree n and Abel sum $[a]$. Its kernel is the divisors with $\deg = 0$ and $\mathfrak{s} = 0$, which by theorem 9.1 are exactly $\operatorname{Prin}(T)$. The first isomorphism theorem then gives $\operatorname{Div}(T)/\operatorname{Prin}(T) = \operatorname{Pic}(T) \cong \mathbb{Z} \oplus T$, matching exercise 5.3.

D.10 Chapter 10

Exercise 10.1. The vertical line $x = c$ is $X - cZ = 0$. On E, $y^2 = 4c^3 - g_2 c - g_3 =: y_0^2$, giving the two affine points $(c, \pm y_0)$; and the line passes through $O = [0:1:0]$ (it contains the direction $[0:1:0]$, since $X - cZ$ vanishes there). So $L \cdot E = (c, y_0) + (c, -y_0) + O$, which has degree three as required. By theorem 10.5 these three sum to O: $(c, y_0) \oplus (c, -y_0) \oplus O = O$, i.e. $(c, y_0) \oplus (c, -y_0) = O$. Thus $(c, -y_0) = -(c, y_0)$, the reflection across the x-axis of section 8.6.

Exercise 10.2. If m, m' both satisfy $m(P), m'(P) \neq 0$, then m/m' is a rational function on E with neither zero nor pole at P, hence $\operatorname{ord}_P(m/m') = 0$. Therefore $\operatorname{ord}_P(\ell/m) = \operatorname{ord}_P(\ell/m') + \operatorname{ord}_P(m'/m) = \operatorname{ord}_P(\ell/m')$, so the multiplicity does not depend on the auxiliary form.

Exercise 10.3. By definition $I_P(L, E) = \operatorname{ord}_P(\ell/m) \geq 1$ exactly when $P \in L$. Choose a local coordinate t on E at P and parametrize E by $\gamma(t)$ with $\gamma(0) = P$. Then $I_P(L, E) \geq 2$ iff $(\ell/m)(\gamma(t))$ vanishes to order ≥ 2, iff $\frac{d}{dt}(\ell/m)(\gamma(t))|_0 = 0$, iff the line $\ell = 0$ contains the tangent direction $\gamma'(0)$ — i.e. L is the tangent line to E at P. When L is tangent, $I_P(L, E) \geq 2$ absorbs two of the three intersections (proposition 10.2), leaving one further point R' (possibly $= P$ at a flex), so $L \cdot E = 2P + R'$.

Exercise 10.4. With $v = 4u^3 - g_2 uv^2 - g_3 v^3$ and $v = O(u^3)$ near 0: substituting the ansatz $v = c_3 u^3 + c_5 u^5 + \cdots$ (only odd powers by the symmetry $u \mapsto -u$, $v \mapsto -v$), the terms $g_2 uv^2$ and $g_3 v^3$ are $O(u^7)$, so to leading order $v = 4u^3 + O(u^5)$; hence $c_3 = 4$ and $\operatorname{ord}_O(Z/Y) = \operatorname{ord}_O v = 3$. Since Y is a unit at O, $\operatorname{ord}_O(Z|_E) = 3$ and $\operatorname{div}_0(Z|_E) = 3O$.

Exercise 10.5. Let R be the third intersection of the line through P and Q with E, so $L \cdot E = P + Q + R$. By theorem 10.5, $P \oplus Q \oplus R = O$, hence $R = -(P \oplus Q)$ and $P \oplus Q = -R$, the reflection of R. When $P = Q$ the chord degenerates; the correct line is the *tangent* to E at P, which meets E at $2P + R$ (exercise 10.3), giving $2P = -R$ — the doubling rule.

D.11 Chapter 11

Exercise 11.1. If $\boxdot$ makes Φ a homomorphism, then $\Phi(u+v) = \Phi(u) \boxdot \Phi(v)$; since Φ is a bijection, every $P = \Phi(u)$, $Q = \Phi(v)$, and $P \boxdot Q = \Phi(u+v)$ is determined — this is $\oplus$. It is abelian because $u + v = v + u$ in $\mathbb{C}$; $P \oplus O = \Phi(u+0) = \Phi(u) = P$; and $\Phi(u) \oplus \Phi(-u) = \Phi(0) = O$, so $\ominus\Phi(u) = \Phi(-u)$.

Exercise 11.2. The line $L = PO$ through $P = (x, y)$ and $O = [0 : 1 : 0]$ is the vertical line $x = x_P$ (the point at infinity O lies on every vertical line). It meets E at (x, y), $(x, -y) = \ominus P$, and O; so $L \cdot E = P + \ominus P + O$. By theorem 10.5, $P \oplus \ominus P \oplus O = O$, which is automatic, and reading the chord–tangent recipe: the third point of PO is $\ominus P$, and $P \oplus O = \ominus(\ominus P) = P$. Hence O is the identity.

Exercise 11.3. By theorem 11.3, $\boxplus = \oplus$, and $\oplus$ is associative because Φ carries it to addition on $\mathbb{C}/\Lambda$, which is associative since addition in $\mathbb{C}$ is: $(P \oplus Q) \oplus S = \Phi((u+v)+w) = \Phi(u+(v+w)) = P \oplus (Q \oplus S)$. It reduces precisely to associativity of $+$ in $\mathbb{C}$. The direct geometric proof is hard because associativity of the chord–tangent rule is a nonobvious incidence statement about nine points and lines (the Cayley–Bacharach/9-point theorem); the torus makes it a triviality by linearizing the group.

Exercise 11.4. By proposition 11.4 the operation $\boxplus$ is defined using only lines and O on E, so it is one and the same regardless of how E was presented. Theorem 11.3 applied to Φ gives $\oplus = \boxplus$, and applied to Φ' gives $\oplus' = \boxplus$; hence $\oplus = \oplus'$. Since both Φ and Φ' are isomorphisms onto the same $(E, \boxplus)$, the composite $\Phi^{-1} \circ \Phi'$ is a group isomorphism $\mathbb{C}/\Lambda' \to \mathbb{C}/\Lambda$, forcing Λ and Λ' to be homothetic (proposition 2.7): the same cubic comes only from homothetic lattices.

Exercise 11.5. Under Φ, the translation $P \mapsto P \oplus A$ corresponds to $z \mapsto z + a$ on T, where $a = \Phi^{-1}(A)$; this is the biholomorphism τ_a of proposition 3.2. Conjugating by Φ, $P \mapsto P \oplus A = \Phi(\Phi^{-1}(P) + a)$ is a composition of biholomorphisms, hence a biholomorphism of E. The addition map $E \times E \to E$ and inversion are holomorphic (transported from T), so E is a complex Lie group acting on itself by holomorphic translations.

D.12 Chapter 12

Exercise 12.1. Substituting $y = mx + k$ into $y^2 = 4x^3 - g_2x - g_3$ gives $4x^3 - m^2x^2 - (g_2 + 2mk)x - (g_3 + k^2) = 0$, a cubic whose three roots are x_P, x_Q, x_R. By Vieta's formula the sum of the roots is the negative of the x^2-coefficient over the leading coefficient: $x_P + x_Q + x_R = m^2/4$, hence $x_R = \frac{m^2}{4} - x_P - x_Q$. The third intersection is $R = (x_R, mx_R + k)$, and $P \oplus Q = -R = (x_R, -(mx_R + k))$.

Exercise 12.2. Differentiating $y^2 = 4x^3 - g_2x - g_3$ implicitly: $2y\,y' = 12x^2 - g_2$, so the tangent slope at $P = (x_P, y_P)$ is $m = (12x_P^2 - g_2)/(2y_P)$. The tangent meets E at P with multiplicity two, so in the cubic of the previous exercise x_P is a double root and $x_P + x_P + x_{2P} = m^2/4$, giving $x_{2P} = \frac{m^2}{4} - 2x_P$. If $y_P = 0$, the slope m is infinite: the tangent is vertical, meeting E at P (doubly) and O, so $R = O$ and $2P = \ominus O = O$, matching that P is two-torsion.

Exercise 12.3. The line $y = 0$ (the x-axis) meets $E : y^2 = 4x^3 - g_2x - g_3$ where $4x^3 - g_2x - g_3 = 0$, i.e. at $(e_1, 0), (e_2, 0), (e_3, 0) = T_1, T_2, T_3$; these are exactly the nonzero two-torsion points. As three points on a line, theorem 10.5 gives $T_1 \oplus T_2 \oplus T_3 = O$. In $E[2] \cong (\mathbb{Z}/2\mathbb{Z})^2$ the three nonzero elements are T_1, T_2, T_3, and indeed the sum of all nonzero elements of $(\mathbb{Z}/2\mathbb{Z})^2$ is $(1,1)+(1,0)+(0,1) = (0,0)$, consistent.

Exercise 12.4. If $3P = O$, then in the divisor language $3[P]-3[O]$ is principal, so by theorem 10.5 (with $P = Q = R$) the line L with $L \cdot E = 3P$ exists — the tangent at P meets E only at P, with multiplicity three, which is the definition of an inflection point. Conversely a flex P has $L \cdot E = 3P$, so $3P\oplus = O$, i.e. $3P = O$. The point O is a flex by proposition 8.3 (the line at infinity meets E at O with multiplicity three), consistent with $3O = O$.

D.13 Chapter 13

Exercise 13.1. With $b = y_1 - mx_1$, $-(mx_3 + b) = -mx_3 - (y_1 - mx_1) = m(x_1 - x_3) - y_1$. The second form is preferable numerically because it uses the input point (x_1, y_1) directly and avoids forming the intercept b as a separate quantity (one fewer subtraction, less cancellation error); it also makes the geometric reading clear — start at P, move by slope m over the run $x_1 - x_3$, then the sign records the reflection.

Exercise 13.2. The slope $m = (y_2 - y_1)/(x_2 - x_1) = (y_1 - y_2)/(x_1 - x_2)$ is unchanged by swapping $P \leftrightarrow Q$. Then $x_3 = m^2/4 - x_1 - x_2$ is symmetric in x_1, x_2. For y_3, note the line intercept $b = y_1 - mx_1 = y_2 - mx_2$ is the same computed from either point (both lie on the line), so $y_3 = -(mx_3 + b)$ is manifestly symmetric. Hence $P \oplus Q = Q \oplus P$.

Exercise 13.3. On E, $y_2^2 - y_1^2 = 4(x_2^3 - x_1^3) - g_2(x_2 - x_1)$. Factoring, $(y_2 - y_1)(y_2 + y_1) = (x_2 - x_1)\big(4(x_2^2 + x_2x_1 + x_1^2) - g_2\big)$, so for $x_2 \neq x_1$,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4(x_2^2 + x_2x_1 + x_1^2) - g_2}{y_2 + y_1}.$$

As $Q \to P$ (so $x_2 \to x_1$, $y_2 \to y_1 \neq 0$) this tends to $(4 \cdot 3x_1^2 - g_2)/(2y_1) = (12x_1^2 - g_2)/(2y_1)$, the tangent slope. Correspondingly Vieta's $x_1 + x_2 + x_R = m^2/4$ becomes $2x_1 + x_R = m^2/4$, so the addition formula limits to the doubling formula.

Exercise 13.4. If $x_1 = x_2$ then $y_1^2 = y_2^2$ (both equal $4x_1^3 - g_2x_1 - g_3$), so $y_2 = \pm y_1$; since $P \neq Q$, $y_2 = -y_1$, i.e. $Q = \ominus P$. The slope $(y_2 - y_1)/(x_2 - x_1)$ has zero denominator and is undefined — the line is vertical — and the third intersection is O, so $P \oplus Q = \ominus O = O$. Pseudocode:

```
add(P, Q):
if P = O: return Q;  if Q = O: return P
if x1 = x2:
if y1 = -y2:  return O  // includes y1 = y2 = 0
else:  return double(P)  // here P = Q, y1 != 0
else:
m = (y2-y1)/(x2-x1)
x3 = m^2/4 - x1 - x2
return (x3, m*(x1-x3)-y1)
```

Exercise 13.5. With $x_{2P} = \frac{25}{24}$ and $y_{2P} = -\frac{35}{24\sqrt{6}}$: $y_{2P}^2 = \frac{1225}{576 \cdot 6} = \frac{1225}{3456}$, while $4x_{2P}^3 - 4x_{2P} = 4 \cdot \frac{15625}{13824} - \frac{100}{24} = \frac{62500}{13824} - \frac{57600}{13824} = \frac{4900}{13824} = \frac{1225}{3456}$. They agree, so $2P \in E$. Its inverse is the reflection $\ominus(2P) = \big(\frac{25}{24}, +\frac{35}{24\sqrt{6}}\big)$.

D.14 Chapter 14

Exercise 14.1. $h = \wp' - A\wp - B$ is a $\mathbb{C}$-linear combination of $\wp', \wp, 1$, hence elliptic. Its poles can only be at lattice points; at 0, $\wp'$ has a pole of order three and $\wp$ of order two, so h has a pole of order exactly three (the $\wp'$ term dominates). Thus h has order three, and by theorem 4.7 exactly three zeros. By the choice of A, B, $h(u) = \wp'(u) - A\wp(u) - B = 0$ and likewise $h(v) = 0$, so u, v are zeros. The case $u \equiv \pm v$ is excluded because then $\wp(u) = \wp(v)$ and the slope A has a zero denominator (for $u \equiv v$ one must instead take the limit, giving the duplication formula; for $u \equiv -v$ the sum $u + v \equiv 0$ is a pole of $\wp$).

Exercise 14.2. Expanding along the first column, the determinant equals

$$\big[\wp(v)(-\wp'(u+v)) - \wp(u+v)\wp'(v)\big] - \big[\wp(u)(-\wp'(u+v)) - \wp(u+v)\wp'(u)\big] + \big[\wp(u)\wp'(v) - \wp(v)\wp'(u)\big] = 0,$$

a linear equation in the row $(1, \wp(u+v), -\wp'(u+v))$. Geometrically it says $(\wp(u+v), -\wp'(u+v))$ lies on the line through P, Q; its x-coordinate is therefore the third root $\frac{A^2}{4} - \wp(u) - \wp(v)$ of the intersection cubic, which is $\wp(u+v)$ — exactly theorem 14.1.

Exercise 14.3. Differentiating $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ gives $2\wp'\wp'' = (12\wp^2 - g_2)\wp'$, so $\wp'' = 6\wp^2 - \frac{1}{2}g_2$. Hence

$$\frac{\wp''(u)}{\wp'(u)} = \frac{6\wp(u)^2 - \frac{1}{2}g_2}{\wp'(u)} = \frac{12\wp(u)^2 - g_2}{2\wp'(u)} = \frac{12x^2 - g_2}{2y},$$

with $x = \wp(u)$, $y = \wp'(u)$ — precisely the tangent slope of theorem 13.2. So the analytic duplication slope and the geometric tangent slope agree.

Exercise 14.4. Swapping $u \leftrightarrow v$ negates both $\wp'(u) - \wp'(v)$ and $\wp(u) - \wp(v)$, leaving the slope A and hence A^2 unchanged; the term $-\wp(u) - \wp(v)$ is symmetric, so theorem 14.1 is symmetric. As $v \to 0$, $\wp(v) \sim v^{-2} \to \infty$ and $\wp'(v) \sim -2v^{-3}$, so $A \sim \wp'(v)/(-\wp(v)) \sim (-2v^{-3})/(-v^{-2}) = 2v^{-1}$ and $A^2/4 \sim v^{-2} \sim \wp(v)$; the divergent $-\wp(v)$ and $+A^2/4$ cancel to leading order, leaving $\wp(u+0) = \wp(u)$.

Exercise 14.5. Replacing v by $-v$ in theorem 14.1 and using $\wp(-v) = \wp(v)$, $\wp'(-v) = -\wp'(v)$,

$$\wp(u-v) = -\wp(u) - \wp(v) + \tfrac{1}{4}\Big(\tfrac{\wp'(u)+\wp'(v)}{\wp(u)-\wp(v)}\Big)^2.$$

Adding to the formula for $\wp(u+v)$, the two square terms combine via $(a-b)^2 + (a+b)^2 = 2a^2 + 2b^2$ with $a = \wp'(u)$, $b = \wp'(v)$:

$$\wp(u+v) + \wp(u-v) = -2\wp(u) - 2\wp(v) + \frac{\wp'(u)^2 + \wp'(v)^2}{2(\wp(u) - \wp(v))^2}.$$

By the differential equation $\wp'(u)^2 = 4\wp(u)^3 - g_2\wp(u) - g_3$ (and likewise for v), the numerator is a polynomial in $\wp(u), \wp(v)$, so the right-hand side is rational in $\wp(u), \wp(v)$.

D.15 Chapter 15

Exercise 15.1. $\big(\frac{28\sqrt{6}}{125}\big)^2 = \frac{784\cdot 6}{15625} = \frac{4704}{15625}$. On the other side, with $x_3 = -\frac{24}{25}$, $4x_3^3 - 4x_3 = 4\big(-\frac{13824}{15625}\big) + \frac{96}{25} = -\frac{55296}{15625} + \frac{60000}{15625} = \frac{4704}{15625}$. They agree, so $P \oplus Q \in E$. Its inverse is $\ominus(P \oplus Q) = \big(-\frac{24}{25}, +\frac{28\sqrt{6}}{125}\big)$.

Exercise 15.2. The tangent at P has slope $m = (12x_1^2 - g_2)/(2y_1)$ and intercept $b = y_1 - mx_1$. It meets E at P doubly, so x_1 is a double root of $4x^3 - m^2x^2 - (g_2 + 2mb)x - (g_3 + b^2) = 0$; matching against $4(x - x_1)^2(x - x_3)$, the x^2-coefficient gives $2x_1 + x_3 = m^2/4$, so $x_3 = \frac{m^2}{4} - 2x_1$. Reflecting the third point, $2P = (x_3, -(mx_3 + b)) = \big(\frac{m^2}{4} - 2x_1,\ -m(\frac{m^2}{4} - 2x_1) - b\big)$, as in theorem 13.2.

Exercise 15.3. By proposition 15.1 with $g_2 = 4$, $g_3 = 0$: $a = -1$, $b = 0$, so $E' : Y^2 = X^3 - X$. In these coordinates $P = (2, \sqrt{6})$, $Q = (-\frac{1}{2}, \frac{\sqrt{6}}{4})$ (halving the y's), and the slope is $M = m/2 = \frac{3\sqrt{6}}{10}$, $M^2 = \frac{54}{100}$. Then $X_3 = M^2 - X_1 - X_2 = \frac{27}{50} - \frac{3}{2} = -\frac{24}{25}$ (the same x) and $Y_3 = M(X_1 - X_3) - Y_1 = -\frac{14\sqrt{6}}{125}$. Mapping back by $y = 2Y$ gives $y_3 = -\frac{28\sqrt{6}}{125}$, matching section 15.2.

Exercise 15.4. *Pitfall 1.* Compute $P \oplus (\ominus P)$ for $P = (2, 2\sqrt{6})$: the line is the vertical $x = 2$, which has *no* third finite intersection. If O is forgotten the sum looks undefined; including O, the line meets E at $P, \ominus P, O$, so $P \oplus (\ominus P) = O$. *Pitfall 2.* In section 15.2 the third intersection is $R = (-\frac{24}{25}, +\frac{28\sqrt{6}}{125})$; reporting R as $P \oplus Q$ gives the reflection of the true answer — wrong by a sign in y. The correct sum is $\ominus R = (-\frac{24}{25}, -\frac{28\sqrt{6}}{125})$. *Pitfall 7.* On the singular cubic $y^2 = 4x^3$ ($g_2 = g_3 = 0$, $\Delta = 0$) the origin $(0, 0)$ is a cusp; the tangent there is not defined and the doubling formula divides by $y = 0$. With $\Delta \neq 0$ (corollary 7.10) no such point exists and every step is well defined.

Exercise 15.5. Take $S = (-1, 0)$, a two-torsion point. From section 15.2, $P \oplus Q = (-\frac{24}{25}, -\frac{28\sqrt{6}}{125})$. Computing with the formulas (slope cases handled as in exercise 13.4):

$$(P \oplus Q) \oplus S = (49, 280\sqrt{6}) = P \oplus (Q \oplus S),$$

so the two bracketings agree, confirming associativity numerically (guaranteed by remark 11.5). Both intermediate sums lie on E, as a direct substitution checks.

D.16 Chapter 16

Exercise 16.1. With $g_2 = 4$, $g_3 = 0$: $\Delta = g_2^3 - 27g_3^2 = 64 - 0 = 64$, and

$$j = 1728\,\frac{g_2^3}{\Delta} = 1728\,\frac{64}{64} = 1728.$$

For a curve with $g_2 = 0$ (and $g_3 \neq 0$), $j = 1728 \cdot 0/\Delta = 0$: these are the curves with $j = 0$, having complex multiplication by $\mathbb{Z}[\zeta_3]$.

Exercise 16.2. A point $[z]$ satisfies $n[z] = [0]$ iff $nz \in \Lambda$ iff $z \in \frac{1}{n}\Lambda$. Hence $E[n] = \frac{1}{n}\Lambda/\Lambda$. The map $\frac{1}{n}\Lambda/\Lambda \to \Lambda/n\Lambda$, $z \mapsto nz$, is an isomorphism, and $\Lambda \cong \mathbb{Z}^2$ gives $\Lambda/n\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^2$. The map $[n] : [z] \mapsto [nz]$ has fibers the cosets of $E[n]$, of which there are $|E[n]| = n^2$; so $[n]$ is n^2-to-one and has degree n^2.

Exercise 16.3. From $G_k(\alpha\Lambda) = \sum_{\lambda \neq 0}(\alpha\lambda)^{-k} = \alpha^{-k}G_k(\Lambda)$: $g_2 = 60G_4 \mapsto \alpha^{-4}g_2$ and $g_3 = 140G_6 \mapsto \alpha^{-6}g_3$. Then $g_2^3 \mapsto \alpha^{-12}g_2^3$, $g_3^2 \mapsto \alpha^{-12}g_3^2$, so $\Delta = g_2^3 - 27g_3^2 \mapsto \alpha^{-12}\Delta$. Finally $j = 1728g_2^3/\Delta \mapsto 1728\,\alpha^{-12}g_2^3/(\alpha^{-12}\Delta) = j$: invariant. In modular-forms language g_2, g_3, Δ, j have weights $4, 6, 12, 0$.

Exercise 16.4. For $E : y^2 = x^3 + 1$ over $\mathbb{F}_5$, the squares are $\{0, 1, 4\}$. Checking each x: $x = 0 \Rightarrow y^2 = 1$, $y = \pm 1$; $x = 1 \Rightarrow y^2 = 2$, none; $x = 2 \Rightarrow y^2 = 9 = 4$, $y = \pm 2$; $x = 3 \Rightarrow y^2 = 28 = 3$, none; $x = 4 \Rightarrow y^2 = 65 = 0$, $y = 0$. The affine points are $(0, 1), (0, 4), (2, 2), (2, 3), (4, 0)$ — five of them — plus O, so $\#E(\mathbb{F}_5) = 6$. Then $a_5 = 5 + 1 - 6 = 0$ and $|a_5| = 0 \leq 2\sqrt{5} \approx 4.47$, satisfying Hasse. The point $(4, 0)$ has order 2; a group of order 6 with an element of order 2 is $\mathbb{Z}/6\mathbb{Z}$, so $E(\mathbb{F}_5) \cong \mathbb{Z}/6\mathbb{Z}$.

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